# The Effects of "Buy American": Electric Vehicles and the Inflation Reduction Act

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**Abstract**

We provide the first ex post microeconomic welfare analysis of the electric vehicle (EV) tax credits in the Inflation Reduction Act (IRA). Relative to pre-IRA policy, the credits generated $1.92 in domestic benefits per dollar of government spending. Relative to having no EV credits, they yielded $1.10 in domestic benefits per dollar of government spending, with subsidy spending of $32,100 per additional EV. A leasing loophole that sidestepped domestic content rules created negative domestic benefits. A prominent example of green industrial policy, the credits harmed foreign countries by shifting surplus to domestic producers and helped them by decreasing $CO_2$ emissions.

**JEL Codes:** F18, H23, L11, Q58.

**Keywords:** Electric vehicles, Inflation Reduction Act, green industrial policy, trade restrictions, environmental taxes and subsidies, trade and the environment.

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# 1 Introduction

The Inflation Reduction Act (IRA) was forecast to cost up to a trillion dollars, making it among the largest potential climate investments in history (Bistline, Mehrotra, and Wolfram 2023). The Biden Administration also designed the IRA to protect domestic manufacturing, secure supply chains, and survive future elections. The IRA used uniform tax credits for low-carbon technologies made in the US and allied countries. Tax credits for electric vehicles (EVs), a central component of the IRA, were forecast to cost $70 to $390 billion over a decade (Bistline, Mehrotra, and Wolfram 2023; CRFB 2023). The One Big Beautiful Bill Act ended the EV credits in September 2025.

The EV tax credits offer a fascinating case study of combining climate and industrial policy. To encourage a transition to EVs while reducing China's dominance of manufacturing and battery supply chains, the IRA restricts tax credits for *purchased* EVs to vehicles assembled in North America that contain sufficient battery inputs from the US or allies. Equally generous credits to companies for *leasing* EVs lack these restrictions, generating a "leasing loophole." The IRA is part of a broader "Buy American" federal policy goal. Yablon (2023) summarizes tensions this created:

> Many of the US's allies like Japan, Canada, and especially the European Union have not been all-in. They see the Biden administration's signature accomplishments – such as the Inflation Reduction Act (IRA) – less as long-awaited efforts to finally make good on promises of climate action and more as a threat to the ability of places like Europe to attract investment themselves. ... After decades of pleading with America to finally take action on issues such as climate, why are our closest partners so annoyed at us now that we're actually doing what they asked?

French President Macron, for example, both lauded the IRA's environmental "common objective" and lamented its "super aggressive" industrial policy (France 24 2022; Rose and Mason 2022).

This paper's primary goal is to characterize the market and welfare effects of actual and counterfactual policy in this context for a social planner that may prioritize domestic over foreign producer surplus. We discuss tradeoffs that this prominent case of green industrial policy creates between trade and the environment and between domestic and foreign interests. The EV credits combine a non-cooperative, national perspective on producer surplus that encourages profit shifting with more cooperative investment in the global public good of $CO_2$ mitigation.

Our analysis uses uniquely extensive data. We combine proprietary transaction-level dealership records from Cox Automotive, monthly national new vehicle registrations by detailed model trim ("submodel") and lease status from Experian, supply conditions data from Edmunds, second choice surveys from Strategic Vision, registration microdata from California and Texas, web scraping records for Tesla, our surveys of over 250 dealerships, air pollution tests for gasoline vehicles (GVs), and credit eligibility lists from the Treasury Department.

We first describe externalities for EV and GV submodels. For each submodel, we calculate $CO_2$ emissions from manufacturing and scrap ("cradle to grave"), $CO_2$ and local air pollution emissions from driving, externalities from fatal accidents, and energy pricing externalities from gasoline taxes

and electricity markups. We assume a $241 social cost of carbon (US EPA 2023) and other marginal damages building on Holland et al. (2016). We also report results under a US domestic social cost of carbon. We find heterogeneity within EV submodels comparable to that within GV submodels. This finding conflicts with the prevailing policy approach of promoting substitution from GVs to EVs (vehicle electrification), without considering which EVs consumers choose.

To evaluate the EV tax credits' short-run impacts, we use event study analyses. EV buyers with incomes below thresholds could receive $7,500 income tax credits for buying eligible vehicles. EVs assembled outside of North America lost eligibility in August 2022. Battery sourcing requirements in April 2023 made purchases of many other vehicles lose eligibility. Beginning in January 2023, leased EVs became eligible for credits, regardless of assembly location or income. Regressions suggest that consumers capture a meaningful share of the credits' incidence, though we interpret estimates for purchase prices cautiously due to ongoing supply chain constraints. Driven by the decreased price of leasing relative to buying, EV markets shifted toward leasing throughout 2023. Vehicles assembled outside of North America, which lost credit eligibility in August 2022 but regained it for leasing in January 2023, had remarkable shifts to leasing of 40 to 45 percentage points.

To study counterfactuals and welfare, we use an equilibrium model that has nested logit demand with heterogeneity by income, exogenous choice sets, constant marginal costs, and static Nash-Bertrand pricing. The model analyzes the short- to medium-run—long enough to ignore temporary inventory constraints but short enough to ignore supply chain adjustments, model entry, or learning-by-doing. The demand system allows flexible substitution on the margins most important for evaluating the EV credits: between EVs and GVs, across vehicle classes, between buying and leasing, and across submodels.[1] Our heterogeneous consumer types match vehicle market patterns by income. We estimate substitution parameters to match moments from leasing event studies, second choice data, and price sensitivity estimates from Grieco, Murry, and Yurukoglu (2024).

We find that repealing the IRA EV credits or replacing them with pre-IRA credits pits trade against the environment—these reforms benefit foreign countries by undoing the IRA's profit shifting incentives, but harm foreign countries by increasing $CO_2$ emissions. Eliminating all EV credits increases foreign firms' share of EV registrations by almost a third (increasing trade) and decreases total EV registrations by nearly a fifth (harming the environment). Returning to pre-IRA credits has smaller magnitudes but some similar patterns. Compared to no credits, the IRA spends $32,100 per incremental EV sold, because 77 percent of credit recipients would have chosen an EV even without the credits (i.e., these credits are inframarginal). From the US perspective, moving from no EV credits to the IRA credits has a marginal value of public funds (MVPF) of 1.10 to 1.23, depending on whether the domestic or global cost of carbon is used. But replacing pre-IRA credits with the IRA credits has an MVPF of 1.92 to 2.13, which is higher partly since pre-IRA policy mostly subsidized foreign vehicles. Relative to pre-IRA policy, the IRA EV credits have a higher

[1] Utilization (i.e., miles traveled) and other heterogeneity particularly matter for other policy questions, like taxing gasoline or miles traveled; we design the model to capture the key substitution patterns for the policies we study.

MVPF from the US than from the global planner's perspective. This is unusual for a climate policy, since investing in a global public good like $CO_2$ mitigation typically benefits the global planner more than the domestic planner. Profit shifting reverses that logic—the incentive to shift producer surplus across borders outweighs the free rider incentive that discourages mitigation.

Since the IRA pursued many objectives, we also calculate the credits' impact on employment and manufacturing. Eliminating the credits or returning to the pre-IRA environment decreases US auto assembly and parts employment by 11,500 to 14,400 jobs, from a baseline of about 870,000 workers, and reduces the number of EV batteries assembled domestically by 240,000 to 280,000. The credits spend $165,000 or more per auto industry job created, so this policy is not particularly cost-effective solely as an employment program. The long-run capacity investment needed to produce the estimated increase in US-assembled EVs is $2.3 to $3.8 billion, roughly one or two large plants. The credits thus advance industrial policy objectives but at high fiscal cost. These outcomes largely reflect the same reallocation of sales toward US production that drives profit shifting.

We also study three other sets of counterfactual changes—to the EV credits' trade restrictions, income restrictions, and credit magnitudes. Counterfactuals that equalize trade restrictions between leases and purchases close most of the leasing loophole, though they preserve the incentive to lease to avoid income restrictions. The leasing loophole performs poorly, with an MVPF of 0.49 to 0.67 for imposing the IRA's purchase restrictions on leases because many people who lease foreign EVs to claim credits through the loophole would have bought a domestic or unsubsidized foreign EV instead. Thus, the loophole transfers producer surplus to foreign firms while increasing EV take-up little. Removing trade restrictions from purchases would reduce domestic welfare when evaluated under a domestic social cost of carbon, but benefit the environment and increase US welfare under the global social cost of carbon. We generally find that "Buy American" provisions decrease the credits' environmental benefits. While broader debates contest how trade affects the environment, here, trade increases the market share of a good (EVs) that decreases $CO_2$ emissions.

Counterfactuals that remove the income restrictions have moderate benefits, though they are regressive. Removing the income restrictions increases US EV sales, transfers surplus to domestic producers, decreases $CO_2$ emissions, and obtains an MVPF of 1.34 to 1.61. This reform also decreases leasing among high-income households, since they leased partly to avoid the income restrictions. We estimate that the income restrictions only drove 14 percent of the aggregate shift to leasing, with 86 percent due to trade restrictions.

Finally, we consider counterfactuals that change the EV credits' magnitudes. We calculate the constrained optimal subsidy that maximizes total surplus in our framework, holding fixed the models eligible for purchase credits. We decompose it into three terms: the net distortion (markup minus negative externality); indirect substitution from non-subsidized vehicles; and profit shifting from foreign to domestic firms. We analyze both submodel-specific and uniform subsidies.

The sign and magnitude of these constrained optimal subsidies depend on political economy. Under a global social cost of carbon, optimal policy would tax all vehicles, since both EVs and GVs have negative externalities that exceed markups. If political economy prevents taxes on

GVs, however, the sign of optimal policy reverses—it becomes optimal to subsidize EVs to induce substitution from GVs, which on average have higher externalities. Under simplifying assumptions, including non-distortionary taxation, a uniform EV subsidy of $9,825 maximizes US total surplus. This exceeds the actual $7,500 subsidy, though not dramatically so. The profit-shifting component accounts for 40 percent of this value, so non-cooperative motivations matter. If taxation generates deadweight loss, the optimal subsidy is much lower. Differentiated subsidies reflecting submodel-specific externalities obtain about 40 percent higher welfare effects than uniform subsidies.

Our paper has several important limitations. Our short- to medium-run analyses do not describe the credits' long-term effects, when automakers might adjust supply chains, introduce new models, or benefit from learning by doing. We also ignore US trading partners' potential retaliatory trade restrictions. Short- to medium-run analysis is important given Congressional budgeting horizons, two- to six-year election cycles, rapid developments in EV policies and investments, and the credits' three-year lifetime before the One Big Beautiful Bill Act cut them. Additionally, while we measure externalities carefully, results reflect our externality assumptions. For example, we do not account for EVs' lower noise externalities or tire-wear pollution from heavier vehicles, though we include other weight-related externalities.

We contribute to a nascent literature studying the IRA and to analyses of prior US tax credits for clean vehicles by providing the first ex post empirical microeconomic welfare evaluation of the IRA's EV tax credits. This literature includes policy overviews (Bown 2023; Buckberg 2023), reduced-form and structural evaluations of previous state and federal credits (Chandra, Gulati, and Kandlikar 2010; Sallee 2011; Gallagher and Muehlegger 2011; Jenn, Azevedo, and Ferreira 2013; Jenn, Springel, and Gopal 2018; Clinton and Steinberg 2019; Sheldon and Dua 2019; Xing, Leard, and Li 2021; Muehlegger and Rapson 2022; Lohawala 2023), and ex ante evaluations of the IRA credits (Bistline, Mehrotra, and Wolfram 2023; Cole et al. 2023; Slowik et al. 2023; Hahn et al. 2026). Four working papers model long-run benefits: Linn (2022) models US EV credits with endogenous vehicle entry, Head et al. (2026) model reallocation of EV supply chains in response to the IRA, and Barwick, Kwon, and Li (2024) and Barwick et al. (2025) study Chinese EV tax credits in models of endogenous attributes or learning-by-doing. Cox and Acosta (2023) and Bombardini et al. (2024) study costs of Buy American provisions in federal procurement. Our estimate of subsidy additionality echoes recent estimates of environmental policy additionality in other settings (Arkolakis and Walsh 2023; Aspelund and Russo 2024; Chen, Ryan, and Xu 2024). More broadly, our analysis, soon after the IRA's passage and effective repeal, echoes timely retrospective analysis of Trump tariffs and COVID lockdowns (Fajgelbaum et al. 2020; Flaaen, Hortaçsu, and Tintelnot 2020; Chetty et al. 2024; Gopinath and Neiman 2026).

We also provide the first theoretical and empirical analysis of profit shifting and the environment. We describe optimal domestic subsidies or taxes for differentiated, traded products in concentrated industries, then empirically calculate these subsidies and decompose them into net distortions, indirect substitution, and profit shifting. We also distinguish the role of profit shifting in welfare analysis of the actual credits and contrast welfare consequences for global versus na-

tional planners, for foreign and domestic producer surplus versus environmental externalities, and for local versus global externalities. Unilateral trade policy in imperfectly competitive markets can transfer surplus from foreign to domestic firms (Brander and Spencer 1981; Brander and Spencer 1984; Venables 1985; Bagwell and Staiger 2012). Quantitative models of trade and the environment typically assume perfect competition (Costinot, Donaldson, and Smith 2016; Shapiro 2021; Kortum and Weisbach 2021; Caliendo et al. 2024); trade-environment models with monopolistic competition do not analyze profit shifting (Nordhaus 2015; Shapiro and Walker 2018; Farrokhi and Lashkaripour 2025), though Levaggi and Panteghini (2023) analyze a simple model of multinationals. Trade-environment overviews have little discussion of profit shifting (Copeland and Taylor 2003; Cherniwchan, Copeland, and Taylor 2017; Copeland, Shapiro, and Taylor 2022; Desmet and Rossi-Hansberg 2024; Balboni and Shapiro 2025). Market power makes optimal environmental taxation differ from Pigouvian taxation (Buchanan 1969). A few studies analyze tradable industries without focus on profit shifting (Fowlie, Reguant, and Ryan 2016; Ganapati, Shapiro, and Walker 2020; Hsiao 2026). Profit shifting may especially matter for trade-environment issues because polluting industries have high trade exposure, returns to scale, shipping costs, capital intensity, and other drivers of concentration (Copeland, Shapiro, and Taylor 2022; Shapiro 2025).

Finally, we contribute to the broader literature on auto market environmental regulation (Bento et al. 2009; Fowlie, Knittel, and Wolfram 2012; Jacobsen 2013; Jacobsen and van Benthem 2015; Knittel and Sandler 2018). Diamond (1973) discusses the theory of homogeneous corrective taxes for heterogeneous externalities. Knittel and Sandler (2018), Jacobsen et al. (2020), and Jacobsen et al. (2023) estimate welfare losses from imperfectly pricing heterogeneous externalities from primarily GVs. Holland et al. (2016; 2019; 2020; 2024) measure differences in EV externalities across space due to different fuels in the electric grid. Building on their work, we incorporate market power distortions, international profit shifting, distortionary taxation, and fatal car accidents, and we analyze the social planner's problems from global and national perspectives. These elements prove important—distortions from market power and profit shifting constitute much of the constrained optimal subsidies we calculate. Our equilibrium model builds on Goldberg (1995), Berry, Levinsohn, and Pakes (1999), Goldberg and Verboven (2001), Miravete, Moral, and Thurk (2018) and other work by analyzing buy-lease substitution and constrained optimal subsidies. Leasing accounts for about one-fifth of new US vehicle registrations, and our detailed dealership data and policy-induced price variation in lease versus purchase prices allow us to study this margin.

# 2 Policy Background

## 2.1 Electric Vehicle Markets

An electric vehicle includes any vehicle with an electric motor and a plug. This covers plug-in hybrid electric vehicles (PHEVs), which have both an electric motor and a gasoline engine, and battery electric vehicles (BEVs), which lack a gasoline engine. Gasoline vehicles include any vehicles with a gasoline engine and no plug, including traditional (non-plug-in) hybrids. We exclude fuel cell

vehicles since they accounted for only 0.02 percent of new vehicle sales in 2022–2023.

EVs represented 18 percent of global new light-duty vehicle sales in 2023, up from 2 percent in 2018 (IEA 2024). In 2023, about 60 percent of global new EV sales were in China, Chinese carmakers made half of all global EVs, and China was the world's largest EV exporter (IEA 2024). China also dominates battery supply chains (Leruth et al. 2022).

The Biden Administration responded to China's dominance in global EV markets by emphasizing domestic manufacturing, industrial policy, and secure supply chains. For example, Biden National Economic Council director Brian Deese said that "[we envision] a twenty-first-century American industrial strategy—a strategy to strengthen our supply chains [and] rebuild our industrial base" (Deese 2021). Similarly, Biden National Security Adviser Jake Sullivan said that "clean-energy supply chains are at risk of being weaponized in the same way as oil in the 1970s, or natural gas in Europe in 2022. So through the investments in the Inflation Reduction Act and Bipartisan Infrastructure Law, we're taking action" (Sullivan 2023). In explaining his pivotal vote for the IRA, Senator Joe Manchin (2022) wrote, "the increased risk of geopolitical uncertainty demands that we turn our focus to increasing US energy production and bringing good paying energy and manufacturing jobs back to America."

## 2.2 Clean Vehicle Credits

Clean Vehicle Credits are non-refundable income tax credits of up to $7,500 for buying new plug-in EVs or fuel cell vehicles under 14,000 pounds. The Energy Improvement and Extension Act of 2008 first established EV tax credits under Internal Revenue Code Section 30D. The 2009 American Recovery and Reinvestment Act (ARRA) limited full eligibility to the first 200,000 EVs each manufacturer sold, after which credits phased down to zero over four quarters. Tesla and General Motors (GM) exceeded the 200,000 limit and lost eligibility in 2018 and 2019. Toyota, Ford, BMW, and Stellantis all exceeded 200,000 by mid-2023. Thus, by mid-2023, the pre-IRA policy would have mostly subsidized purchases from foreign manufacturers, since all major domestic EV manufacturers had reached the cap.

Both individual and corporate taxpayers could receive the 30D credits. For example, a business could claim the credit for buying vehicles for its motor pool or for a vehicle it leases to an individual. In January 2024, after our data conclude, buyers became able to claim the credit at point of sale rather than on their tax returns.

The IRA, which became law on August 16, 2022, changed 30D eligibility requirements for both taxpayers and vehicle models. For taxpayers, the IRA required that individual buyers have Adjusted Gross Income (AGI) below $300,000 for married couples filing jointly, $225,000 for household heads, or $150,000 for all other taxpayers, starting January 2023.

Figure 1 illustrates how the IRA changed eligibility of different vehicle models. The first column shows that Tesla and GM vehicles received no credit pre-IRA, since they had exceeded the 200,000 vehicle sales limit (GM owns the brands Chevrolet, Cadillac, GMC, and Buick). The remaining columns reflect the dates of three changes due to the IRA.

The first major change we analyze is that starting August 17, 2022, credit eligibility required final assembly in North America. This excluded the European and Asian models in red in Figure 1. We call these the Excluded August 2022 group. Second, starting January 1, 2023, the policy eliminated the 200,000 vehicle sales limit, which re-included the Tesla and GM vehicles colored in blue; we call these the Included January 2023 group. Contemporaneously, eligibility began requiring Manufacturer's Suggested Retail Prices (MSRPs) below $55,000 (for cars) and $80,000 (for SUVs and trucks), which excluded the Lucid Air, Mercedes-Benz EQS, and Tesla Models S and X.

Third, beginning April 18, 2023, policy added a battery component and minerals requirement beyond the North American assembly requirement. We call the submodels thereby excluded the Excluded/Reduced April 2023 group. This reform split the credit into two parts: $3,750 for satisfying the critical minerals requirement and another $3,750 for satisfying the battery component requirement. Eligibility required at least 40 percent of battery minerals to be extracted or processed in the US or a free trade partner, or recycled in North America, and at least 50 percent of battery components to come from North America. The required percentages increased in 2024 and were scheduled to rise further. In 2023, these requirements reduced or fully eliminated credits for many Ford, Jeep, Rivian, Audi, BMW, and Nissan models, colored in orange. These sourcing requirements illustrate that the credits' trade-related features did not uniformly favor US-owned firms: foreign battery sourcing reduced credits for models of the US-owned Ford, Jeep, and Rivian brands, while the August 2022 assembly requirement had preserved eligibility for North American-assembled models of foreign-owned firms, including certain Nissan, BMW, and Volkswagen vehicles. The trade-related features thus target production location and sourcing rather than ownership per se. Because substituting battery suppliers is generally infeasible in the short run, our analysis timeframe largely excludes such substitution.

Besides amending 30D, the IRA also established a new "Commercial Clean Vehicle Credit" under Internal Revenue Code Section 45W, effective January 1, 2023. Section 45W also offered $7,500 credits for new plug-in EVs under 14,000 pounds with no final assembly, battery content, or buyer income requirements.[2] In late December 2022, the US Treasury announced that businesses could claim the 45W credit for leasing EVs to individuals. Analysts refer to this as the "leasing loophole": individuals who would not qualify for the 30D credit to *purchase* a given vehicle, due to vehicle eligibility or income restrictions, could instead *lease* that same vehicle, and the leasing company would qualify for the 45W credit. The IRA originally scheduled the 30D and 45W credits to sunset after 2032 and did not cap credits before then. The One Big Beautiful Bill Act accelerated the credits' end to September 30, 2025.[3]

[2]The Section 45W credits imposed additional limits in the form of "incremental cost" and "percentage of basis" clauses; however, the IRS effectively nullified the former throughout 2023 via a safe harbor clause, and the latter was generous enough to only meaningfully affect a small minority of EVs (Internal Revenue Service 2023a). We abstract from these details.

[3]We focus on the 2022-2023 period given our data availability. Anticipation of the expiring credits in September 2025 led to substantial EV registrations just before then.

# 3 Data

This section describes our main data; Appendix A.1 provides further details, including on other datasets we use. Our primary dataset is a submodel-by-month panel of registrations and prices from January 2022 through December 2023 for all vehicles below 10,000 pounds. We define a "submodel" as a make × model × trim × powertrain (GV, BEV, PHEV) combination. For example, the Nissan Leaf has five submodels (S, SV, S Plus, SV Plus, and SL Plus). The four Tesla models in our data (S, X, Y, and 3) have a single trim and powertrain, so each constitutes one submodel.

The registrations data come from a nationwide panel of new vehicle registrations by month and submodel that we acquired from Experian. Purchases and leases are recorded separately. We measure the lease share as the ratio of leases to total registrations. We include only registrations for personal use or lease, excluding other sales to businesses. We also exclude heavy-duty vehicles.

Our price data combine three sources, since none is complete alone: lease terms from dealership transaction microdata from Cox Automotive, supplemental lease terms for Tesla vehicles from the Tesla website and other sources, and purchase prices from registration microdata from the California Department of Motor Vehicles (DMV).

The Cox data include purchase prices and lease terms for 6.5 million new vehicle transactions in 2022 and 2023, representing 29 percent of national transactions in those years. Brand-level coverage relative to Experian varies but mostly ranges from 10 to 50 percent; see Appendix Figure A3. Cox's coverage depends on its relationships with dealers, so the data lack sales by direct-to-consumer brands Tesla, Rivian, and Lucid. For leases in the Cox data, we observe lease duration, annual percentage rates (APR), rebates, and monthly payments.[4] We augment the lease terms in the Cox data with Tesla lease terms for base model configurations from online scraping.[5]

The California DMV data cover all new vehicles registered in the state in 2022 and 2023. For purchases in the California DMV data, the reported price is the full price excluding sales tax, license fees, or financing costs. For leases, the California DMV data report no rebates or other contract terms beyond the price. Instead, we combine the lease terms in the Cox and Tesla data with the California purchase prices to compute lease prices that are directly comparable to observed purchase prices for all vehicles. Analyses using Cox purchase prices where available (about 93 percent of registrations in our cross-section) find similar results, as the California and Cox purchase prices have a submodel-by-month correlation of 0.91.

We construct a lease price variable for each transaction, reflecting the discounted lease payments plus the residual, i.e., the car's resale value at the lease term's end. Index transactions by $i$, and let $k = k(i)$ and $t = t(i)$ index the submodel and month for transaction $i$. Define $T_i$ as the lease term in months. Define $d_i$ as the down payment, $m_i$ as the monthly payment, $p_{kt}$ as submodel $k$'s average purchase price, $D_T$ as the percent depreciation over $T$ months, and $\delta_t$ as the discount

[4] Our data record lease rebates, though possibly not those paid directly by manufacturers, whose share we cannot measure.

[5] See Appendix A for a description of these sources. We do not collect lease data for Rivian or Lucid submodels. Rivian does almost no leasing during this period, while Lucid only enters our sample near the end of the panel.

factor. We observe $T_i$, $d_i$, $m_i$ in the Cox data, and take $p_{kt}$ from the California DMV data. For depreciation $D_T$, we follow standard industry estimates that a vehicle loses 20 percent of its value in the first year and 15 percent annually thereafter (Capital One 2024). We construct the discount factor $\delta_t$ from the interest rate on new vehicle loans in month $t$ from Federal Reserve Bank of St. Louis (2024b).[6] Let $s$ index the series of monthly payments. The lease price equals

$$L_i = \underbrace{d_i + \sum_{s=1}^{T_i} \delta_t^{s-1} m_i}_{\text{lease payments}} + \underbrace{\delta_t^{T_i}(1 - D_{T_i})p_{kt}}_{\text{residual}}. \tag{1}$$

Equation (1) lets us compare lease prices to purchase prices, as both reflect a price over a vehicle's life, and compare lease prices across lease terms.[7]

In the submodel-by-month panel, the purchase price and lease price variables represent the mean across transactions. We define a relative lease price variable equal to the submodel-by-month difference between the constructed lease price and the observed purchase price.

Appendix Figure A4 shows the full list of EV submodels and the months in which they appear in our data. We have data on a total of 62 EV submodels as of January 2022, and another 108 entered the market between January 2022 and December 2023.[8] To address compositional effects from submodel entry, many of our analyses look at changes within submodels over time. Many submodels have phase-in or phase-out periods with unusually low registrations when they enter or exit, so our analysis sample excludes months when a submodel's registrations fall below half of its sample average, for both EVs and GVs (Appendix Figure A5 and Appendix Table A4).[9]

Table 1 presents descriptive statistics for the submodel-by-month dataset. Our data cover 1,165 EV or GV submodels across all months. The mean submodel-by-month observation has 1,070 registrations, a purchase price of $51,029, a lease price of $48,497, and a lease share of 25 percent.

In addition to the submodel-by-month panel, we use several other datasets. First, to measure supply chain constraints, we use monthly "days-to-turn" from Edmunds (2024), i.e., the mean duration that vehicles sold in that month were available in the dealership's inventory before being sold. The Edmunds data exclude Tesla, so we collect delivery wait times originally reported on the Tesla website (Tesla 2023; Pritchard 2023; The Internet Archive 2023).

Second, in summer 2023, we surveyed dealerships to determine EV market wait times, prices,

[6] This approach abstracts from loan-level variation in interest rates.

[7] We performed similar analyses using a simpler (though less economically founded) measure, computed as the undiscounted sums of payments while restricting to 36-month leases, the most common duration in our data. This removes the dependence on a choice of discount factors and the residual calculation. We find qualitatively similar results to those that follow from equation (1), particularly for the Excluded August 2022 group's response to the 45W change in January 2023, which serves as an input to the model estimation.

[8] Timing makes it unlikely that the IRA's passage affected the introduction of new models in 2023, for two reasons. Model planning typically requires multiple years' lead time. Additionally, in 2021 and up through August 2022, auto manufacturers faced substantial uncertainty about the likelihood of the IRA's passage or the details of EV tax credits conditional on IRA passage.

[9] The resulting regression coefficients, which our model targets in estimation, differ by less than a percent if we do not filter on phase-in or phase-out.

and other market conditions. Research assistants collected data on the earliest date they could drive home popular EV models. This survey obtained 681 data points from 258 dealerships across eight metro areas and 20 brands, excluding Tesla and Rivian. These interviews guide our discussions of evolving supply chain conditions throughout 2022-2023 and lead the equilibrium model to focus on summer 2023 as a baseline. Appendix A.3 describes the survey in detail.

Third, to identify substitution patterns in our demand model, we use second choice data from the New Vehicle Experience Study (NVES), administered by Strategic Vision. Strategic Vision surveys a large sample of US new vehicle buyers soon after their purchase or lease, and has a response rate of about five percent. Among other questions, the survey asks whether buyers considered any vehicle other than the one they acquired, and if so what model. The survey also asks for their household income. We have 339,560 survey responses from the respondents who acquired a new vehicle in 2022 and 2023.

Fourth, we use a few data sources to measure how changes in EV markets affect domestic auto industry employment, EV battery assembly, and EV plant investment. We calculate these effects by combining our model-implied marginal cost estimates, the US input-output table, and wage data together with assumptions grounded in prior work on the cost breakdown between parts and final assembly; Appendix A.4 provides details.

Finally, we use several sources to measure externalities; Appendix A.2 provides details. Our main analysis assumes that vehicles generate four externalities: $CO_2$ emissions from manufacturing and scrap, $CO_2$ and local air pollution emissions from driving, fatal accidents, and pecuniary and fiscal externalities from energy price distortions arising from explicit or implicit taxes on gasoline and electricity. For $CO_2$ emissions from the value chain of manufacturing plus scrapping vehicles, we use estimates from Argonne National Laboratory (Kelly et al. 2022), which distinguishes three powertrains (BEV, PHEV, GV). We rescale emissions from the manufacturing of EV batteries by the size of each submodel's battery. For $CO_2$ emissions and local air pollution damages from driving vehicles, we combine estimates of local air pollution damages and local grid emission intensities from Holland et al. (2024) with US EPA (2024a) exhaust test and fuel efficiency records. We assume that 63 percent of PHEV miles traveled are on gasoline and 37 percent are on electricity, following Plötz et al. (2020). Since EVs can benefit from future improvements in grid emissions intensity, we conduct a sensitivity analysis using long-run marginal emissions estimates from the National Renewable Energy Laboratory, which reduces estimated EV environmental damages relative to our baseline assumptions. For the mortality cost of car accidents due to each submodel's weight relative to the lightest vehicle, we use regression estimates from Anderson and Auffhammer (2014) and apply the US Department of Transportation (2024) $13.2 million value of a statistical life. For positive energy pricing externalities, we use the utility-specific markups on residential electricity above private marginal cost calculated by Borenstein and Bushnell (2022) for EVs, as well as federal and state gas taxes for GVs. For all externalities, we assume that vehicles have a useful life of 150,000 miles, following US EPA (2014).[10] We inflate all values to July-August 2023 dollars

[10] Available evidence is limited but supports our assumption that EVs and GVs have comparable useful lifetimes.

using the CPI for All Urban Consumers (Federal Reserve Bank of St. Louis 2024a). Our main analysis assumes that externalities including marginal damages do not change over time in real terms, so we do not discount externalities from driving later in a vehicle's life.

We assume a global social cost of carbon of $241 (in July-August 2023 dollars), following US EPA (2023). The "domestic SCC" uses domestic (instead of global) damages from $CO_2$ emissions, equal to $28 per ton, computed using the ratio of the US SCC to global SCC from Ricke et al. (2018) times our CPI-inflated global SCC.

The global SCC represents damages to all regions, years, and pathways due to climate change. The domestic SCC describes the damages to a country resulting from an additional ton of $CO_2$. US regulatory practice itself has varied: the Obama and Biden Administrations conducted regulatory analysis using a global SCC, the first Trump Administration used a domestic SCC (Aldy et al. 2021), and the second Trump Administration effectively uses an SCC of zero. This variation in practice motivates reporting results under both the domestic and global SCC. Using the global SCC for policy design may generate larger environmental benefits than using the domestic SCC, since cooperative use of the global SCC efficiently addresses cross-border externalities (Barrett 1994; Weitzman 2014; Kotchen 2018). Because climate policy is a repeated game across countries, the domestic SCC is likely below the optimal carbon tax for a planner focused only on US welfare. We use the domestic SCC to report domestic costs of $CO_2$ emissions and the global SCC to report global costs of $CO_2$ emissions. This allows us to decompose global benefits of $CO_2$ mitigation into the shares accruing to the US and to foreign countries. We also separately report how each counterfactual affects quantities of $CO_2$, which can be evaluated using any SCC of interest. We view this measurement decision as distinct from the normative decision about whether a country ought to design policy using the domestic or global SCC.

# 4 Descriptive Facts

## 4.1 Moments of Vehicle Data

This section presents descriptive facts, each of which plays a specific role in our analysis. Many of our descriptive figures using the submodel-by-month dataset present fixed-weight arithmetic indexes (see Appendix A.1), so that submodel entry and market share changes do not drive fluctuations in the series.

**Tesla has dominated the US EV market.** Tesla represents 53 percent of US EV sales in 2022 and 2023 (Appendix Figure A1, Panel (a)). Tesla's dominance leads US EVs to have lower externalities and higher markups than other vehicles.

**EV prices and inventory constraints peaked in mid-2022.** EV prices increased in early 2022 and decreased in late 2022. Tesla cut prices in late 2022; other manufacturers followed. Media

Nguyen-Tien et al. (2024) estimate a survival model of 30 million vehicles registered in Great Britain. Their preferred specification indicates that the median lifetime of BEVs in years is 1.5 percent below the median lifetime of GVs, and that the median lifetime miles for BEVs are about 7.5 percent greater than for GVs.

reports attribute the price cuts to easing pandemic-related supply chain bottlenecks and softening demand partly reflecting rising interest rates (Boudette 2023; Cao 2023; Shepardson and Nair 2023). These market-wide price trends, unrelated to the IRA, lead us to use GVs as controls in our event studies in Section 4.2 and to interpret price results cautiously.

These price trends reflect inventory constraints (Figure 2). The average non-Tesla EV sold in July 2022 had sat at the dealership for about 20 days, while the mean vehicle sold in July 2023 had been at the dealership for over 50 days (Panel (a)). Days-to-turn of 15 to 20 may be near just-in-time sales for a typical car, while days-to-turn of 50 represents less severe supply chain constraints.[11] The mean Tesla ordered in July 2022 would be delivered in about 200 days, while the mean Tesla ordered in July 2023 would be delivered in about 15 days (Panel (b)). Our dealership survey found that by summer 2023, market share-weighted EV wait times were below one month for 95 percent of EVs sold via dealerships and 91 percent of EVs were available immediately. Our event study analyses in Section 4.2 therefore consider the relevance of supply constraints, which were tighter for the August 2022 eligibility change than for the April 2023 change. This evidence also leads us to base our equilibrium model in Section 5 around July–August 2023, when supply chain conditions had eased.

**EV buyers have incomes that are high but often below the eligibility limit**. The IRA's 30D credits required buyer incomes below $300,000 for married couples filing jointly, $225,000 for household heads, or $150,000 for all other taxpayers. The NVES data show that 57 percent of EV buyers in 2022 and 2023 reported household income below $200,000, and 78 percent were below $300,000 (Appendix Figure A6); IRS Statistics of Income (SOI) data show similar patterns. Our equilibrium model therefore incorporates income eligibility and income-dependent EV demand.

**Assembly locations shifted little after the 30D eligibility requirement**. Most vehicles excluded from eligibility did not change their assembly sites in our analysis timeframe.[12] This guides our focus on the short- to medium-run, which we treat as a period of fixed international sourcing, so that in the model the credits reallocate sales and profits between existing US- and foreign-assembled vehicles rather than relocating production.

**The EV credits largely represent a "Buy American" policy.** Appendix Figure A2 shows that about 90 percent of EVs assembled in North America are US-assembled. The comparable share for GVs is much lower, at about two-thirds. While the Section 30D EV credits required North American assembly, in practice this meant US assembly in the short run, leading us to interpret this setting as more "Buy American" than "Buy North American."

**Strategic rapid transactions are rare.** Several dealers in our survey proposed a strategic transaction to evade Section 30D restrictions—a customer leases the vehicle, then soon buys out the

[11] For reference, our full data show the average days-to-turn was above 50 for both powertrains all throughout 2019 and 2020, only dipping below this threshold in the supply-constrained period of 2021 and 2022.

[12] A notable exception is the Volkswagen ID.4, which transitioned assembly from Germany to Volkswagen's Chattanooga, Tennessee plant, though this shift began before the IRA's passage. Hyundai, Kia, and Nissan also announced and began constructing US facilities for their EVs around this time. Hyundai's investment in Georgia was announced before the IRA; Kia's Georgia facility was announced in mid-2023. Both began production by 2025. Nissan's announcement had 2026 as the target date for production, but Nissan later postponed it to 2028.

lease. This would obtain the $7,500 Section 45W leasing credit for what was effectively a purchase, a "loophole within a loophole" (Davis 2023; Bernstein 2024). Texas registration data, where we can track vehicles over time, show that this is rare enough to ignore; see Appendix B.1.

**EVs have somewhat lower externalities than GVs**. Table 2 summarizes lifetime submodel-level externality values by powertrain. Using the global SCC, the mean EV imposes about $14,800 in lifetime negative externalities, while the mean GV imposes $19,900 in negative externalities. The gap is smaller than one might expect because, while EVs have $6,600 lower $CO_2$ externalities from driving, they have higher $CO_2$ externalities from manufacturing ($1,200), slightly higher local pollution externalities via the electric grid ($200), and higher accident externalities due to EVs' greater weight ($2,400), though this is offset by more positive energy pricing externalities ($2,200). These values enter our model's welfare analysis directly.

The small gap partially reflects composition within EVs between BEVs and PHEVs. Unlike BEVs, PHEVs on average have $CO_2$ externalities only marginally lower than those of GVs. They also have $3,300 higher accident externalities than GVs, and $1,000 higher accident externalities than BEVs, due to their greater weight. How much this contributes to the total externality difference with GVs depends on the chosen SCC. Under the global SCC, PHEVs have similar negative externalities to GVs, but PHEVs have *higher* average externalities under the domestic SCC. Table 2 also shows that environmental externalities from $CO_2$ exceed those from local air pollution under either SCC, so our subsequent discussions of the "environment" primarily reflect $CO_2$.

EVs and GVs have different distributions across vehicle classes. For example, 40 percent of EVs are large luxury vehicles, compared to 24 percent of GVs.[13] At the same time, short- to medium-run changes to EV subsidy policy may especially affect market shares for the classes of vehicles already in the market. Thus, our comparisons of EV and GV externalities reflect differences in these powertrains' attributes during our period of analysis. We conjecture that in the long run, EVs may approach GVs' distribution across vehicle classes.

**EVs have large heterogeneity in externalities and similar heterogeneity to GVs**. Figure 3 visualizes the substantial variation in externalities across EV submodels, which reflects their different weights and electricity use per mile. For both EVs and GVs, valued at the global SCC, most vehicles have lifetime externalities of $5,000 to $30,000, while valued at the domestic SCC, most vehicles have externalities of $0 to $15,000. Dispersion within EVs is similar to that within GVs. For example, the 90th percentile submodel among EVs generates externalities of $23,700 over its lifetime, while the 10th percentile submodel among EVs generates $10,500 in externalities; the comparable statistics for GVs are $28,200 and $14,500. We also note one other type of heterogeneity: the mostly foreign-assembled vehicles that the 30D credits exclude have slightly larger negative externalities than 30D-eligible submodels ($847 on average), which affects our evaluation of trade restrictions in Section 6.[14]

[13]These statistics use data from the Wards Automotive Yearbook for 2022, which is available at model level, so these represent the share of models but not registrations (Wards Intelligence 2022). We define large luxury vehicles using the provided Wards segmentation as any Luxury Sport/Utility or Luxury Cross/Utility vehicle.

[14]This largely reflects Tesla, and excluding Tesla vehicles reverses the sign. Teslas have lower externalities than the average EV both because they are BEVs rather than PHEVs and because the Tesla Model 3, the most popular

Table 3 shows two measures of the dispersion in externalities across submodels—the coefficient of variation, which equals the standard deviation divided by the mean; and the interdecile ratio, which equals the log of the 90th percentile externality divided by the 10th percentile externality. The table shows that, surprisingly, EVs and GVs have broadly similar dispersion in externalities, driven in part by the variation within EVs across BEVs and PHEVs. Tests fail to reject that dispersion is equal for the two powertrains, other than the coefficient of variation of carbon damages, where EVs have more dispersion than GVs. This finding also holds for total externalities at the global SCC when separately comparing BEVs and PHEVs to GVs, as well as for several components of externalities individually.

The standard deviation of carbon damages across EV submodels is just under twice that across states, averaging across EV submodels (Holland et al. 2016). Geographically, the least carbon-intensive region has about half the carbon emissions per kWh of the most carbon-intensive region; across vehicles, the most carbon-intensive submodels have over three times the carbon damages per mile as the least carbon-intensive submodels.

This similar dispersion matters because many policies globally move consumers from GVs to EVs ("vehicle electrification"). Such policies largely ignore whether consumers purchase high- or low-externality EVs. Policies that make consumers substitute from low-externality GVs to high-externality EVs (e.g., Priuses to Cybertrucks) can increase externalities. Our results show that EV policies may benefit from encouraging substitution from high- to low-externality EVs, just as many GV policies encourage substitution from dirty to clean GVs. Additionally, policies like the IRA that subsidize some but not all EVs could inadvertently increase emissions if they subsidize dirty EVs. The counterfactual analysis in Section 6 revisits these ideas.

## 4.2 Event Studies Around Credit Eligibility Changes

### Methodology

Our event study analyses focus on lease shares and to a lesser extent prices. Estimates for registration quantities were potentially more related to supply conditions and other market trends; see Appendix C.2. We present two graphical analyses: descriptive trends in the fixed-weight index and event study estimates (the event studies use actual values, not the index). The event studies use two-way fixed effects regressions with GVs as controls. In equilibrium, policy changes for any vehicle can affect all vehicles, so the event study estimates capture changes relative to controls, not absolute effects against an unaffected control group.

To formalize the methodology, define $y_{kt}$ as the lease share for submodel $k$ in month $t$. We index tax credit eligibility groups by $e = e(k)$ and define $t^e$ as the month when group $e$'s eligibility changes. We let $s$ index months in event time (relative to $t^e$), and we define $\mathcal{S}^e$ as the set of months in event time over the 24-month sample, excluding month $s = -1$. Let $D_k^e$ represent an indicator for whether submodel $k$ in eligibility group $e$ experienced an eligibility change.[15] We define $\phi_k$ and

EV sedan, is smaller and lighter than typical EVs, many of which are crossovers or SUVs.

[15] When considering purchase prices and registrations, we replace $D_k^e$ with the absolute value of the change in

$\nu_t$ as submodel and month-of-sample fixed effects. We estimate event study graphs using

$$y_{kt} = \sum_e \sum_{s \in \mathcal{S}^e} \gamma_s^e D_k^e \cdot 1\{t - s = t^e\} + \phi_k + \nu_t + \varepsilon_{kt}. \tag{2}$$

We cluster standard errors by model to account for correlation across submodels and over time. As in the descriptive figures, we weight all observations of submodel $k$ by the mean monthly registrations in the months when this submodel is available. Unweighted results would be sensitive to idiosyncratic outcomes for low-volume submodels and to the exact definition of a submodel.

We focus on the evolution of outcomes within four groups of EVs: the three eligibility groups defined in Section 2.2—Excluded August 2022, Included January 2023, and Excluded/Reduced April 2023—and "Other EVs," which had no or other eligibility changes (i.e., they were always eligible, never eligible, or newly excluded in January 2023 due to MSRP caps). In some cases, such as our analysis of the impact of changes to the 45W lease credit, we include vehicles with April 2023 eligibility changes in the "Other EVs" group.

We considered several alternative specifications. "Doubly robust" specifications where the GV control group is reweighted to match the EV pre-IRA average price give similar estimates; see Appendix C.3. Alternative specifications using only EVs as controls give considerably noisier estimates that are mostly not statistically distinguishable from zero.[16]

**Prices and Economic Incidence**

This subsection discusses purchase prices around 30D tax credit eligibility changes. These results provide some evidence on the economic incidence of the credits, with the strong caveat that trends unrelated to the IRA in the new vehicle (and especially new EV) market, including a serious supply crunch due to chip shortages and other supply chain constraints, affect our price event studies. Additionally, the pass-through our estimates capture is local to consumers who continue to purchase while many substitute to leasing (see Figure 5). Furthermore, ineligibility of some households (Appendix Figure A6) and the non-refundability of tax credits may decrease take-up. Section 5 discusses non-refundability in more detail. Because buyers could claim the 30D credit later on their income taxes, constant purchase prices around an eligibility change would imply that consumers bear the full economic incidence, ignoring confounding supply chain conditions.

Appendix C.2 presents results. The data show modest changes in transaction prices when vehicles lose credit eligibility, consistent with consumers bearing much of the credits' economic incidence, though the caveats above lead us to interpret these estimates cautiously.

We also study lease prices as a measure of economic incidence on consumers choosing to lease. Our empirical analyses focus on the relative lease price (i.e., lease price minus purchase price) for

purchase credit available to submodel $k$, divided by \$7,500. For example, a vehicle that becomes eligible for the full \$7,500 tax credit has $D_k^e = 1$ and a vehicle where eligibility falls from \$7,500 to \$3,750 has $D_k^e = 0.5$. This primarily affects the vehicles which experienced a partial reduction in credit eligibility in April 2023.

[16] New difference-in-differences approaches that focus on addressing staggered treatment timing (Roth et al. 2023; Roth 2024) are less relevant in our setting because equation (2) estimates a separate set of coefficients $\{\gamma_s^e\}$ for each eligibility group.

each submodel-by-month observation, to difference out potentially confounding pricing trends.

Before and after the IRA, Section 30D allowed firms that lease vehicles to claim corporate income tax credits for leasing eligible vehicles to any buyer. Thus, firms leasing vehicles in the Excluded August 2022 group could claim credits under 30D through August 2022, lost eligibility from August to December, and could again claim credits under 45W beginning January 2023. Tesla and GM could not claim tax credits for leasing in 2022 and the several preceding years, but they could claim credits under 45W (or 30D, for eligible models) starting January 1, 2023. Firms leasing all other vehicles could always claim tax credits for leasing EVs.

The lease tax credit effectively reduces the seller's marginal costs for leasing a vehicle relative to selling it. In a Nash-Bertrand pricing model like the one Section 5 describes, if leasing and purchasing the firm's other vehicles experience approximately proportional cross-price effects, one would expect newly eligible firms to reduce the price of leasing relative to purchasing by roughly $7,500, the amount of the tax credit.

Appendix C.2 discusses relative lease price event studies. Figure 4 shows the largest decline occurred for the Excluded August 2022 group, whose relative lease prices fell by about $5,000 compared with December 2022; declines for other EV groups were smaller or emerged later.[17]

**Lease Shares**

Purchase-to-lease substitution governs the size of the leasing loophole's effects. One might expect an increase in EV lease shares in 2023 for two reasons. First, a price effect—as the previous subsection discusses, relative lease prices decreased for some EVs in 2023. Second, a buyer eligibility effect—the new Section 30D income limits raised the relative price of purchasing 30D-eligible vehicles for income-ineligible buyers in January 2023.

Panel (a) of Figure 5 presents the fixed-weight indexes of lease shares for the same three groups plotted in Figure 4. For the Included January 2023 group, the lease share is about 10 percent until the end of 2023, when it begins rising coincident with the timing of Tesla's lease rebates. For both the Excluded August 2022 group and all other EVs, lease shares decreased moderately in 2022 and then increased markedly in the first half of 2023. Appendix Figure A9 shows NVES trends in lease shares for EVs eligible and ineligible for the purchase credit, split by household income eligibility.

Panel (b) of Figure 5 presents the corresponding event study estimates, where we again define the "event" as occurring in January 2023. Consistent with Panel (a), we see a 40 to 45 percentage point lease share increase for the Excluded August 2022 group compared to the GV trend since December 2022, little change for the Included January 2023 group until the end of 2023, and a 20 to 30 percentage point increase for all other EVs.

[17] We estimated qualitatively similar but slightly larger effects for the Excluded August 2022 group when conducting the analysis using lease prices in levels (without subtracting new-vehicle prices). The estimates for the Included January 2023 group are less similar. However, the larger estimates may simply reflect the decline in vehicle prices for that product group. Additionally, we analyzed payments for 36-month leases without including a residual value. Panels (a) and (b) of Appendix Figure A13 present lease price event studies using these two alternative measures, respectively.

These leasing increases blunt the Section 30D trade and income restrictions and create deadweight loss, since tax incentives induce leasing by consumers who would not otherwise lease. They also lead our equilibrium model to treat leasing and purchasing as separate options.

# 5 Equilibrium Model

## 5.1 Model Setup

**Demand** We assume a demand system with heterogeneous types corresponding to income levels. Within each type, we describe a continuum of consumers with idiosyncratic preferences following a four-level nested logit. The nests capture four margins of substitution that are relevant for evaluating EV purchase and lease credits: between purchases and leases of the same submodel, across submodels within a vehicle class, across vehicle classes within a powertrain, and from EVs to GVs.[18] Incorporating income allows us to model the household income restrictions to credit eligibility. It also lets us match realistic purchasing patterns—how average vehicle price and the likelihood of acquiring a new vehicle, an EV, and a lease vary across the income distribution. In addition, recent work has shown that demographic interactions help generate a demand system flexible enough to allow a range of pass-throughs (Miravete, Seim, and Thurk 2023).

Consumers have quasilinear utility and unit demand, so they select exactly one good plus a continuous amount of the numeraire. Consumers choose between submodels and transaction types (purchase or lease). Consumers can purchase any submodel and can lease any submodel that offers a lease option. We index each submodel-by-transaction type choice by $j$. Each choice corresponds to a submodel $k(j)$, class $c(j)$, and gasoline or electric powertrain $g(j)$. We define class as a combination of vehicle segment (e.g., sedan, SUV, etc.) and powertrain, so an EV sedan is a different class than a GV sedan.[19] There are $J$ total inside goods, plus we include $j = 0$ to index the outside option (e.g., buying a used vehicle).

The market has $M$ consumers, indexed by $i$. Consumer $i$ belongs to one of a finite set of consumer types $h(i)$, with population shares $w_h$. Each consumer selects the choice $j$ that maximizes their utility. Let $y_{h(i)}$ be the representative income of consumer $i$, let $p_j$ represent the purchase price or lease price, and let $\tau_{h(i)j}$ be the available 30D tax credit accounting for income eligibility.[20]

[18] We estimate versions where the middle nest represents either vehicle class or domestic versus foreign production. Our baseline estimates use class because our externality calculations, which are central to the paper, strongly relate to class attributes like weight and size. Appendix D.4 discusses estimates that instead nest on domestic versus foreign firms, with results that yield similar conclusions. Conditional second choice probabilities in the NVES guide our decision to place powertrain as the top nest. Consumers are more likely to report a second choice with the same powertrain as their first choice powertrain than they are to report the same vehicle class or domestic status. This suggests that consumers have stronger preferences for powertrain than either alternative.

[19] Our data have nine classes with enough registrations to include in estimation. GVs include SUVs, sedans, pickup trucks, minivans, and coupes/convertibles. EVs include SUVs, sedans, pickup trucks, and minivans. Informed by the patterns in the second choice surveys, we combine hatchback vehicles with sedans and crossover vehicles with SUVs.

[20] The credit $\tau_{h(i)j}$ reflects income eligibility but assumes that eligible purchasing households claim the full credit. During our sample period the 30D credit was non-refundable, so households with federal income tax liability below the credit amount could claim only part of it (Borenstein and Davis 2025). We do not observe buyers' tax liability, so we do not model partial claimability directly. Non-refundability binds mainly at lower incomes, while EV buyers

Indirect utility from choice $j$ is given by

$$U_{ih(i)j} = \xi_j - \alpha_{h(i)} \left( p_j - \tau_{h(i)j} \right) + \boldsymbol{\beta}_{h(i)}^{'} \boldsymbol{X}_j + \epsilon_{ij}. \tag{3}$$

Here, $\alpha_{h(i)}$ represents the marginal utility of money and $\boldsymbol{\beta}_{h(i)}$ represents heterogeneous preferences for specific vehicle characteristics in $\boldsymbol{X}_j$. Common preferences for vehicle and purchase option attributes are captured by $\xi_j$. Finally, $\epsilon_{ij}$ is an idiosyncratic preference distributed type-1 extreme value, which we discuss below. We normalize $U_{ih(i)0} = \epsilon_{i0}$ for the outside option. Bold indicates vectors, so $\boldsymbol{\xi}$ is the vector of $\xi_j$ parameters, and $\boldsymbol{X}_j$ is a vector of binary inside good indicators, binary EV indicators, and binary lease indicators.

We allow $\xi_j$ to differ freely between the purchase and lease option of the same submodel. While the physical vehicle is the same, contract-specific differences determine the attractiveness of leasing relative to purchasing. Manufacturers push specific lease incentives, calculate residual values, decide whether to allow lease buyouts, and provide financing terms that vary by submodel. Consumers may therefore perceive the relative value of leasing differently across submodels.

We parameterize heterogeneous preferences across types by[21]

$$\boldsymbol{\beta}_{h(i)}^{'} \boldsymbol{X}_j = (\beta_0 \times 1_j\{\text{Inside good}\} + \beta_{EV} \times 1_j\{\text{EV}\} + \beta_{Lease} \times 1_j\{\text{Lease}\}) \times y_{h(i)}.$$

We parameterize price sensitivity across types by

$$\alpha_{h(i)} = \exp\left(\alpha_0 + \alpha_y y_{h(i)}\right),$$

which maintains the property that demand is downward-sloping. Here, $\alpha_y < 0$ would imply that low-income households are more price elastic.

We assume a nested logit error structure for $\epsilon_{ij}$. Appendix Figure A16 shows the nesting structure. The nested logit model differs from a standard multinomial logit by allowing the idiosyncratic error to be correlated across vehicles that share a nest. Specifically, we define three sets of random coefficients $\left\{\zeta_{ik(j)}^{k}, \zeta_{ic(j)}^{c}, \zeta_{ig(j)}^{g}\right\}$ representing idiosyncratic preferences common to all choices within a group. For example, $\zeta_{i,GV}^{g}$ represents a random coefficient common to all choices within GVs. This commonality generates correlation in preferences across choices. Three parameters $\left\{\sigma^k, \sigma^c, \sigma^g\right\} \in [0, 1)$ capture the dispersion of these sets of random coefficients.[22] Then $\epsilon_{ij}$ takes a standard nested logit form:

$$\epsilon_{ij} = \zeta_{ig(j)}^{g} + (1 - \sigma^g)\zeta_{ic(j)}^{c} + (1 - \sigma^c)\zeta_{ik(j)}^{k} + (1 - \sigma^k)\tilde{\epsilon}_{ij}, \tag{4}$$

skew strongly toward high-income households (Appendix Figure A6).

[21] Appendix Figure A18 shows that, empirically, lease shares are similarly flat in income for both EVs and GVs prior to the IRA and the main difference between EV and GV lease rates is in the overall level. Therefore, we do not include a further income interaction specific to EV leases.

[22] Cardell (1997) and Galichon (2022) study this formulation of the nested logit model using the representation in equation (4). In our setup, for any two choices $j$ and $r$, $\text{Cor}(\epsilon_{ij}, \epsilon_{ir}) = 1 - \left(1 - \sigma^{\ell(j,r)}\right)^2$, where $\ell(j, r) \in \{k, c, g\}$ is the lowest nest that the two choices share, and the correlation is zero if they share none.

where $\tilde{\epsilon}_{ij}$ is i.i.d. type-1 extreme value, uncorrelated across $j$. To be consistent with random utility maximization, we require $1 > \sigma^k \geq \sigma^c \geq \sigma^g \geq 0$, though we do not impose this in estimation (McFadden 1978). If $\sigma^k = \sigma^c = \sigma^g = 0$, this reduces to standard logit. As the dispersion parameters approach 1, the within-nest preference correlation increases, making consumers more likely to select alternatives that share a nest.

Concretely, consider a consumer whose first choice is purchasing an EV sedan submodel. A larger $\sigma^k$ implies more probable substitution to a lease of the same submodel than to a different submodel, a larger $\sigma^c$ implies more probable substitution to another vehicle of the same class than to a different class, and a larger $\sigma^g$ implies more probable substitution to another EV than to a GV or to the outside option.

The market share for any choice among each type, $s_j^h$, and the aggregate market demand summing over types, $q_j$, are given by

$$\begin{aligned} s_j^h &= \int 1\left\{U_{ihj} \geq \max_{k=0,\ldots,J} U_{ihk}\right\} dF(\boldsymbol{\epsilon}_i) \\ q_j &= M \sum_h w_h s_j^h. \end{aligned} \tag{5}$$

Appendix D.1 provides specific functional forms under the nested logit assumption. The substitution between any two choices $j$ and $r$ can be compactly expressed in terms of conditional choice shares within nests for each type. Writing the market share in equation (5) as the product of conditional shares via $s_j^h = s_{j|k(j)}^h s_{k(j)|c(j)}^h s_{c(j)|g(j)}^h s_{g(j)}^h$ (where, for example, $s_{c(j)|g(j)}^h$ represents the conditional market share of choice $j$'s class within its powertrain $g(j)$ among consumers of type $h$) delivers

$$\begin{aligned} \frac{\partial q_j}{\partial p_r} = M \sum_h w_h \alpha_h s_j^h \Bigg[ & s_r^h + \left(\frac{1}{1-\sigma^g} - 1\right) s_{r|g(j)}^h g(j,r) + \left(\frac{1}{1-\sigma^c} - \frac{1}{1-\sigma^g}\right) s_{r|c(j)}^h c(j,r) \\ & + \left(\frac{1}{1-\sigma^k} - \frac{1}{1-\sigma^c}\right) s_{r|k(j)}^h k(j,r) - \left(\frac{1}{1-\sigma^k}\right) 1(j,r) \Bigg] \end{aligned}, \tag{6}$$

where $g(j,r)$, $c(j,r)$, $k(j,r)$, and $1(j,r)$ are indicators if $j, r$ are respectively the same powertrain, class, submodel, and choice. Equation (6) shows that, when $1 > \sigma^k \geq \sigma^c \geq \sigma^g \geq 0$, two different choices ($j \neq r$) are closer substitutes the more nests they share, since each term in the expression is positive and more indicator terms equal one. The own-price effect ($j = r$) is negative due to the final subtracted term.

**Supply** Auto manufacturing firms indexed by $f$ each offer submodel-by-transaction type choices among their portfolio $\mathcal{J}_f$. Firms set their prices to maximize surplus in static Nash-Bertrand competition. We assume no interactions with the federal fuel economy and greenhouse gas emission

standards.[23] The terms $\boldsymbol{p}$ and $\boldsymbol{\tau}$ represent the vectors of prices and 30D credits faced by consumers for all choices, $c_j$ is marginal cost, and $\kappa_j$ represents the 45W credit available to firms for leasing choice $j$. In practice, $c_j$ could be interpreted as marginal *opportunity* cost, which could include both physical production costs and cost reductions due to the dynamic benefits from learning-by-doing as in Barwick et al. (2025). The function $q_j\left(p; \boldsymbol{p}_{-j}, \boldsymbol{\tau}\right)$ represents choice $j$'s total quantity demanded from equation (5) as a function of its price, given the price of all other products in the market and demand subsidies available. Firm $f$ then solves the following maximization problem for all products in its portfolio:

$$\max_{\{p_j\}_{j\in\mathcal{J}_f}} \sum_{j\in\mathcal{J}_f} \left(p_j - c_j + \kappa_j\right) \times q_j\left(p_j; \boldsymbol{p}_{-j}, \boldsymbol{\tau}\right), \tag{7}$$

where $(p_j - c_j + \kappa_j)$ is the 45W credit-inclusive markup over marginal costs, which we will denote using $\mu_j$ when computing producer surplus.[24] Firm $f$'s first-order condition with respect to an arbitrary choice $j$ in $\mathcal{J}_f$ is

$$[p_j]: \quad q_j + \sum_{r\in\mathcal{J}_f} \left(p_r - c_r + \kappa_r\right) \frac{\partial q_r}{\partial p_j} = 0. \tag{8}$$

We define firms $f$ at the level of the parent company, such as Stellantis (which owns Chrysler, Dodge, Jeep, and other brands). Our data have 17 firms.

**Equilibrium** The equilibrium describes a set of prices $\boldsymbol{p}$ such that any firm $f$ facing demand for its products as in equation (5) and taking prices of competitor firms as given has, for each $j$ in its portfolio, a best response $p_j$ that satisfies equation (8) in the pricing game and is consistent with the price vector $\boldsymbol{p}$. Nocke and Schutz (2018) show that a unique price equilibrium exists with nested logit demand and multi-product firms whenever each firm supplies products in only one nest. Garrido (2024) extends this result to deeper nesting structures but shows that it no longer holds under consumer heterogeneity, the same difficulty that heterogeneity introduces in mixed logit models with multi-product firms. Our baseline model satisfies neither condition: firms sell products across multiple nests, and consumers are heterogeneous in income. Imposing downward-

[23] Automakers had many banked compliance credits in 2023, though many transacted over the period we study (US EPA 2024b). The standards did not bind for that year at the industry level, but were designed to generally bind over a longer future period. Our assumption would exactly reflect reality if a future administration set non-binding standards or set future stringency to equate marginal benefits and marginal costs of future compliance. This occurred under the second Trump administration, which substantially weakened these standards (Domonoske 2025). Our setting would violate this assumption if more EV sales in 2023 make it easier to comply with the standards in future years. This would generate a "waterbed effect," through which EV tax credits would have smaller net effects on $CO_2$ emissions.

[24] For robustness, we consider an alternative conduct assumption where all EV manufacturers except Tesla price EVs at marginal cost (i.e., zero markups). This "penetration pricing" check reflects a scenario where manufacturers seek to maximize production subject to nonnegative profits, which can increase dynamic benefits in a setting with learning-by-doing through production. One implication is that EV credits to these firms are fully passed through to consumers and the primary market inefficiency comes from externalities, rather than market power for most EVs (Tesla and GVs do retain market power). Appendix Table A9 presents those results.

sloping demand helps, and in practice we do not encounter multiple equilibria in estimation.

We report two robustness analyses that relate to the pricing equilibrium. The first removes consumer heterogeneity and has each firm price separately within each powertrain-class nest, ignoring within-firm cannibalization across nests, which satisfies Nocke and Schutz (2018)'s conditions for uniqueness. This produces lower and less dispersed markups, with a share-weighted markup correlation of around 60 percent. The second specification preserves consumer heterogeneity but replaces the nine powertrain-by-class nests with four powertrain-by-foreign/domestic brand nests, a substantially different ownership structure across nests, with a share-weighted markup correlation of over 99 percent. Neither suggests that the baseline model is a distant local equilibrium or that results are extremely sensitive to the definition of product ownership.

### 5.2 Estimation, Inference, and Model Fit

**Estimation Procedure** We estimate the following sets of parameters: the two price response parameters $\boldsymbol{\alpha} = \{\alpha_0, \alpha_y\}$, three preference parameters $\boldsymbol{\beta} = \{\beta_0, \beta_{EV}, \beta_{Lease}\}$, and three nested logit parameters $\sigma = \left\{\sigma^k, \sigma^c, \sigma^g\right\}$; the $J$-vector of non-price attributes $\boldsymbol{\xi}$; and the $J$-vector of marginal costs $\boldsymbol{c}$. First, for given values of $\{\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\sigma}\}$ and observed market quantities $q_j$, we back out the unique vector of non-price attributes $\boldsymbol{\xi}$ using the Berry (1994) contraction mapping. Second, given values of $\boldsymbol{\xi}$, we use minimum distance gradient-based optimization to find new values of $\{\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\sigma}\}$ that match an exactly identifying set of moments described below. We iterate these steps until convergence. Third, we construct the demand slopes $\frac{\partial q_j}{\partial p_r}$ implied by the demand parameters from the first and second steps, and back out the marginal costs $\boldsymbol{c}$ implied by the system of first-order conditions from equation (8).

Our baseline scenario considers the market as of July and August 2023, with the choice set including submodels available and IRA credit eligibility as in those months. Section 4 documents that supply chain constraints had relaxed by this period; for example, July 2023 was the first month since early 2022 that mean days to turn for EVs exceeded 50. Moreover, including months after August would introduce subsequent changes to vehicle eligibility (e.g., a price cut brought the Tesla Model X below the 30D MSRP cap) and to the set of vehicles available due to new submodel introductions.

We set market quantities $q_j$ equal to six times their total for July plus August, representing annualized values, so we abstract from seasonality in vehicle markets. To model heterogeneous individuals by income, we define 12 representative households based on the AGI bins provided in the IRS Statistics of Income for 2023, with population weight $w_h$ equal to that bin's share of taxpayers and income equal to the average AGI within that bin. We define the 30D tax credit income eligibility threshold at the \$300,000 AGI cap for married couples.[25] We assume that each of

[25] We take two additional steps in modeling income. First, we transform $y_{h(i)}$ into z-scores of log income, estimating the mean and standard deviation of the log-income distribution using an interval regression on the binned IRS SOI data. This means we can also interpret $\xi_j$ as the mean vehicle preference for an average-income household (having a z-score of zero). Second, to explicitly account for the \$300k threshold, we split the provided \$200k-\$500k bin in the IRS SOI table into a \$200k-\$300k bin and a \$300k-\$500k bin using the interval regression to impute population

the 131 million US households purchases a vehicle every six years as in Coşar et al. (2018), giving a potential market size of $M = 21.9$ million households per year. This is larger than the annual number of new vehicle sales to capture that some households consider a new vehicle but ultimately choose the outside option.

**Targeted Moments** We find $\{\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\sigma}\}$ that match eight empirical moments, summarized in Table 4. Together, they discipline the substitution margins that are most important for our policy counterfactuals: overall price sensitivity, substitution between purchasing and leasing the same submodel, substitution across vehicle classes and powertrains, and heterogeneity in vehicle demand by household income. We match these moments jointly, but discuss them in order.

First, we match the market share-weighted model-level own-price demand elasticity of -5.36 that Grieco, Murry, and Yurukoglu (2024) estimate for their most recent reported year.

Second, we match the semi-elasticity of lease shares estimated in Section 4.2—for the Excluded August 2022 group between October–December 2022 and July–August 2023, lease prices decreased by $5,538 relative to purchase prices, and lease shares increased by 39 percentage points (Appendix Table A2). We simulate an equivalent reduction in lease prices to match the increase in lease shares for that group of models. Intuitively, these first two moments are jointly informative about $\alpha_0$ and $\sigma^k$. The own-price elasticity defines how many consumers substitute away from a choice, and the lease share change how many turn to the immediate purchase or lease alternative.

Third and fourth, we match two second-choice moments from the NVES data. Among respondents who named a second choice, 63.7 percent of EV owners report another EV as their second choice. Additionally, 40.9 percent of EV owners report another EV within the same class as their second choice. These two moments are jointly informative about $\sigma^c$ and $\sigma^g$, the correlation in preferences within a class and powertrain.

Finally, we match four moments from the NVES that describe heterogeneity in purchase behavior within the vehicle market. For each, we compare households with incomes above $300,000 to households below $300,000. In the survey, we calculate that high-income households buy cars that are on average $17,193 more expensive (informative of $\alpha_y$), are 1.99 times more likely to buy any vehicle (informative of $\beta_0$), are 13.2 percentage points more likely to buy an EV (informative of $\beta_{EV}$), and are 1.6 percentage points more likely to lease a vehicle (informative of $\beta_{Lease}$) than households with incomes below $300,000. Appendix D.2 details how we calculate these statistics.

**Calibration and Inference** In the terminology commonly used in macroeconomics, this exactly identified moment-matching procedure is a calibration (Hansen and Heckman 1996; Kydland and Prescott 1996; Cocci and Plagborg-Møller 2025). It differs from a conventional fully estimated differentiated-products demand model in one important respect: we do not identify the overall level of price sensitivity in our monthly panel using excluded instruments. Instead, we match the elasticity from Grieco, Murry, and Yurukoglu (2024) because their instruments (based on exchange

weights and average income under the assumption of a log-normal distribution.

rate shocks) are plausible but require many years of data. Their estimate also reflects an annual own-price demand elasticity with flexible supply, whereas the quantity variation in our complete monthly time-series data partly reflects supply constraints.

The remaining seven target moments are constructed from our event-study and NVES data. Thus, the procedure combines internally constructed moments with an externally calibrated price-elasticity target. The approach is also related to the identified-moments strategy of Nakamura and Steinsson (2018), in which a structural model is disciplined by reduced-form or quasi-experimental evidence on the particular behavioral responses relevant to the counterfactual exercise. We compute standard errors using the delta method. Following Cocci and Plagborg-Møller (2025), we additionally compute and report associated worst-case standard errors that are valid under any admissible correlation structure between our targeted moments. These represent sharp upper bounds; Appendix D.3 provides details.

To explore the potential importance of modeling uncertainty, we estimate several alternative specifications—a model with no income heterogeneity, an average own-price elasticity target of -4, an alternative nesting structure with domestic and foreign brands, Nocke-Schutz restrictions on supply, and a conduct assumption that firms (except Tesla) price their EVs at marginal cost—and present the results in Appendix Table A9.

**Estimates** Table 5 presents the scalar parameter estimates. Since we have as many moments as parameters to estimate, we match the targeted moments exactly. The standard errors are small; for example, we estimate the average price response parameter as $\alpha_0$ =-0.514 with a standard error of (0.019), or (0.186) under worst-case bounds. Low-income households are more price elastic via $\alpha_y$ =-0.374 with a standard error of (0.011), or (0.148) under conservative bounds. Other parameter estimates have similar levels of precision.

Appendix D.4 discusses eight additional pieces of evidence on model fit. The nested logit parameters in our demand system are consistent with utility maximization; the model matches heterogeneity throughout the income distribution (though the model only targets binary income partitions); the model fits second-choice moments that the model does not formally target—gasoline vehicles, miles per gallon equivalents, US firm ownership, and domestic assembly shares; and the estimated aggregate vehicle demand elasticities are in line with existing published values. In addition, we discuss the magnitudes and interpretation of estimated marginal costs alongside benchmarks of estimated percentage markups against industry reports of gross profit margins.

While model fit is good along most of these dimensions, fit is weaker along one dimension relevant for counterfactuals: conditional on a first choice from US firms, second-choice shares of US firms' vehicles are higher in the NVES than in the model. Appendix D.4 details this comparison and re-estimates the model with a foreign/domestic nest that matches these moments; Section 6 discusses the implications for our counterfactual estimates.

### 5.3 Welfare and Measures of Policy Effectiveness

Social welfare has four components: consumer surplus, producer surplus, government spending, and externalities. We compute consumer surplus using the Small and Rosen (1981) log-sum formula adjusted for nested logit demand and aggregating over consumer types, denoted by $CS(\boldsymbol{p}, \boldsymbol{\tau})$. We compute producer surplus as the sum of 45W credit-inclusive markups, $\mu_j$.[26] Government subsidies have a fiscal cost that depends on the marginal cost of raising public funds (MCPF), $\eta$.[27] Externalities arise from the production and use of choice $j$ and consist of carbon and non-carbon components under an assumed social cost of carbon, $\phi_j = SCC \times \phi_j^{CO_2} + \phi_j^{Other}$. Appendix A.2 provides details of how we calculate these externalities.[28]

Global welfare under a given subsidy policy $(\boldsymbol{\tau}, \boldsymbol{\kappa})$ equals

$$W(\boldsymbol{\tau}, \boldsymbol{\kappa}) = \underbrace{CS(\boldsymbol{p}, \boldsymbol{\tau})}_{\text{consumer surplus}} + \underbrace{\sum_{j \in \mathcal{J}} q_j \mu_j}_{\text{producer surplus}} \underbrace{- \eta \sum_{j \in \mathcal{J}} q_j (\tau_j + \kappa_j)}_{\text{government spending}} - \underbrace{\sum_{j \in \mathcal{J}} q_j \phi_j}_{\text{negative externalities}} \tag{9}$$

Equation (9) takes the perspective of a global planner who internalizes the effects of a policy on all firms and uses a global social cost of carbon. Section 6 considers alternatives, denoted $W^{US}$, that describe how a domestic planner may evaluate a policy. For example, the US social planner may place no weight on foreign producer surplus, which shows up as a restricted set $\mathcal{J}_{US} \subset \mathcal{J}$ in the value of producer surplus. Or, the domestic planner may assign different values to $\phi_j$, which we denote $\phi_j^{US}$, such as assuming a domestic social cost of carbon or placing additional value on domestic job creation within $\phi_j^{Other}$:

$$W^{US}(\boldsymbol{\tau}, \boldsymbol{\kappa}) = CS(\boldsymbol{p}, \boldsymbol{\tau}) + \underbrace{\sum_{j \in \mathcal{J}_{US}} q_j \mu_j}_{\text{domestic profits}} - \eta \sum_{j \in \mathcal{J}} q_j (\tau_j + \kappa_j) - \underbrace{\sum_{j \in \mathcal{J}} q_j \phi_j^{US}}_{\text{domestic externalities}} \tag{10}$$

We use these welfare measures to highlight the tradeoffs in green industrial policy between trade and the environment. Because policymakers also cite industrial policy goals, we separately report effects on manufacturing outcomes like employment, which are not part of our primary welfare calculation. We separate producer surplus into components from US- versus foreign-owned firms.

[26] While producer markups are endogenous to policy, we assume that the outside option (e.g., used vehicles) is competitively sold at zero markup. Since the mean indirect utility of the outside option is normalized to zero, the absolute level of utility cannot be measured. We can, however, report changes in surplus relative to a baseline scenario.

[27] For clarity of exposition, we omit notation corresponding to household credit eligibility for purchase credits, but we account for it in our results. That is, we write $\tau_j$ in the formulas below rather than summing over credits with household income eligibility $\tau_{h(i)j}$.

[28] Conceptually, for each choice, we want to compute uninternalized externalities, not directly affected by the policy changes we consider, over the life of those vehicles in the locations where they are operated. Changes in new vehicle prices induce substitution to used vehicles, taxis, ride sharing, public transportation, reduced overall mobility, and other options. We assume that the bulk of this effect involves driving used vehicles more and delaying their scrap. Since most used vehicles today are GVs, we assume that the outside option has the same externality as the average GV, except with no manufacturing $CO_2$ emissions.

For calculating these statistics, we define products owned by "US firms" ($\mathcal{J}_{US}$) as Tesla, Rivian, Lucid, Ford, GM, and the former Chrysler Group brands (Chrysler, Dodge, Jeep, and Ram). We define all others as "foreign firms."

Production location and ownership are distinct margins. The 30D credits condition on where a vehicle undergoes final assembly and where its battery inputs originate, not on nationality of ownership: a foreign-owned firm assembling in North America, such as Volkswagen with the US-assembled ID.4, can qualify for credits and support domestic employment, while ownership influences where profits accrue.[29] Our accounting accordingly tracks both dimensions: we divide producer surplus by ownership, and we divide employment, battery assembly, and investment by production location. Producer surplus and manufacturing outcomes capture related but distinct objectives: the reallocation of sales toward US firms and US assembly that shifts profits toward domestic producers also supports domestic employment, battery assembly, and investment, so the profit shifting and industrial policy motives largely move together across our counterfactuals. Section 6.4 incorporates a policymaker value for domestic manufacturing jobs directly into the planner's objective. Reporting these objectives separately lets policymakers who weight them differently evaluate the same policy under their preferred welfare weights.

In addition to welfare, our counterfactual analyses discuss two other aggregate metrics for policy evaluation: the welfare cost per ton of $CO_2$ abated and the marginal value of public funds (MVPF, see Hendren and Sprung-Keyser (2020) and Hahn et al. (2026)). We define these as

$$\frac{\text{Cost}}{\text{ton}\text{CO}_2} = \frac{\Delta CS + \Delta PS - \Delta G - \Delta E^{\text{non-CO}_2}}{\Delta \text{CO}_2} \tag{11}$$

$$\text{MVPF} = \frac{\Delta CS + \Delta PS - \Delta E^{\text{non-gas tax}}}{\Delta G - \Delta \text{gas tax revenue}} \tag{12}$$

where $CS$ and $PS$ are consumer and producer surplus, $G$ is government spending on EV subsidies, $\Delta E^{\text{non-gas tax}}$ are negative externalities excluding changes in gas tax revenue, and $\Delta E^{\text{non-CO}_2}$ are negative externalities excluding changes in carbon damages. For the MVPF, we place the fiscal externality from gas taxes in the denominator because it flows into government budgets, and the MVPF seeks to compare the welfare effects of a particular policy against other fiscal policies. We keep retail electricity markups in the numerator. The majority of electricity consumers in the US are served by private entities, and these firms have a positive willingness-to-pay for energy markups to cover, for example, long-run costs of infrastructure.

For counterfactuals that decrease government spending, the MVPF may be interpreted as evaluating the higher-spending alternative. For example, when comparing the IRA to its repeal, a

[29] Appendix Table A12 assesses how much additional flow profit manufacturers were "leaving on the table" by not shifting production facilities to North America. One at a time, we take the ten top-selling EV models below the MSRP cap without full 30D credits in our baseline equilibrium and grant them full eligibility while leaving marginal costs unchanged. The range in annual flow profit increases is \$25 to \$288 million, or 1.1 percent to 5.6 percent of baseline profits, for the major brands. Rivian receives large percentage benefits from obtaining the full credit for the R1S (its battery has insufficient qualifying content in the baseline to qualify for the full credit), partly because Rivian's baseline surplus is small.

higher MVPF means that the additional spending on the IRA is more valuable. The MVPF can help compare against alternative estimates of the MCPF. The MCPF is 1 in our model with lump-sum taxes, but it exceeds 1 in a more realistic case where governments raise revenue through distortionary taxes. The cost per ton abated helps compare against alternative estimates of the social cost of carbon, though it has limitations (Hahn et al. 2026). For example, it is negative for policies which reduce carbon emissions but are welfare-positive even when not considering carbon damages.

### 5.4 Constrained Optimal Policy

This subsection discusses our methodology for estimating constrained optimal uniform and differentiated EV subsidies, subject to coverage restrictions.

**Theory** Our setting has two potential market failures: over-consumption due to negative externalities $\phi_j$ and under-consumption due to markups $\mu_j$. We refer to the difference between price and social marginal cost, $\mu_j - \phi_j$, as the "price distortion." Below we present results when the MCPF is 1, but Appendix E shows corresponding derivations under a general MCPF, when financing subsidies generates additional distortions. We also solve for the constrained optimal consumption subsidy $\boldsymbol{\tau}$ that applies equally to purchases and leases; this counterfactual imposes no income eligibility restrictions and does not separately consider the producer subsidy $\boldsymbol{\kappa}$. In a standard model where statutory and economic incidence are independent, $\boldsymbol{\tau}$ and $\boldsymbol{\kappa}$ are perfect substitutes and infinite combinations of the two yield the same welfare (Weyl and Fabinger 2013).

Appendix E.1 derives the standard result that the optimal policy for a global planner (maximizing $W$) implements a vector of choice-specific Pigouvian subsidies or taxes that exactly offset the price distortion: $\tau_j^P = (\mu_j - \tilde{\phi}_j)$, where $\tilde{\phi}_j \equiv \phi_j - \phi_0$ is the net externality relative to the competitively supplied outside option.[30] Recall that $\tau_j < 0$ represents a tax. In practice, the IRA and analogous subsidies may deviate from optimal policy for several reasons. National policymakers may maximize domestic welfare, as in $W^{US}$. Additionally, institutions or political economy may limit policymakers to subsidizing only EVs or another subset of choices, such as vehicles meeting 30D requirements. Furthermore, policymakers may choose uniform rather than vehicle-specific subsidies.

Our expression for the uniform subsidy amount that maximizes US welfare, $\tau^{SB,U}$, requires additional notation. Let $\mathcal{S}$ denote the set of subsidized choices and let $\backslash\mathcal{S}$ denote its complement, including the outside option. Define $\mathcal{J}_{For}$ as the set of choices among foreign-owned firms, each of which receives producer surplus equal to quantity sold times markup, $\pi_j = q_j\mu_j$. Appendix E.2 derives the expression for the vector of constrained optimal differentiated subsidies that applies only to vehicles in $\mathcal{S}$ but does not restrict the subsidy to be equal across vehicles.

[30] When the MCPF exceeds 1, the global planner equates the cost of the transfer to the government, $\eta\tau_j^P$, with the combined distortions in the economy from unpriced externalities and the transfers themselves. An additional term arises from transfers to inframarginal consumers, each of whom receives a marginal unit of income at a cost of $\eta$ to taxpayers.

The uniform subsidy $\tau^{SB,U}$ instead restricts the subsidy to be equal for all choices in $\mathcal{S}$. Here, demand-response weighted averages provide sufficient information to determine the subsidy level. Define $\bar{\mu}_{\mathcal{S}} \equiv (\sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}\mu_j)/(\sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau})$ as the demand-response weighted average markup among $\mathcal{S}$ and $\bar{\phi}_{\mathcal{S}} \equiv (\sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}\phi_j)/(\sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau})$ as the demand-response weighted average externality among $\mathcal{S}$ (correspondingly, $\bar{\mu}_{\setminus\mathcal{S}}$ and $\bar{\phi}_{\setminus\mathcal{S}}$ among unsubsidized goods).[31]

**Proposition 1.** *At an MCPF of one, constrained optimal uniform subsidies for a fixed subset of choices, $\mathcal{S}$, are*

$$\tau^{SB,U} = \underbrace{\left(\bar{\mu}_{\mathcal{S}} - \bar{\phi}_{\mathcal{S}}\right)}_{\textit{price distortion}} - \underbrace{\left(\bar{\mu}_{\setminus\mathcal{S}} - \bar{\phi}_{\setminus\mathcal{S}}\right)}_{\textit{indirect substitution}} - \underbrace{\left(\sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}\right)^{-1}\left(\sum_{j\in\mathcal{J}_{For}} \frac{d\pi_j}{d\tau}\right)}_{\textit{profit shifting}} \tag{13}$$

*Proof.* Appendix E.3. □

When only EVs receive a subsidy, the constrained optimal uniform EV subsidy equals the difference in demand-response weighted average price distortions between EVs and non-EVs, minus the marginal surplus shifted to foreign firms per marginal electric vehicle at the optimal subsidy level. When implementing this formula empirically, we sometimes aggregate the first two terms and report a two-part decomposition: externality and markup distortions, and profit shifting away from foreign firms.

We solve for constrained optimal subsidies by treating the equation as a contraction mapping and re-computing markups, profits, and derivatives in each iteration of subsidies. Also, in the more general setting with an MCPF above one, there is an additional term in the decomposition of equation (13) corresponding to tax distortions.

**Descriptive Patterns of Price Distortions** The distribution of baseline (negative) price distortions across EVs and GVs (Figure 7) contributes to the constrained optimal subsidies that the next section recovers. Figure 7 echoes the histogram of negative externalities in Figure 3, but it includes the model-derived markup to infer social marginal cost.

Figure 7, Panel (a), shows that at a *global SCC*, negative externalities exceed markups for the vast majority of submodels. This implies that if policy applied to all vehicles, an EV tax rather than subsidy would be optimal. Dispersion within powertrains reflects differences across submodels in fuel economy, weight, local pollution, and markups. This dispersion implies that vehicle-specific differentiated taxes or subsidies may outperform uniform EV subsidies. The market share-weighted average distortions are about \$3,400 for EVs and \$11,700 for GVs. This implies that a lower tax or higher subsidy for EVs will have indirect benefits through encouraging substitution from GVs. Furthermore, within EVs, Section 30D-eligible and Section 30D-ineligible vehicles have average price distortions of \$3,100 and \$4,200, respectively. The strong within-EV substitution our model

[31] Specifically, $\bar{\mu}_{\setminus\mathcal{S}} \equiv (\sum_{j\in\setminus\mathcal{S}} \frac{dq_j}{d\tau}\mu_j)/(\sum_{j\in\setminus\mathcal{S}} \frac{dq_j}{d\tau})$ and $\bar{\phi}_{\setminus\mathcal{S}} \equiv (\sum_{j\in\setminus\mathcal{S}} \frac{dq_j}{d\tau}\phi_j)/(\sum_{j\in\setminus\mathcal{S}} \frac{dq_j}{d\tau})$. We present these expressions for generic $\phi_j$. In our main results, for the US planner, we value externalities at $\phi_j^{US}$.

estimates means that relaxing the 30D trade restrictions will tend to subsidize foreign EVs that have larger negative externalities.

Panel (b) of Figure 7 shows the same distributions, but valued at a *domestic SCC*. This change to the cost of $CO_2$ emissions brings externalities below markups for almost all vehicles, implying that most vehicles are priced above their domestic social marginal cost. Absent profit shifting concerns, this suggests that for a US social planner using the domestic SCC, optimal policy is a subsidy to correct for market power distortions, rather than a tax. An alternative avenue for policy, outside the scope of our analysis, would be more aggressive antitrust policy.

# 6 Effects of Counterfactual Policies

## 6.1 Repealing IRA EV Credits

Tables 6 and 7 describe the market and welfare impacts of repealing or modifying the IRA. Column 1 describes baseline outcomes under the IRA in July–August 2023; columns 2 and 3 describe the two alternative scenarios.

**EV registrations.** Eliminating the EV credits decreases EV registrations, especially from US firms, and decreases leasing (Table 6, column 2). EV registrations fall by 18 percent, or 216,000 EVs annually (rows 4 and 7). By removing the leasing loophole, this repeal also decreases EV lease rates by 20 percentage points, or two-thirds of baseline leasing (row 8). Consistent with the IRA's "Buy American" goal, the decrease is concentrated among US firms' EVs, which fall by more than a fourth. Registrations of foreign firms' EVs slightly increase (rows 5–6). Ultimately, the US assembly share among all EVs falls by 10 percentage points, or 14 percent of baseline (row 9).

Reverting from the IRA to pre-IRA policy causes smaller decreases in EV registrations and leasing than an IRA repeal does. The composition of EVs changes dramatically, however, since pre-IRA policy mostly subsidized foreign EVs (Table 6, column 3).[32] Registrations of foreign firms' EVs increase by 70 percent (rows 4 and 6). The US assembly share among EVs falls by 20 percentage points. Reverting to pre-IRA policy also decreases the leasing rate by 19 percentage points, again due to removing the leasing loophole (row 8).[33]

The IRA requires substantial government spending per additional EV registered because most

[32]In this scenario, \$7,500 purchase and lease credits without income eligibility restrictions were available for the first 200,000 EVs each manufacturer sold and then phased out in the year after the manufacturer reached that limit. As of July and August 2023, Tesla and GM would have received $\tau = \kappa =$ \$0, Toyota and Ford would have received $\tau = \kappa =$ \$1,875, BMW and Stellantis would have received $\tau = \kappa =$ \$3,750, and all other firms would have received $\tau = \kappa =$ \$7,500. By 2024, additional manufacturers would have reached their 200,000 limit, and pre-IRA policy would have soon implied no tax credits available for any major manufacturer.

[33]One benchmark for these magnitudes comes from Germany, which in December 2023 removed a \$4,900 EV subsidy. Annualized EV sales in the first 10 months of 2024 in Germany fell by 26.6 percent (European Automobile Manufacturers' Association 2024). While the setting differs, Germany's decline is in the ballpark of our projection that repealing the IRA EV credits would decrease EV registrations by 18 percent. An alternative US benchmark comes from the removal of EV credits under the One Big Beautiful Bill Act. EV sales in October 2025 were 30 percent lower than one year prior, though that may be especially large because dealers advertised the end date of the credits and many consumers bought EVs in the preceding months in anticipation of credit expiry (Cox Automotive 2025).

credits go to individuals who would have purchased an EV anyway (Table 6, column 2). Relative to a scenario with no EV credits, the IRA spends $32,100 of government revenue per additional EV registration, far above the $7,500 credit value. The additionality rate, defined as the number of marginal purchases and leases divided by the change in the number of credit-eligible vehicles registered, is only 23 percent.

**Auto employment, battery manufacturing, and investment**. Repealing the EV credits or returning to pre-IRA policy slightly decreases US auto manufacturing employment, by 11,500 to 14,400 jobs (Table 6, columns 2–3, row 11).[34] One-third of these jobs assemble vehicles and two-thirds manufacture auto parts (rows 12–13). These changes represent a small fraction of the baseline total industry employment of 870,000. Returning to pre-IRA policy decreases employment more than eliminating EV credits does because of pre-IRA policy's preference for foreign vehicles.

The EV credits are not especially cost-effective when evaluated solely as a labor market policy (Table 6, columns 2–3). Relative to pre-IRA policy, the government spends $165,000 on IRA EV credits per US auto industry job created. Relative to no credits, spending is $604,000 per job (row 15). For comparison, these values are far larger than the $10,700 per job that Slattery (2025) reports for the average discretionary government subsidy awarded per promised job in their data.[35] The EV credits have a high cost per job created because most subsidized vehicles would have sold anyway, and each additional US-assembled vehicle embodies only a fraction of a job-year.

Repealing the IRA also decreases the number of US-assembled EV batteries by about a quarter million annually (Table 6, row 14). The two types of repeal have fairly similar impacts on battery assembly. These magnitudes resemble the change in US firms' EV registrations because many US firms assemble EV batteries domestically.

The additional US EVs produced due to the IRA are roughly equivalent to the capacity of one or possibly two additional moderately large assembly plants. In the short- to medium-run, firms could have used excess production capacity at existing plants to meet this increased demand for EVs, but sustaining this production in the long run might require capacity investment. Appendix D.7 compares the increased EV registrations estimated in Table 6 to a few large, recent investments in vehicle assembly plants. This back-of-the-envelope comparison, which takes capacity costs per vehicle from two recent projects as representative and assumes no baseline surplus assembly capacity, suggests that the increase in US-assembled EVs caused by the IRA corresponds to roughly between $2.3 and $3.8 billion of additional investment in EV assembly.

[34]These predictions combine the model's counterfactual registrations with per-vehicle labor requirements. As Appendix A.4 details, we allocate the model-implied production cost of each submodel between final assembly and parts using an industry-level 20–80 percent split, apply the submodel's US assembly share and North American parts share, and convert the implied payroll into workers at industry average wages; the division of jobs between assembly and parts does not necessarily equal the 20–80 division of costs. This approach assumes that employment scales proportionally with production, an assumption which implicitly abstracts from plant capacity constraints and economies of scale. We assume all submodels have the same division of costs between final assembly and parts. The resulting estimates indicate the job-years associated with domestically sold vehicles rather than counts of permanent jobs.

[35]These numbers represent *job-years*, as they map the annual number of registrations to the number of workers involved. The $10,700 figure from Slattery (2025) is per job-year.

Across counterfactuals, manufacturing and profit shifting move together: policies that raise domestic producer surplus also raise domestic employment and battery assembly, because both reflect the same reallocation of sales toward US firms and US assembly. These outcomes do not move one-for-one because profits follow ownership while manufacturing follows production location.

**Social welfare.** Repealing the IRA credits or returning to pre-IRA policy increases both foreign producer surplus and foreign negative externalities, a tradeoff at the heart of our paper (Table 7, columns 2–3). Under an IRA repeal, foreign total surplus falls, because the increase in foreign externalities exceeds the increase in producer surplus (rows 4 and 8). Under a return to pre-IRA policy, in contrast, foreign total surplus increases, due to foreign preferences in pre-IRA policy.

Both repealing the IRA credits and returning to pre-IRA policy decrease US total surplus because the decline in US consumer and producer surplus and rise in US externalities exceed the decline in government spending (Table 7, columns 2–3). The magnitudes are larger under a global SCC, though the result also holds with a domestic SCC. Under a repeal, consumers lose more than producers, and under a return to pre-IRA policy, the opposite occurs (rows 1 and 3). In either case, the loss of US producer and consumer surplus dwarfs the increase in externalities (row 7).

Since the IRA's EV credits raise domestic surplus while reducing $CO_2$ emissions, the cost per ton from the US planner's perspective is negative (Table 7, row 18), implying a positive assessment of policy under any level of $CO_2$ damages. From the global planner's perspective, the marginal abatement cost is positive, at \$73 per ton relative to no EV credits and \$7 per ton relative to pre-IRA policy (row 19). These cost per ton metrics miss benefits to inframarginal credit recipients, who received most of these credits (Hahn et al. 2026).

These policies have modest to moderate MVPFs. From the global planner's perspective, the IRA credits generate \$1.11 to \$1.23 in benefits per dollar of government spending (Table 7, row 15). This is not a large MVPF, especially relative to estimates of the MCPF.

Relative to pre-IRA policy, the US planner has a higher MVPF for the EV credits than the global planner does (Table 7, rows 15–17). This is surprising because in typical climate change settings, the global planner values greenhouse gas mitigation more than the national planner does. Profit shifting in the IRA EV credits relative to pre-IRA policy explains this reversal. This asymmetry between the IRA's domestic and foreign impacts also reflects domestic political economy: passing the IRA involved support from domestic manufacturing and labor interest groups, who valued the IRA's trade restrictions, and from environmental interest groups, who valued its climate change mitigation. From the US planner's perspective, the pre-IRA credits have an MVPF of 0.66 relative to no EV credits. This low MVPF, representing a welfare loss of pre-IRA relative to no credits, reflects the prioritization of pre-IRA credits toward foreign brands.[36]

**Sensitivity analyses**. The estimated confidence intervals are reasonably tight (Appendix Tables A6 and A7; methodology in Appendix D.3). Treating the demand elasticity as known obtains

[36]The pre-IRA versus no-credits MVPF cannot be obtained by taking the simple difference between the MVPFs for the two counterfactuals individually (IRA relative to pre-IRA, and IRA relative to no credits), because each uses a different change in government spending as its denominator, so we calculate it directly.

worst-case standard error bounds from Cocci and Plagborg-Møller (2025) that are numerically similar to the delta-method standard errors. Calibrating uncertainty on the demand elasticity to match one result from Grieco, Murry, and Yurukoglu (2024) implies more uncertainty, although most results retain a fair degree of precision. For example, our main estimate is that the US planner using a domestic SCC obtains an MVPF from returning to pre-IRA policy of 1.92; our standard estimate for the 95% confidence interval is [1.88, 1.95], but the worst-case bound with the calibrated uncertainty for the Grieco, Murry, and Yurukoglu (2024) elasticity obtains a confidence interval of [1.72, 2.11]. Similarly, we estimate that the IRA credits, relative to pre-IRA policy, increased the number of US-assembled EV batteries by 284,000 annually; the standard 95% confidence interval is [279,000, 289,000] and the worst-case confidence interval under the calibrated Grieco, Murry, and Yurukoglu (2024) elasticity is [185,000, 383,000]. We therefore interpret statistical uncertainty as an important but not overwhelming source of uncertainty in our counterfactuals.

We also consider uncertainty from assumed parameter values and model structure. Alternative SCC or MCPF values change results in sensible ways (column 1 of Appendix Table A8). With an MCPF equal to 1.4 (Finkelstein and Hendren 2020), compared to no EV credits, the IRA credits decrease US and global welfare. Lower SCC values produce lower US or global surplus from the IRA, and a lower US MVPF. Under any SCC or MCPF values we consider, the change in US surplus is more positive than the change in global surplus, due to profit shifting.

Additionally, we consider five alternative assumptions on demand and supply (Appendix Table A9, columns 2–6): eliminating income heterogeneity in the model, replacing a vehicle class nest with a foreign/domestic nest, assuming an own-price model-level elasticity of -4, modifying supply assumptions to satisfy the Nocke-Schutz restriction, and assuming non-Tesla EVs price at cost. Several key qualitative conclusions hold across these alternatives. The homogeneous demand model and the foreign/domestic nest change results relatively little, and so suggest less modeling uncertainty than the statistical uncertainty from confidence intervals (Appendix Tables A6 and A7). Assuming a demand elasticity of -4 somewhat attenuates estimated impacts of the IRA on several key outcomes, while imposing the Nocke-Schutz restriction somewhat amplifies them. Assuming non-Tesla EVs price at cost changes outcomes the most, though we view this scenario as extreme. The range of outcomes from these last three scenarios is slightly greater than the statistical uncertainty from standard confidence intervals, though less than the statistical uncertainty from the worst-case confidence intervals. MVPFs for all these model-based alternatives are similar except for assuming non-Tesla EVs price at cost, which decreases the MVPFs by about 0.2. Appendix D.6 discusses further.

### 6.2 Modifying Trade Restrictions

We consider scenarios that revise the IRA's trade restrictions. We first consider closing the leasing loophole by imposing assembly and content requirements on leases, in line with the IRA's purchase credit restrictions (Tables 6 and 7, column 4). We also consider removing trade restrictions for both purchases and leases (column 5). Both cases treat leases and purchases nearly symmetrically;

income eligibility remains the only difference. These two scenarios have a roughly equal magnitude impact on subsidy spending, though with opposite signs.

Both trade counterfactuals reduce the EV lease share from the IRA baseline by half to two-thirds (Table 6, row 8). Closing the leasing loophole decreases the EV market share by 0.4 percentage points, while removing trade restrictions on purchases increases the EV market share by 1.1 percentage points (row 7). Closing the leasing loophole modestly increases US auto manufacturing employment, by less than 1,000, while removing trade restrictions on purchases decreases US auto employment by 4,300. The changes in US battery production are similarly small (row 14). For example, closing the leasing loophole increases the number of US batteries produced by 13,000.

Modifying the trade restrictions has a much smaller welfare impact than reverting to the pre-IRA baseline does (Table 7). Under the domestic SCC, the IRA's trade restrictions improve welfare; under the global SCC, the trade restrictions decrease US welfare. The difference between the two assessments reflects the different impacts of industrial policy goals (relatively more important under the domestic SCC) versus environmental policy goals (more important under the global SCC). The MVPF for the leasing loophole is below 0.5 for the US planner under the domestic SCC (row 17). Because closing the leasing loophole is a lower spending scenario relative to the IRA, this MVPF suggests that the loophole is not an effective use of government spending.

Overall, industrial policy here aligns with profit shifting but works against the environment—the trade restrictions that shift profits and support domestic manufacturing also raise emissions. Liberalizing trade benefits US consumers and increases EV registrations, reducing global negative externalities, but harms domestic producers and reduces domestic auto employment and battery assembly.

### 6.3 Removing Income Restrictions

The IRA incorporated income restrictions on credits partly to address the credits' regressivity and limit their fiscal impact. In years before the IRA, households with annual incomes above $200,000 received a majority of EV credits (Borenstein and Davis 2025). Our data reveal similar patterns (Appendix Figure A6), with just over half of EV buyers reporting household incomes at or below $200,000. The leasing loophole relaxes the credits' income restrictions, since high-income buyers who leased could still qualify for credits.

We study a counterfactual which preserves the trade aspect of the leasing loophole, but eliminates income restrictions for purchase credits, thereby eliminating the incentive for households with high incomes to lease. Removing the income restrictions slightly increases EV registrations and decreases leasing (Table 6, column 6). Removing the income threshold has a high cost per additional EV registered ($59,900) because higher-income households are less price-sensitive and more likely to buy an EV without a subsidy, so the credits they receive more often act as transfers.

The income restrictions have mixed welfare effects. Removing income restrictions leads richer households to purchase lower externality domestic EVs. This reform decreases $CO_2$ emissions and has a global MVPF of 1.34 (rows 5 and 15). The US planner has a higher MVPF, due to profit

shifting (Table 7, rows 16–17). Removing the income restrictions, however, increases the credits' regressivity (Table 6, row 17). Under the IRA, 44 percent of households earning above $300,000 who registered an EV could receive a credit, via the leasing loophole. Without income restrictions, this share is 77 percent. The income restrictions therefore decreased the credits' regressivity and decreased government spending, but that redistribution decreased climate benefits.

Income restrictions explain little of the observed increase in leasing; trade restrictions explain most of it. Imposing trade restrictions on leases while maintaining the income restrictions lowers the EV lease share by 16.9 percentage points (Table 6). Eliminating EV credits lowers the lease share by 19.7 percentage points, almost the same amount as the combined effect of the income and trade restrictions. We conclude that the trade restrictions account for about 86% (=16.9/19.7) of the observed total change in leasing, and the income restrictions account for 14%.

### 6.4 Constrained Optimal Policies

This subsection quantifies three types of constrained optimal subsidies, following Section 5.4. We do not believe that policymakers in 2022 purposefully chose the $7,500 credit value, which they left unchanged in nominal terms from earlier legislation; we nonetheless assess its optimality within our model.

Our first counterfactual preserves the IRA's restrictions on submodel eligibility for credits, but replaces the $7,500 IRA credit with the single, uniform subsidy value across submodels that maximizes US total surplus using the domestic SCC.[37] Our second counterfactual adds an explicit value on auto jobs to this objective. Our third counterfactual lets the planner assign different subsidy amounts to each eligible submodel. We calculate uniform subsidies in the first two counterfactuals using equation (13), and differentiated subsidies using equation (34) from Appendix E.2. Comparing the first and third counterfactuals reveals the gains from differentiation.

The US-optimal uniform subsidy in our baseline model is 34 percent above the IRA's actual subsidy (Table 8, column 2).[38] Relative to the IRA baseline, this higher subsidy increases annual EV registrations by 93,000. Since the uniform subsidy treats purchases and leases symmetrically, the higher subsidy decreases EV leasing rates by almost two-thirds (row 8). US welfare increases from the IRA baseline by around $1.5 billion (Table 9, rows 11–12). The decomposition in Appendix Table A11 reveals that profit shifting incentives account for 40 percent of the uniform subsidy value. Welfare increases relative to the IRA for two reasons: first, mechanically, the subsidy level is optimal given the constraints, and second, the policy instrument differs by closing the leasing

[37] We apply these restrictions to the submodel as a whole and allow both purchases and leases to claim the subsidy if a submodel is eligible, regardless of household income.

[38] The actual mean IRA credit amount, shown in column 1, is slightly below $7,500 because some vehicles only qualify for $3,750 in credits due to battery or critical mineral supply chains only satisfying one of the two requirements imposed in April 2023. We compute the mean credit amount conditional on 30D or 45W credit receipt, assuming the full credit amount is claimed by taxpayers. In reality, households with lower tax liability may claim less than the full amount of the credits due to their non-refundability (Borenstein and Davis 2025). We account for income eligibility by omitting high-income households purchasing 30D-eligible vehicles from the average, though high-income households who lease EVs are included.

loophole. Differentiating credits among vehicles as in column 4 increases US welfare by $2 to $3 billion annually relative to the IRA, with the range depending on the SCC. Uniform subsidies therefore capture one-half to two-thirds of these gains.

Our second counterfactual lets the US planner value domestic auto manufacturing for EVs and GVs at $10,700 per job, as in Slattery (2025). Comparing policy with and without value for domestic manufacturing jobs quantifies the importance of labor-related political economy considerations. In a neoclassical model of an economy at full employment, creating jobs in a sector does not affect aggregate welfare, since it merely reallocates employment between sectors. Under other assumptions, however, labor market frictions or changes in local economic activity may generate welfare consequences to labor in particular regions or sectors (Bartik 2015).

We find that subsidy dollars are a blunt instrument for creating auto jobs. Although domestic manufacturing employment provided a key rationale for the credits, the constrained optimal subsidy changes little when accounting for policymaker preference for employment (Table 8, column 3). Under the domestic SCC and assuming an MCPF of 1, these considerations raise the constrained optimal EV subsidy by about 5 percent (row 1). Compared to that baseline uniform subsidy, this counterfactual increases government spending on subsidies by $0.9 billion annually (Table 9, row 8) and increases domestic auto manufacturing jobs by only 1,300 (Table 8, row 11). Relative to no credits, the fiscal cost per additional job is $495,000, more than an order of magnitude greater than the $10,700 job value used in the planner's objective (Table 8, row 13).

This echoes the finding in Slattery (2025), albeit from a completely different setting, that the number of new jobs from industrial policy only explains a small share of observed subsidies. The marginal subsidy dollar mostly transfers surplus to inframarginal buyers, and the additional vehicles sold are built with limited domestic labor and some imported parts. Furthermore, inframarginal buyers have disproportionately high incomes, so subsidy dollars which create the fewest jobs also flow disproportionately to high-income households.

The primary qualitative conclusions persist under several alternative parameter assumptions, though some magnitudes vary (Appendix Tables A8 and A11). Assuming that the MCPF is 1.4 rather than 1.0 shrinks the constrained optimal subsidy substantially. Even under a global SCC, where the optimal policy applied to all vehicles would be a tax, the constrained optimal policy for eligible EVs remains a subsidy, because the planner cannot tax GVs, and because EV subsidies pull buyers away from GVs. Long-run marginal emission rates imply much cleaner EV charging but barely move the optimal subsidy, because accident and manufacturing externalities account for much of the wedge. Appendix D.5 discusses details further.

# 7 Conclusion

The IRA was among the largest potential climate change policies ever. It proved short-lived, as the One Big Beautiful Bill Act reversed most of its provisions after three years. We provide the first ex post microeconomic welfare analysis of a central component of the IRA—tax credits to subsidize

the purchase of new EVs, partly conditional on requirements that supply chains locate in the US or allied countries. We present descriptive facts on the market, event studies around implementation, and an equilibrium model of supply and demand to assess welfare impacts.

Our event studies show that in the IRA's inaugural year, the credits substantially increased leasing, in line with trade, income, and price restrictions on eligibility. Our short-run event study analysis also finds suggestive evidence that consumers received most of the credits' price impact. We interpret this evidence cautiously given supply conditions at this time. Our equilibrium model implies that the IRA's EV credits have an MVPF of 1.1 to 2.1, by shifting profits from foreign to domestic firms and decreasing negative externalities, though several scenarios have substantially different consequences from the global versus US planner's perspective. We highlight heterogeneity in externalities among EVs. Many countries encourage vehicle electrification without focusing on which EVs consumers adopt. Externalities vary about as much within EVs as within GVs, with considerable overlap between the two, so failing to subsidize (or tax) different EVs in proportion to their different externalities misses opportunities to increase welfare.

We also highlight tradeoffs in green industrial policy between foreign and domestic interests, between trade and the environment, and between these objectives and industrial policy goals like domestic manufacturing. The manufacturing benefits we quantify—jobs, battery assembly, and plant investment—largely track the same reallocation toward US production that shifts profits. Trade policy is traditionally either cooperative (e.g., Most Favored Nation tariffs) or non-cooperative (e.g., "Column 2" tariffs on US imports from North Korea). Green industrial policy has both elements: the industrial policy component non-cooperatively exploits profit shifting to relocate clean energy production domestically, while the green component invests in the global public good of $CO_2$ mitigation that partly benefits foreign countries, and may even use a cooperative perspective on the environmental externality (e.g., the global SCC).

We leave several questions unanswered. Longer-run impacts of even temporary subsidies can relocate supply chains, generate learning-by-doing, and create interactions with EV charging infrastructure with potentially important welfare and distributional effects. Our welfare accounting—separating producer surplus by ownership, tracking manufacturing outcomes by production location, decomposing environmental benefits into domestic and foreign components, and pricing policymaker preferences for manufacturing employment—may be useful for evaluation of other industrial policies. Additionally, in response to the IRA, some US trading partners are considering or implementing their own green industrial policies. Just as research has studied Nash tariffs by nesting quantitative trade models within a game between countries (Ossa 2014), future work could study the incentives for green industrial policy as a strategic choice among countries. Finally, why do countries design climate policy that benefits foreign countries, but via industrial policy, which non-cooperatively shifts profits? International competition in industrial policy is to some extent a zero-sum game, while the climate is a global public good. In evaluating their priorities, what weight do policymakers in one country effectively put on environmental benefits abroad? Regular international negotiations encourage weights above non-cooperative zero on foreign climate bene-

fits, but perhaps not equal to one. Domestic political economy likely contributes to these decisions, though its exact mechanisms are unclear. Interest groups also play a role—organized lobbies like auto firms benefit from industrial policy and create domestic political pressure for non-cooperative policy. Our analysis studies this leading example of green industrial policy, but it cannot explain all of the tensions that we have highlighted.

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Table 1: **Summary Statistics**

| | Mean | Std. dev. | Min. | Max. | Obs. |
|---|---|---|---|---|---|
| Registrations | 1,070 | 1,882 | 26 | 38,781 | 18,924 |
| Purchase price ($000s) | 51.0 | 19.1 | 12.7 | 96.7 | 17,586 |
| Lease price ($000s) | 48.5 | 21.1 | 16.6 | 374.1 | 15,726 |
| Percent leased | 25.1 | 17.7 | 0 | 100 | 18,924 |

Notes: This table presents summary statistics for our submodel-by-month panel of new vehicle sales and prices for 2022 and 2023. Registration counts are from Experian. Purchase prices are from the California DMV; lease prices are computed from combining lease term data from Cox Automotive with purchase prices from the California DMV.

Table 2: **Share-Weighted Average Externalities by Vehicle Powertrain**

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Electric Vehicles | | | | |
| | All EVs | Battery electric vehicles | Plug-in hybrids | Gasoline vehicles | Structural model outside option (used GVs) |
| Manufacturing + scrap $CO_2$ (global SCC) | 2,974 | 3,142 | 2,156 | 1,815 | – |
| Driving $CO_2$ (global SCC) | 6,958 | 6,042 | 11,389 | 13,578 | 13,578 |
| Driving local pollutants | 559 | 558 | 561 | 377 | 377 |
| Excess weight in accidents | 10,039 | 9,862 | 10,895 | 7,636 | 7,636 |
| Positive energy pricing externality | (5,764) | (5,807) | (5,558) | (3,515) | (3,515) |
| Total negative externality | | | | | |
| Global SCC | 14,765 | 13,798 | 19,443 | 19,890 | 18,076 |
| Domestic SCC | 5,977 | 5,671 | 7,457 | 6,270 | 6,061 |

Notes: This table presents market share-weighted average lifetime externalities across submodels within a powertrain, weighting submodels by average monthly sales in months when the submodel was available. Units are $/vehicle. Positive energy pricing externalities from gas taxes and electricity markups are shown in parentheses to emphasize they are opposite-signed to damages.

Table 3: **Variation in Externalities by Powertrain**

| | Coefficient of variation | | | | Interdecile ratio | | | |
|---|---|---|---|---|---|---|---|---|
| | EVs | | | GVs | EVs | | | GVs |
| | BEVs | PHEVs | all | | BEVs | PHEVs | all | |
| Total neg. externalities (global SCC) | 0.29 | 0.23 | 0.29 | 0.25 | 0.66 | 0.53 | 0.82 | 0.66 |
| | | | [0.103] | | | | [−0.06, 0.22] | |
| Total neg. externalities (US SCC) | 0.56 | 0.31 | 0.48 | 0.48 | 1.28 | 0.77 | 1.22 | 1.26 |
| | | | [0.949] | | | | [−0.34, 0.20] | |
| Carbon damages | 0.14 | 0.21 | 0.22 | 0.19 | 0.33 | 0.61 | 0.49 | 0.46 |
| | | | [0.050] | | | | [−0.02, 0.15] | |

Notes: All quantities are computed at the submodel level, unweighted. Interdecile ratio is the log of the 90th sample percentile divided by the 10th sample percentile of externalities. BEVs are battery electric vehicles; PHEVs are plug-in hybrid electric vehicles. Bracketed values in the Coefficient of variation column are $p$-values for the difference in coefficients of variation between all EVs and GVs, computed following Feltz and Miller (1996); bracketed values in the Interdecile ratio column are bootstrapped 95 percent confidence intervals for the difference in interdecile ratio between all EVs and GVs.

Table 4: **Parameter Assumptions and Empirical Moments**

| Description | Source | Target Value | Model Value |
|---|---|---|---|
| Market size (million/year) | Federal Reserve Bank of St. Louis (2024c) | 21.9 | – |
| Population shares and average income | Internal Revenue Service (2023b) | – | – |
| Model-level demand elasticity | Grieco, Murry, and Yurukoglu (2024) | -5.36 | -5.36 |
| Share of EV buyers whose 2nd choice is an EV | NVES | 63.7% | 63.7% |
| Share of EV buyers whose 2nd choice is an EV in the same vehicle class | NVES | 40.9% | 40.9% |
| Difference in average transaction prices, high- vs. low-income households | NVES | $17,193 | $17,193 |
| Ratio of inside good choice probability, high- vs. low-income households | NVES, Internal Revenue Service (2023b) | 1.99 | 1.99 |
| Difference in EV choice propensity, high- vs. low-income households | NVES | 13.2pp | 13.2pp |
| Difference in lease choice propensity, high- vs. low-income households | NVES | 1.6pp | 1.6pp |
| Excluded Aug 2022 group lease price change | Appendix Table A2 | -$5,538 | – |
| Excluded Aug 2022 group lease share effect | Appendix Table A2 | 39pp | 39pp |

Notes: This table presents empirical moments we match and other parameter assumptions used in the demand estimation in Section 5. High-income households are those that report income above $300,000 in the survey; low-income households are those that report income below $300,000 in the survey. Refer to Appendix D.2 for additional details on computing the moments.

Table 5: **Parameter Estimates**

| Parameter | Description | Value | Standard error |
|---|---|---|---|
| $\alpha_0$ | Price response parameter | -0.514 | (0.019) |
| $\alpha_y$ | Price response × income interaction | -0.374 | (0.011) |
| $\beta_0$ | Inside good × income interaction | 0.299 | (0.056) |
| $\beta_{EV}$ | EV × income interaction | 0.498 | (0.015) |
| $\beta_{Lease}$ | Lease × income interaction | 0.043 | (0.003) |
| $\sigma^g$ | EV-GV nest parameter | 0.499 | (0.009) |
| $\sigma^c$ | Class nest parameter | 0.613 | (0.007) |
| $\sigma^k$ | Submodel nest parameter | 0.914 | (0.014) |

Notes: This table presents parameter estimates from the demand estimation in Section 5. Standard errors are calculated using the delta method. In Appendix Table A5, we also report sharp worst-case bounds on standard errors following Cocci and Plagborg-Møller (2025), which are valid under any correlation structure across our three blocks of data moments that is consistent with the within-block covariances we can estimate; see Appendix D.3 for details.

Table 6: **Effects of Counterfactual Policies on Market Outcomes**

| | | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|---|
| | | | Repeal IRA | | Modify Trade Restr. | | |
| | | IRA (baseline) | Eliminate EV Credits | Return to Pre-IRA 30D with phaseout | IRA, add trade restrictions on leases | IRA, remove trade restrictions | IRA, remove income threshold |
| | **Panel (a): Market Aggregates under Counterfactual Scenario** | | | | | | |
| 1. | Vehicle registrations (000s/year) | 10,556 | 10,458 | 10,529 | 10,539 | 10,609 | 10,557 |
| 2. | US firms (000s/year) | 3,952 | 3,748 | 3,669 | 3,961 | 3,893 | 3,961 |
| 3. | Foreign firms (000s/year) | 6,604 | 6,710 | 6,860 | 6,578 | 6,716 | 6,596 |
| 4. | EV registrations (000s/year) | 1,184 | 968 | 1,136 | 1,142 | 1,304 | 1,193 |
| 5. | US firms (000s/year) | 835 | 590 | 542 | 836 | 797 | 847 |
| 6. | Foreign firms (000s/year) | 349 | 378 | 594 | 307 | 507 | 346 |
| 7. | EV market share (%) | 11.2 | 9.3 | 10.8 | 10.8 | 12.3 | 11.3 |
| 8. | Lease share, within EVs (%) | 29.3 | 9.6 | 10.2 | 12.4 | 13.8 | 27.5 |
| 9. | US assembly share, within EVs (%) | 70.5 | 60.3 | 50.0 | 73.4 | 60.8 | 71.0 |
| 10. | Cost per additional EV ($000s/EV) | - | 32.1 | 49.2 | 38.4 | 13.8 | 59.9 |
| | **Panel (b): Employment and Battery Production Effects Relative to IRA Baseline (000s/year)** | | | | | | |
| 11. | ΔUS auto manufacturing jobs | - | -11.5 | -14.4 | 0.7 | -4.3 | 0.4 |
| 12. | ΔUS auto parts jobs | - | -7.3 | -9.2 | 0.5 | -2.8 | 0.3 |
| 13. | ΔUS auto assembly jobs | - | -4.2 | -5.2 | 0.2 | -1.5 | 0.1 |
| 14. | ΔUS-assembled EV batteries | - | -241 | -284 | 13 | -45 | 10 |
| 15. | Cost per additional US job ($000s/job) | - | 604 | 165 | -2,141 | -388 | 1,314 |
| | **Panel (c): Allocation of EV Credits** | | | | | | |
| 16. | Share of EVs with credits (%) | 80.0 | 0.00 | 66.3 | 66.7 | 87.9 | 86.5 |
| 17. | Among HHs >$300k (%) | 44.2 | 0.00 | 64.7 | 22.4 | 40.4 | 77.0 |

Notes: This table presents counterfactual simulation results for the IRA and alternative policies. In column 2, all 30D and 45W credits are set to zero. In column 3, we simulate EV credits as they would have been in July-August 2023 had the IRA not passed. Under the pre-IRA Section 30D, all EVs (purchased or leased) are eligible for credits until the 200,000-vehicle cap. Given cumulative sales volumes through mid-2023, we assume no credit for GM and Tesla, 1/4 credit for Toyota and Ford, and 1/2 credit for BMW and Stellantis. In column 4, we apply the Section 30D trade restrictions to all leases, although the buyer income restrictions still do not apply to leases. In column 5, all vehicles are eligible under Section 30D regardless of assembly location or battery sourcing, although buyer income restrictions and MSRP caps still apply. In column 6, we remove the $300,000 income restriction on purchases. New vehicle registrations include only leases and purchases by individuals, not vehicles sold to organizations. "US firms" denotes vehicles produced by US-owned firms, including the traditional US brands in Stellantis. "US assembly" denotes vehicles assembled in the US, regardless of firm ownership. Rows with Δ refer to changes relative to the IRA baseline. Cost per additional EV is the ratio of Δ government spending to Δ new EVs registered relative to the IRA. Cost per additional US job is the ratio of Δ government spending to Δ US manufacturing jobs relative to the IRA.

Table 7: **Effects of Counterfactual Policies on Welfare Outcomes**

| | | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|---|
| | | | Repeal IRA | | Modify Trade Restr. | | |
| | | IRA (baseline) | Eliminate EV Credits | Return to Pre-IRA 30D with phaseout | IRA, add trade restrictions on leases | IRA, remove trade restrictions | IRA, remove income threshold |
| | **Panel (a): Surplus Effects Relative to IRA Baseline ($billion/year)** | | | | | | |
| 1. | ΔUS consumer surplus | - | -4.86 | -0.63 | -1.16 | 2.61 | 0.60 |
| 2. | ΔGlobal producer surplus | - | -1.66 | -1.60 | 0.19 | -0.29 | 0.03 |
| 3. | ΔUS producer surplus | - | -2.62 | -3.97 | 0.41 | -1.23 | 0.18 |
| 4. | ΔForeign producer surplus | - | 0.96 | 2.38 | -0.22 | 0.94 | -0.15 |
| 5. | Δ$CO_2$ emissions (million tons/year) | - | 4.88 | 2.61 | 0.37 | -2.05 | -0.28 |
| 6. | ΔGlobal neg. exter. (global SCC) | - | 1.99 | 1.03 | 0.18 | -0.86 | -0.10 |
| 7. | ΔUS neg. exter. (domestic SCC) | - | 0.95 | 0.48 | 0.10 | -0.42 | -0.04 |
| 8. | ΔForeign neg. exter. (foreign SCC) | - | 1.04 | 0.56 | 0.08 | -0.44 | -0.06 |
| 9. | ΔUS subsidy spending | - | -6.94 | -2.37 | -1.60 | 1.66 | 0.51 |
| 10. | ΔGas tax revenue | - | 0.76 | 0.27 | 0.11 | -0.30 | -0.03 |
| 11. | ΔGlobal surplus | - | -0.82 | -0.61 | 0.57 | 1.21 | 0.19 |
| 12. | ΔUS total surplus (global SCC) | - | -1.78 | -2.99 | 0.79 | 0.27 | 0.33 |
| 13. | ΔUS total surplus (domestic SCC) | - | -0.74 | -2.43 | 0.87 | -0.17 | 0.28 |
| 14. | ΔForeign total surplus (foreign SCC) | - | -0.08 | 1.82 | -0.30 | 1.38 | -0.09 |
| | **Panel (b): MVPF of Higher-Spending versus Lower-Spending Scenario** | | | | | | |
| 15. | Global MVPF | - | 1.11 | 1.23 | 0.67 | 1.62 | 1.34 |
| 16. | US MVPF (global SCC) | - | 1.23 | 2.13 | 0.54 | 1.14 | 1.61 |
| 17. | US MVPF (domestic SCC) | - | 1.10 | 1.92 | 0.49 | 0.91 | 1.50 |
| | **Panel (c): Welfare Cost per Ton of $CO_2$ Abated** | | | | | | |
| 18. | US cost/ton $CO_2$ abated ($/ton) | - | -124 | -904 | 2,361 | 109 | -965 |
| 19. | Global cost/ton $CO_2$ abated ($/ton) | - | 73.3 | 6.60 | 1,769 | -349 | -427 |

Notes: This table presents counterfactual simulation results for the IRA and alternative policies. In column 2, all 30D and 45W credits are set to zero. In column 3, we simulate EV credits as they would have been in July-August 2023 had the IRA not passed. Under the pre-IRA Section 30D, all EVs (purchased or leased) are eligible for credits until the 200,000-vehicle cap. Given cumulative sales volumes through mid-2023, we assume no credit for GM and Tesla, 1/4 credit for Toyota and Ford, and 1/2 credit for BMW and Stellantis. In column 4, we apply the Section 30D trade restrictions to all leases, although the buyer income restrictions still do not apply to leases. In column 5, all vehicles are eligible under Section 30D regardless of assembly location or battery sourcing, although buyer income restrictions and MSRP caps still apply. In column 6, we remove the $300,000 income restriction on purchases. Rows with Δ refer to changes relative to the IRA baseline. "Global surplus" equals the sum of consumer surplus, US producer surplus, foreign producer surplus, and gas tax revenue less subsidy spending and negative externalities. "US total surplus" (under the global SCC) equals global surplus minus foreign automakers' producer surplus. "Foreign SCC" is global SCC minus domestic SCC. The marginal value of public funds (MVPF) equals (Δ consumer surplus + Δ producer surplus - Δ negative externalities)/(Δ subsidy spending - Δ gas tax revenue). The "US MVPF" values are computed with Δ US producer surplus in the numerator, while the "Global MVPF" uses Δ global producer surplus in the numerator. Costs of abatement are in terms of social welfare; they are negative when a policy change results in lower carbon emissions and increased social welfare disregarding the direct effect of carbon damages.

Table 8: **Counterfactual Simulation Results: Optimal Subsidies, Market Outcomes**

| | | (1) | (2) | (3) | (4) |
|---|---|---|---|---|---|
| | | IRA | US-optimal uniform EV subsidy, 30D restrictions | US-optimal uniform EV subsidy, 30D restrictions, valuing jobs | US-optimal differentiated EV subsidy, 30D restrictions |
| | **Panel (a): Market Aggregates under Counterfactual Scenario** | | | | |
| 1. | Mean EV subsidy in this counterfactual | $7,328 | $9,825 | $10,327 | $12,625 |
| 2. | (Standard deviation) | (785) | (0) | (0) | (3,368) |
| 3. | Vehicle registrations (000s/year) | 10,556 | 10,594 | 10,604 | 10,623 |
| 4. | EV registrations (000s/year) | 1,184 | 1,277 | 1,299 | 1,329 |
| 5. | US firms | 835 | 1,013 | 1,040 | 1,112 |
| 6. | Foreign firms | 349 | 264 | 259 | 217 |
| 7. | EV market share (%) | 11.2 | 12.1 | 12.3 | 12.5 |
| 8. | Lease share, within EVs (%) | 29.3 | 11.4 | 11.5 | 9.8 |
| 9. | US assembly share, within EVs (%) | 70.5 | 77.6 | 78.3 | 79.9 |
| | **Panel (b): Employment and Battery Production Effects** | | | | |
| 10. | Cost per additional EV ($000s/EV) | 32.1 | 32.8 | 33.1 | 38.4 |
| 11. | ΔUS auto manufacturing jobs (000s) | 11.5 | 20.8 | 22.1 | 20.5 |
| 12. | ΔUS-assembled EV batteries | 241 | 370 | 394 | 489 |
| 13. | Cost per additional US job ($000s/job) | 604 | 487 | 495 | 676 |

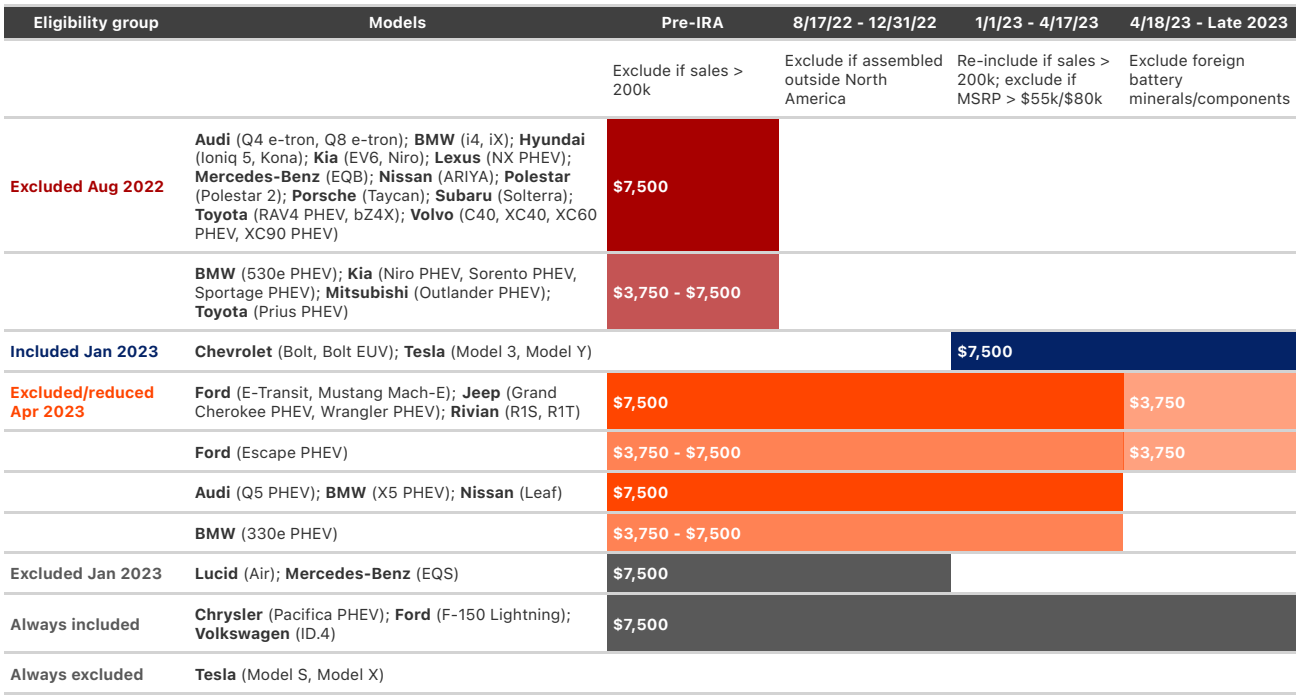
Notes: This table presents counterfactual simulation results for the IRA and alternative policies, relative to a common baseline of no EV subsidies (column 2 in Tables 6 and 7). In column 2, we simulate the uniform EV subsidy, subject to Section 30D trade restrictions, that maximizes US total surplus with the domestic SCC. In column 3, we assume the planner values annual US jobs at $10,700 per job. In column 4, we simulate the choice-specific differentiated EV subsidy that maximizes US total surplus with the domestic SCC. New vehicle registrations include only leases and purchases by individuals, not vehicles sold to organizations. Mean and standard deviation of subsidies are sales-weighted across EV models conditional on positive 30D or 45W credit amount, accounting in column 1 for income-eligibility using the model-implied income distribution. Cost per additional EV is the ratio of Δgovernment spending to Δnew EVs registered.

Table 9: **Counterfactual Simulation Results: Optimal Subsidies, Welfare Outcomes**

| | | (1) | (2) | (3) | (4) |
|---|---|---|---|---|---|
| | | IRA | US-optimal uniform EV subsidy, 30D restrictions | US-optimal uniform EV subsidy, 30D restrictions, valuing jobs | US-optimal differentiated EV subsidy, 30D restrictions |
| | **Panel (a): Surplus Effects Relative to No EV Subsidy Baseline ($billion/year)** | | | | |
| 1. | ΔUS consumer surplus | 4.86 | 7.46 | 7.97 | 8.28 |
| 2. | ΔGlobal producer surplus | 1.66 | 2.47 | 2.64 | 4.96 |
| 3. | ΔUS producer surplus | 2.62 | 4.61 | 4.90 | 7.57 |
| 4. | ΔForeign producer surplus | -0.96 | -2.14 | -2.26 | -2.61 |
| 5. | ΔGlobal neg. externalities (global SCC) | -1.99 | -2.49 | -2.67 | -4.17 |
| 6. | ΔUS neg. externalities (domestic SCC) | -0.95 | -1.16 | -1.24 | -2.11 |
| 7. | ΔForeign neg. externalities (foreign SCC) | -1.04 | -1.33 | -1.42 | -2.06 |
| 8. | ΔUS subsidy spending | 6.94 | 10.1 | 11.0 | 13.9 |
| 9. | ΔGas tax revenue | -0.76 | -1.04 | -1.12 | -1.32 |
| 10. | ΔGlobal surplus | 0.82 | 1.24 | 1.21 | 2.23 |
| 11. | ΔUS total surplus (global SCC) | 1.78 | 3.39 | 3.47 | 4.84 |
| 12. | ΔUS total surplus (domestic SCC) | 0.74 | 2.06 | 2.05 | 2.78 |
| 13. | ΔForeign total surplus (foreign SCC) | 0.08 | -0.81 | -0.84 | -0.55 |
| | **Panel (b): Impacts on $CO_2$** | | | | |
| 14. | Δ$CO_2$ emissions (million tons/year) | -4.88 | -6.24 | -6.68 | -9.67 |
| 15. | Global cost/ton $CO_2$ abated ($/ton) | 73.3 | 41.7 | 59.2 | 10.7 |
| 16. | US cost/ton $CO_2$ abated ($/ton) | -124 | -302 | -279 | -260 |
| | **Panel (c): MVPF versus No EV Subsidy** | | | | |
| 17. | Global MVPF | 1.11 | 1.11 | 1.10 | 1.15 |
| 18. | US MVPF (global SCC) | 1.23 | 1.30 | 1.29 | 1.32 |
| 19. | US MVPF (domestic SCC) | 1.10 | 1.18 | 1.17 | 1.18 |

Notes: This table presents counterfactual simulation results for the IRA and alternative policies, relative to a common baseline of no EV subsidies (column 2 in Tables 6 and 7). In column 2, we simulate the uniform EV subsidy, subject to Section 30D trade restrictions, that maximizes US total surplus with the domestic SCC. In column 3, we assume the planner values annual US jobs at $10,700 per job. In column 4, we simulate the choice-specific differentiated EV subsidy that maximizes US total surplus with the domestic SCC. The marginal value of public funds (MVPF) equals (Δconsumer surplus + Δproducer surplus - Δnegative externalities)/(Δ subsidy spending - Δ gas tax revenue). The "US MVPF" values are computed with ΔUS producer surplus in the numerator, while the "Global MVPF" uses Δglobal producer surplus in the numerator. "US total surplus" (under the global SCC) equals global surplus minus foreign automakers' producer surplus. Costs of abatement are negative when a policy change results in lower carbon emissions but is welfare-positive even when disregarding carbon damages.

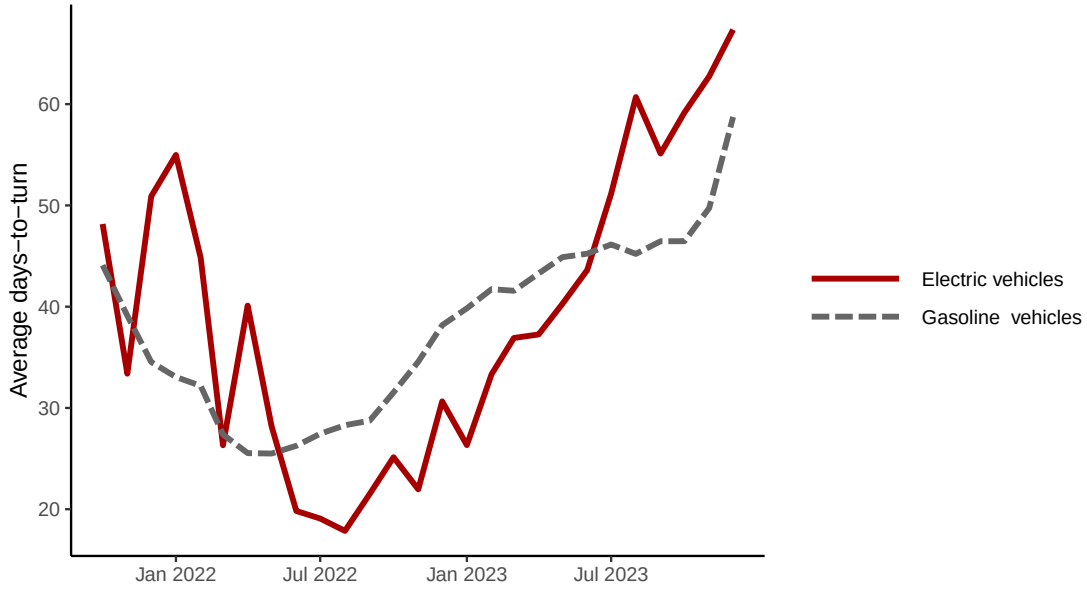
Figure 1: **Section 30D Purchase Credit Eligibility**



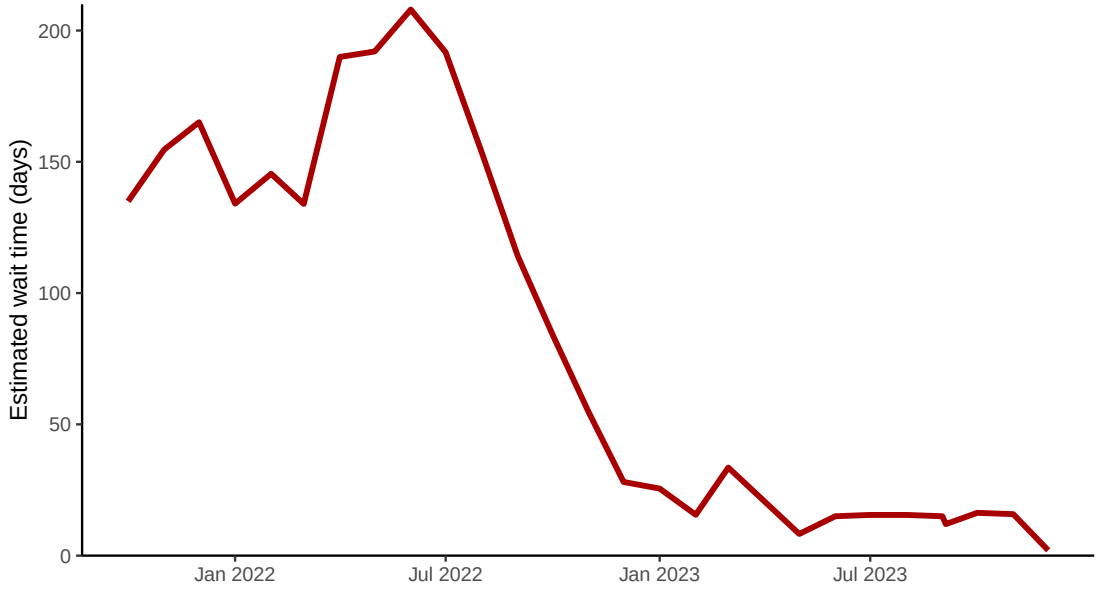

Notes: This figure shows which models are eligible for the IRA Section 30D purchase credit by time period. The figure includes models averaging more than 300 sales per month in 2022 and 2023. The shading intensity indicates the credit amount that the model is eligible for. The red, blue, and orange colors indicate eligibility changes that we study in Section 4.2 below. This figure is inspired by Figure 3 of Buckberg (2023). The final column header says "Late 2023" because several vehicles were retroactively made eligible in late 2023, and the Tesla Model X became eligible in late 2023 as its MSRP decreased.

Figure 2: **Electric Vehicle Supply Constraints**

(a) **Average Days-to-Turn (Not Including Tesla)**



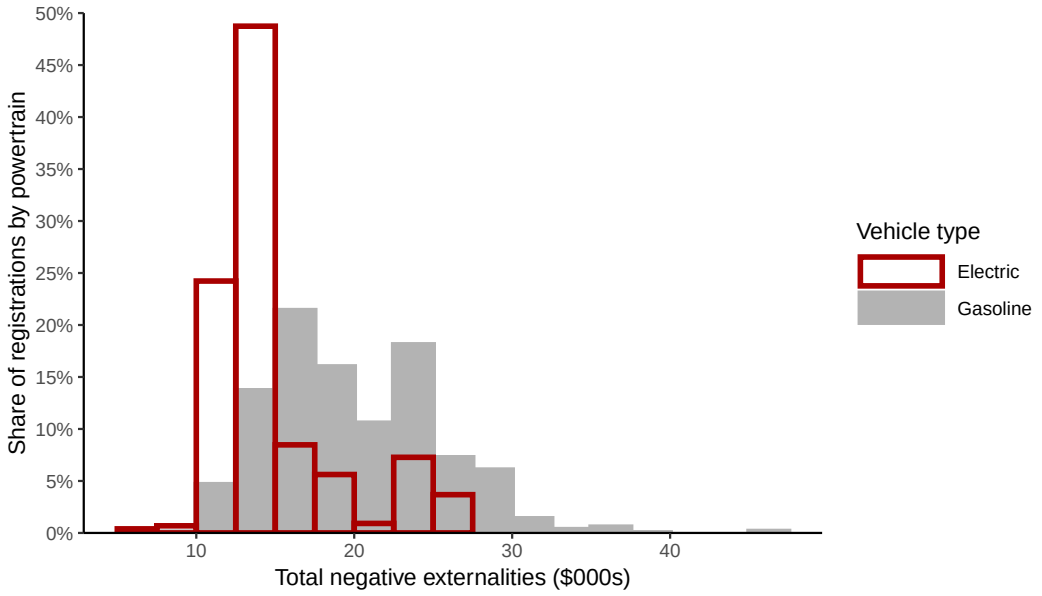

(b) **Tesla Estimated Wait Times**



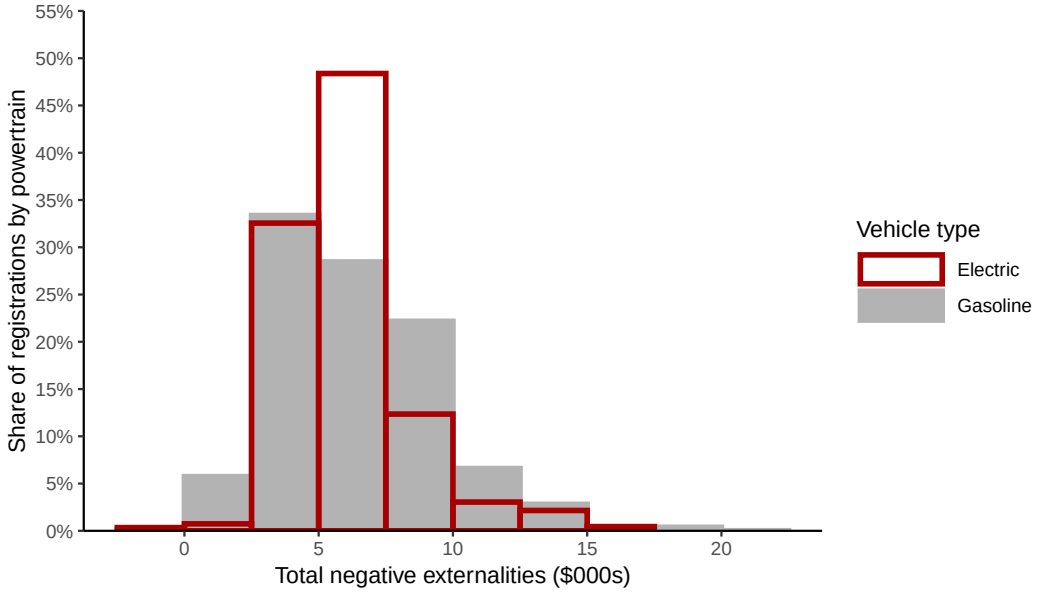

Notes: Panel (a) presents the average days-to-turn by powertrain, i.e., the average time that vehicles sold in that month were available in the dealership's inventory before being sold; data are from Edmunds (2024). The Edmunds data exclude Tesla. Panel (b) presents the average delivery wait time reported on the Tesla website, weighting across models by their market share in July and August of 2023 (Tesla 2023; Pritchard 2023; The Internet Archive 2023).

Figure 3: **Distribution of Negative Externalities Across Submodels**

(a) **Global SCC**



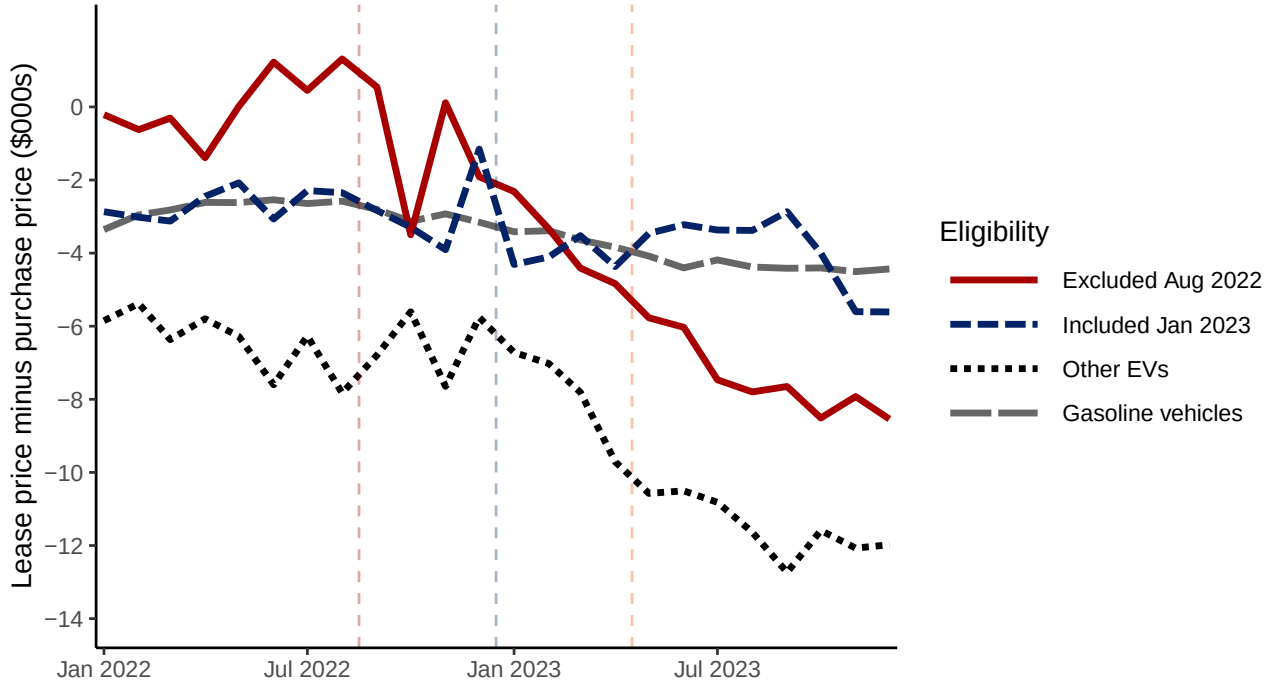

(b) **Domestic SCC**



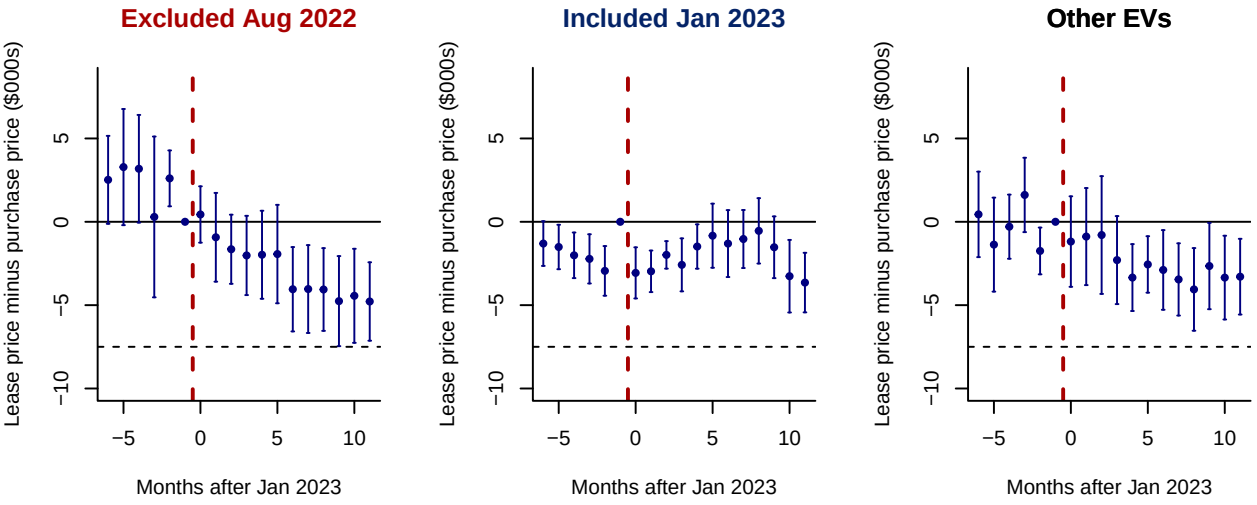

Notes: This figure shows the distribution of total negative externalities across submodels, weighting submodels by registrations in July and August 2023. Carbon damages are evaluated at the global SCC of $241 in panel (a), and at the domestic SCC of $28 in panel (b).

Figure 4: **Relative Lease Price Trends Associated with Eligibility Changes**

(a) **Trends by Eligibility Group**



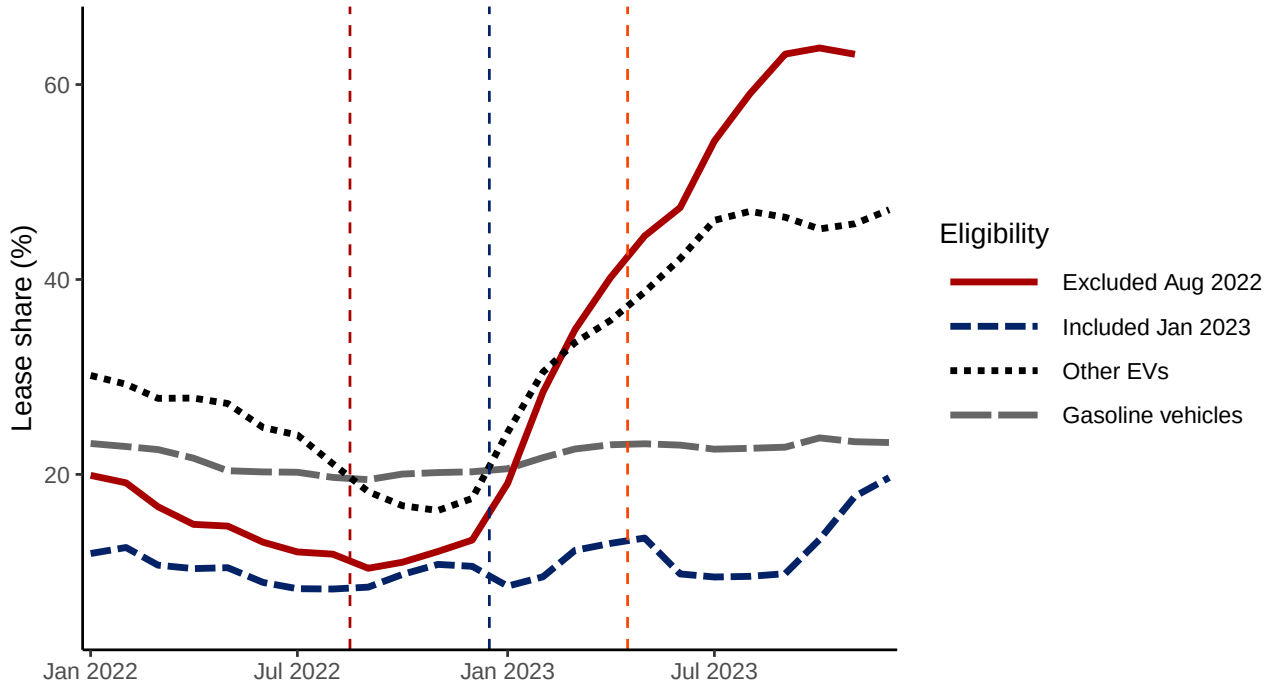

(b) **Event Study Estimates**



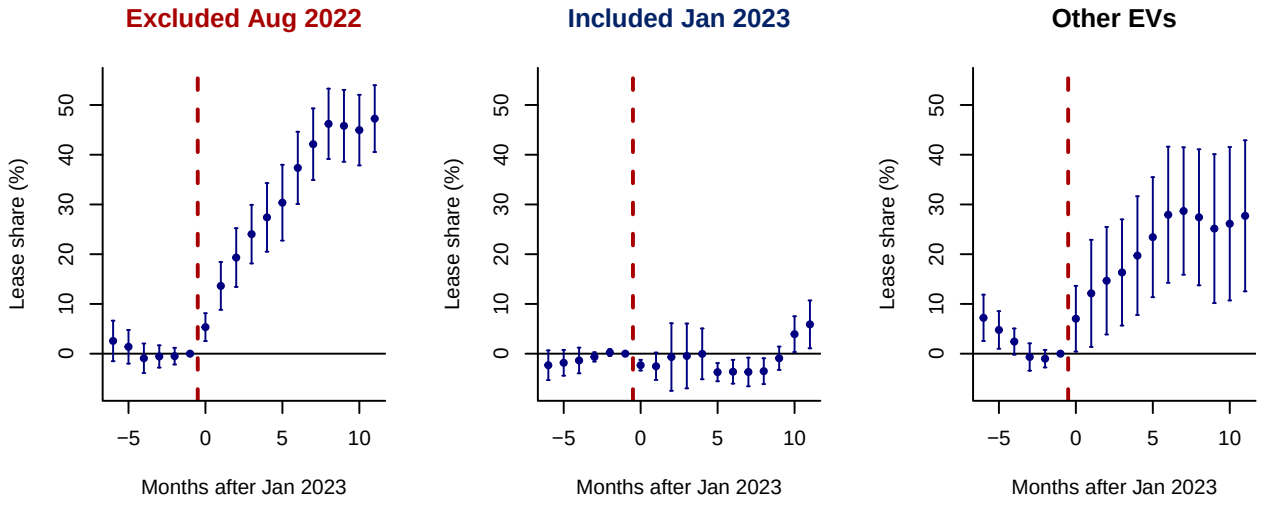

Notes: Panel (a) presents lease price minus purchase price indexes constructed by computing the January 2023 weighted averages for each eligibility group and then recursively adding the sales-weighted average changes for all submodels available in each previous or subsequent month. Panel (b) presents the $\gamma_s^e$ coefficients and 95 percent confidence intervals from equation (2). Eligibility groups are described in Figure 1; the Other EVs group includes all EVs not part of either the Excluded August 2022 or the Included January 2023 groups. In both panels, we weight submodels by average monthly sales in months when the submodel was available.

Figure 5: **Lease Share Trends Associated with Eligibility Changes**

(a) **Trends by Eligibility Group**

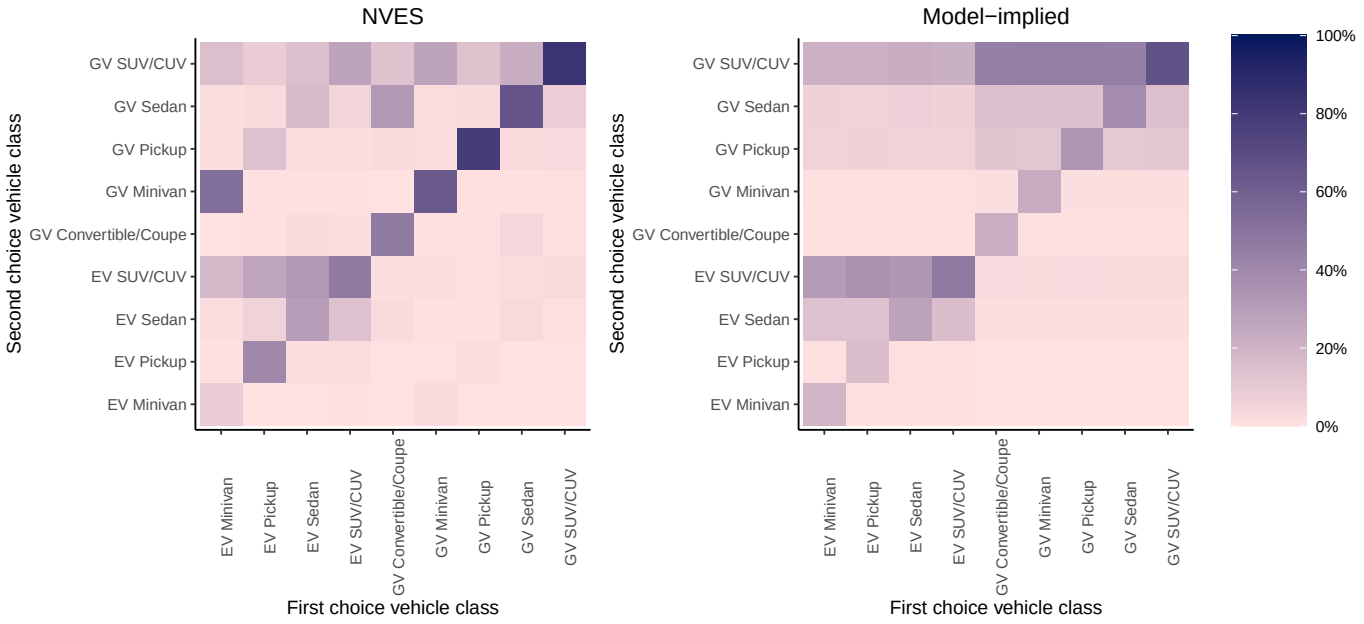



(b) **Event Study Estimates**




Notes: Panel (a) presents lease share indexes constructed by computing the January 2023 weighted averages for each eligibility group and then recursively adding the sales-weighted average changes for all submodels available in each previous or subsequent month. Panel (b) presents the $\gamma_s^e$ coefficients and 95 percent confidence intervals from equation (2). Eligibility groups are described in Figure 1; the Other EVs group includes all EVs not part of either the Excluded August 2022 or the Included January 2023 groups. In both panels, we weight submodels by average monthly sales in months when the submodel was available.

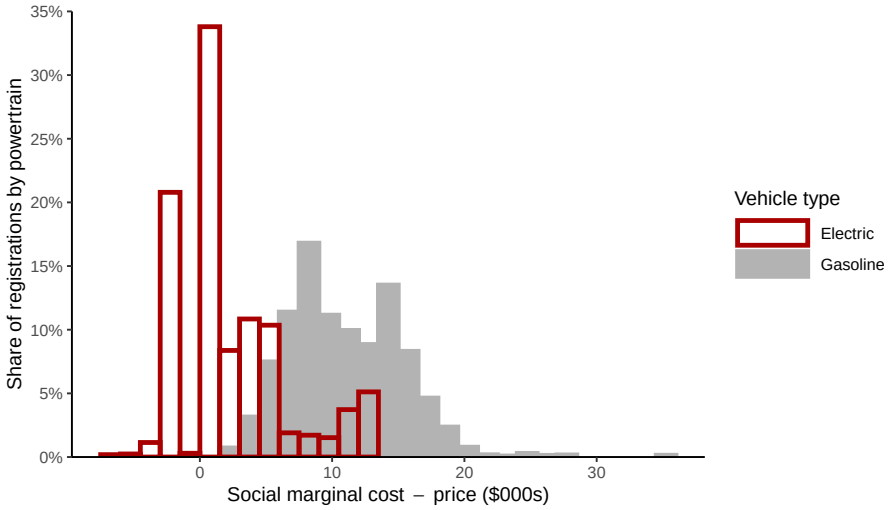
Figure 6: **Heat Map of Second Choices by Vehicle Class**



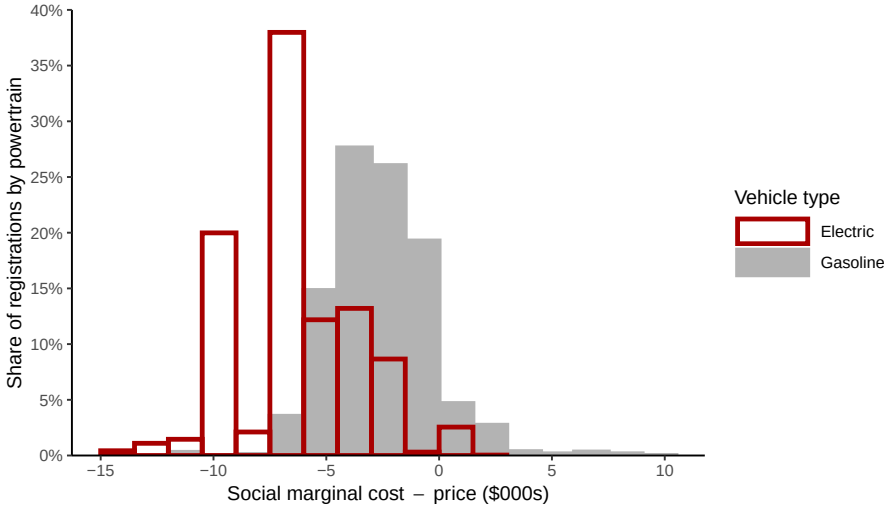

Notes: This figure presents the conditional shares of vehicle class × powertrain second choices; the left panel shows these second choice probabilities from the NVES data, while the right panel shows the model-implied shares. The fact that 0 percent of EV Minivan buyers in the NVES select another EV Minivan as a second choice arises because the NVES aggregates all trims of the Chrysler Pacifica PHEV (the only electric minivan available) with their gasoline counterparts for the second choice survey. Since the minivan segment is a small share of vehicle sales overall, this minimally changes our model fit statistics.

Figure 7: **Distribution of Social Marginal Cost Minus Price Across Submodels**

(a) **Global SCC**




(b) **Domestic SCC**




Notes: This figure shows the distribution across submodels (both purchased and leased) of social marginal cost minus price, weighting submodels by registrations in July and August 2023. Social marginal cost is the inferred marginal production cost $c_j$ plus negative externality $\phi_j$; panel (a) computes negative externalities at the global SCC, whereas panel (b) computes negative externalities at the domestic SCC. Price is the unsubsidized market price $p_j$.

# Online Appendix

## The Effects of "Buy American": Electric Vehicles and the Inflation Reduction Act

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# A Data Appendix

## A.1 Panel Construction

Our core panel dataset consists of data from Experian (quantities), Cox Automotive (lease prices), and the California DMV (purchase prices), together with a collection of supplemental sources.

The Experian data have complete national coverage of new vehicle sales, both purchases and leases, aggregated to the monthly level. To clean these data, we first restrict to considering light-duty vehicles, consider only vehicles purchased for personal use or leased, and exclude fuel cell vehicles. Because vehicle quantities are reported at somewhat different levels of aggregation across observations (e.g. combining similar trims some, but not all, of the time), we aggregate to a "lowest common denominator" definition of submodel, which is consistent across observations and across time.

The Cox Automotive data have national coverage, but are not exhaustive as they depend on there being a business relationship between Cox and the firms/dealerships. The dataset is at the transaction level, which allows us to obtain VIN-level information (e.g. assembly location). Additionally, Cox uniquely includes the details of lease contracts, surfacing down payments, monthly payment, and duration. We clean these data by filtering out observations with implausible monthly payments for the vehicles we consider (i.e., below $200 or above $1,500) and outlier lease durations (less than 12 months or more than 60 months).

The California DMV data have coverage only for California, and are provided in a sequence of cross-sectional snapshots of currently-registered vehicles. The snapshots we use are from October 2022, July 2023, October 2023, and April 2024. The variables associated with a registration record include the prefix of the vehicle identification number (VIN), self-reported price, vehicle make, series, model, model year, date of the most recent registration, most recent odometer reading and reading date, year and month of each ownership transfer, and an indicator for whether the vehicle is leased. The VIN prefix is the first 11 digits, excluding the ninth (check digit). We obtain purchase prices from the California DMV data, which, combined with lease terms from the Cox data, allow us to compute lease prices which are comparable to directly observed purchase prices. We have also run our analyses using purchase prices from Cox for observations where they are available (approximately 93 percent of registrations in our cross-section) and found similar results, which we do not include in their entirety for the sake of brevity. For comparison, using Cox prices results in counterfactual outcomes which are typically much closer to our baseline specification in column (1) of Appendix Table A9 than any of the other robustness checks.

To transform this sequence of snapshots into a single panel, we employ the following algorithm. First, we check the odometer reading; if it is less than 250 miles, we use the corresponding odometer reading date as the initial registration date.[39] For vehicles with odometer readings that are either missing or greater than 250 miles, we then check the latest registration date. Because we observe

[39]In California, the primary registration form for new vehicles (*Form REG-343*) requires reporting the odometer reading "upon date of purchase in California."

the model year of the vehicle, we can check whether the date is plausibly an initial registration date (i.e., falls between July 1 of the year prior to the model year and April 1 of the year after the model year). If it is, we use the most recent registration date as the initial registration date. If a vehicle doesn't satisfy our requirements for inferring initial registration date on the basis of odometer or latest registration information, we assume we cannot identify the initial registration date for the vehicle, and so drop it from our sample. In practice, 96 percent of initial registration dates in our panel are derived from the first step, using odometer reading/reading date. The remaining 4 percent may be subject to some errors, as California's requirement for annual re-registration of a vehicle could confound successful identification of the initial registration date. For example, a vehicle could be purchased in September of the year before its model year and be re-registered in September of its model year; if this vehicle's observation is also missing a valid odometer reading, or if its ownership was transferred (and hence a new, higher odometer reading was recorded), it would enter our panel as having been initially acquired in September of its model year. We check that our findings are robust to using only the vehicles whose initial registration dates are inferred from their odometer reading.

The main analysis sample makes two other exclusions. We exclude any observations with fewer than 25 registrations. Additionally, we exclude any purchase or lease price observations based on fewer than 10 transactions.

We construct fixed-weight price indices as follows. Define $x_{kt}$ as some variable for submodel $k$ in month $t$, define $\bar{q}_k$ as submodel mean monthly sales in the months when firms sell it, and define $t = 0$ as January 2023. The fixed-weight index in month $t$ equals

$$\bar{x}_t = \frac{\sum_k \bar{q}_k x_{k0}}{\sum_k \bar{q}_k} + 1\,(t > 0) \sum_{r=1}^{t} \frac{\sum_k \bar{q}_k \,(x_{k,r} - x_{k,r-1})}{\sum_k \bar{q}_k} + 1\,(t < 0) \sum_{r=t}^{-1} \frac{\sum_k \bar{q}_k \,(x_{k,r} - x_{k,r+1})}{\sum_k \bar{q}_k}.$$

As described in the text, $x_{kt}$ is missing for some $kt$, so the sums over $k$ implicitly include only non-missing observations. This prevents the computed series from displaying fluctuations entirely driven by panel imbalance.

We check the trend of daily new registrations over time based on two cross-sections. We find a delay of roughly 45 days for all registrations to appear in the DMV record; that is, the number of observed registrations falls sharply relative to the trend within the 45 days prior to the date the snapshot was taken. Given the possibility that later registrations overwrite earlier ones, we use the earlier cross-section to approximate the initial registration date of earlier-registered vehicles. The final dataset is constructed using the October 2022 cross-section for registrations between January 1, 2022 and August 15, 2022; the July 2023 cross-section for registrations between August 16, 2022 and April 30, 2023; the October 2023 cross-section for registrations between May 1 and August 31, 2023; and the April 2024 cross-section for registrations between September 1 and December 31, 2023.

In addition to these core data sources, we use the National Highway Traffic Safety Administration (NHTSA)'s VIN decoder to append information on powertrain type (e.g. GV vs. PHEV vs.

BEV) and assembly location. Further combining vehicle names and model years with eligibility information from FuelEconomy.gov, we generate a submodel × month panel, with information on number of monthly registrations (including purchases and leases separately) from Experian, vehicle characteristics including powertrain and assembly location, average recorded purchase price from the California DMV, average lease price and lease price minus purchase price from merging Cox and California DMV data, and credit eligibility. This final merge is achieved by leveraging the presence of VIN-prefix in both the California DMV and Cox datasets, and through a manual crosswalk between our vehicle names and those in Experian.

For Tesla price data, we augment the above panel using the following sources. From Tesla's website, via the Internet Archive's Wayback Machine, we obtain base configuration lease terms and purchase prices for most months in our sample. For those which are inaccessible via the Wayback Machine, we additionally collect price data from contemporaneous reporting (InsideEVs 2024) and the enthusiast-run project Tesla Car Price History (Bautista 2024). This allows us to consistently compare lease prices and purchase prices for the base configuration of each model across time.

Finally, we used new EV registration data from EV-Volumes to inform our dealership inventory survey (Appendix A.3) and compute market shares to weight the resulting wait times. We also used vehicle registration data from Texas to investigate the frequency of consumers exploiting the "loophole within a loophole" as described in Appendix B.1. These registration data are not as straightforward as those from California's DMV to convert into a panel in part due to a lack of an explicit lease flag, but do feature full vehicle VINs and addresses. These features allow us to track changes in ownership over time, which is crucial for identifying lease buyouts.

For measuring credit eligibility, in the few instances of variation in eligibility within a submodel-by-month, we assign maximal eligibility to all observed transactions. This affects a small number of submodels, such as the BMW 330e (which was being assembled in both Mexico and Germany for the US market in late 2022) and the Volkswagen ID.4 (which moved assembly to Tennessee in late 2022, resulting in some model year 2022 German-assembled ID.4s being registered after the IRA passed).

## A.2 Externalities

This subsection provides additional detail on a few components of our estimates of each submodel's externalities.

**$CO_2$ and local air pollution emissions from driving**. For EVs, we compute the damages from generating electricity to charge the vehicles, using hourly-seasonal-regional short-run marginal emission rates from Holland et al. (2024). These estimates include emissions of $CO_2$ and of $SO_2$ and $NO_x$. To aggregate across geographies to the national level, we average across ZIP codes within each state, weighting by each ZIP's total vehicle miles traveled, following Holland et al. (2016), and we average across states weighting by each state's powertrain-specific new vehicle sales in 2023. Since EVs have higher market shares in states with cleaner electricity generation, the state-by-powertrain weighting implies lower marginal damages from EVs than if we assumed that

the marginal EVs and GVs had the same sales everywhere.

As a sensitivity analysis, we also consider the impact of estimating carbon emissions from EV charging using the long-run marginal $CO_2$ emissions estimates from the National Renewable Energy Laboratory's Cambium "MidCase" scenario, averaged across the estimated values for 2025, 2030, and 2035 (Gagnon et al. 2024). This results in a reduction in $CO_2$ from charging of 58 percent relative to our baseline numbers (comparing Appendix Table A1 to Table 2).

For GVs, we compute the harms from tailpipe emissions. We construct $CO_2$ emissions using each submodel's fuel economy rating. We compute submodel-specific local pollution emission factors from US EPA 2024 test data, using submodel-specific certified emission rates for the vehicle's full useful life as in Jacobsen et al. (2023), combined with regulatory standards. Each submodel's emissions for each pollutant are computed as the average between the test value and regulatory value. The regulatory standards are likely to over-estimate true emissions, as vehicles over-comply to avoid recalls; the exhaust tests may under-estimate true on-road emissions, as emissions control systems depreciate with age and auto manufacturers optimize vehicle design towards the emissions control test; averaging the two provides a middle ground. Using these emissions rates, we again compute the weighted average marginal damage across ZIP codes using marginal damage estimates for local pollutants from the AP3 integrated assessment model (Clay et al. 2019). We assume that 63 percent of PHEV miles traveled are on gasoline and 37 percent are on electricity, following Plötz et al. (2020).

We calculate the US SCC by taking the ratio of the US to global SCC in Ricke et al. (2018), averaging across models and scenarios, then multiplying by our CPI-inflated global SCC value.

**Accident externalities**. Following Anderson and Auffhammer (2014), we estimate the expected increased mortality cost from accidents of each submodel relative to the lightest vehicle available. The mortality cost is the product of the accident probability, the incremental death probability from driving a heavier vehicle, and the value of a statistical life (VSL). Our estimate of the accident probability comes from a regression of accident counts on registered vehicles. The regression coefficient yields an estimate of the probability that a vehicle is involved in a multi-vehicle accident each year; we convert this into a lifetime probability while assuming that the same vehicle is not in more than one multi-vehicle accident over its lifetime. We use the estimate obtained from this approach, 29 percent, and follow Anderson and Auffhammer (2014) for the remainder of the calculation.

**Positive energy pricing externalities**. For EVs, we use the utility-specific markups on residential electricity above private marginal cost calculated by Borenstein and Bushnell (2022), weighting utilities within a state by sales and weighting states by new EV sales in 2023. The resulting weighted average markup is 12 cents per kWh. For GVs, we use federal and state gas taxes, weighting states by new GV sales in 2023. The resulting weighted average gas tax is 55 cents per gallon. A submodel's total positive energy pricing externality depends on those markup or tax amounts and the submodel's electricity use or fuel economy.

## A.3 Dealership Inventory Survey

Between July 7 and August 4, 2023, we conducted a survey targeting dealerships selling popular EVs in a number of geographically diverse markets with relatively large EV adoption rates. We collected responses from the West Coast (Los Angeles, San Diego, San Jose, and San Francisco), the East Coast (New York and Philadelphia), the Midwest (Minneapolis), and the South (Houston). These are in addition to a brief pilot, whose results are unreported here, conducted in St. Louis in early July prior to the other surveys.

We compiled a list, from EV-Volumes data, of the EV models that were most popular in 2022 and early 2023. Some vehicles were omitted due to having been discontinued by the time of our survey, and others were omitted due to being largely or entirely sold direct-to-consumer (namely Rivian and Tesla). RAs then searched for dealerships selling each make, and contacted them by phone. For each model, they then asked

> Hello, my name is _____ and I'm interested in purchasing a [*model name*]. I would be paying cash for the vehicle. When would be a feasible delivery time for the vehicle? How much should I expect to pay? How about delivery times for your other electric or plug-in hybrid vehicles?

The RAs then recorded this information for each of the EV models which the dealership in question sold. If the model was not in stock, up to two more dealerships were contacted in the city; our analysis is conducted on the minimum wait time across contacted dealerships in the city. Additional information concerning the exact trim and sales price was recorded, though it is not directly used here; we did not observe systematic markups relative to MSRP. In total, we recorded wait times for 681 dealership-model combinations.

For each dealership-model combination with non-zero wait time, we assign the midpoint of the predicted window given as that dealership-model wait time. We then take the minimum of these within cities to obtain a city-model dataset. These data are merged with EV-Volumes registration records from July and August, and we compute the proportion of wait times that are zero days, within 30 days, and within 60 days both weighted (by national market share) and unweighted.

Many dealerships in our survey mentioned the EV credits. Many also ran advertising campaigns highlighting $7,500 rebates on leases; Appendix Figure A7 provides one example.

## A.4 Employment Intensity

To compute the domestic employment effects of vehicle production, we partition the automotive production sector into two components: final assembly and parts. Cuenca, Gaines, and Vyas (1999) estimate that parts account for $\sim 80\%$ of total costs and final assembly for $\sim 20\%$.[40] We

[40] We considered obtaining these statistics from input-output tables but chose not to for a few reasons. Specifically, we examined the cost structure of the automobile manufacturing industry (NAICS code 336111) in the 2017 Use table after redefinitions at producer prices from the Bureau of Economic Analysis. The partition of intermediate goods into "automotive parts" and other industries in the Use table is likely prone to measurement error. The auto parts

then compute the number of US workers in the final assembly and parts manufacturing according to the following formulas:

$$l_j^{\mathrm{a}} = \frac{c_j \times \mathrm{US}_j^{\mathrm{a}} \times 0.2 \times \mathrm{r}^{\mathrm{a}}}{w^{\mathrm{a}}} \text{ and } l_j^{\mathrm{p}} = \frac{c_j \times \mathrm{NA}_j^{\mathrm{p}} \times 0.8 \times \mathrm{r}^{\mathrm{p}}}{w^{\mathrm{p}}}$$

Using these values, we calculate the following baseline aggregates:

$$L^{\mathrm{a}} = 6 \times \sum_j R_j l_j^{\mathrm{a}} \approx 98,000 \text{ and } L^{\mathrm{p}} = 6 \times \sum_j R_j l_j^{\mathrm{p}} \approx 227,000$$

Here $l_j^{\mathrm{a}}$ and $l_j^{\mathrm{p}}$ represent the per-vehicle number of US workers associated with the production of submodel $j$ in the assembly and parts industries, respectively. From our model, we take submodel-specific estimates of per-vehicle production costs $c_j$. We use ancillary data on the share $US_j^a$ of submodel $j$ assembled in the US, and the share $\mathrm{NA}_j^p$ of submodel $j$ parts from the US or Canada. We calculate the ratio of payroll to value added $r^{\mathrm{a}}$ for assembly, and the ratio of payroll to output $r^{\mathrm{p}}$ for parts. We divide these by the mean wages $w^{\mathrm{a}}$ and $w^{\mathrm{p}}$ of workers in the assembly and parts industries, respectively. To calculate the baseline aggregates, we use data on registrations $R_j$ of each submodel in our two-month panel. We use the share of payroll in value added in the assembly industry since we interpret the 20 percent share as representing the value added from the final assembly process beyond the value of the input parts, abstracting from intermediates besides auto parts. We use the share of payroll in output for the parts industry since the 80 percent share is from the perspective of the final assembler, while expenditures on parts are revenue from the perspective of firms in the parts industry. The aggregate employment figures are calculated analogously across the counterfactual scenarios we consider, and we then compute the change in employment in each industry relative to the baseline under the IRA.

The aggregate employment figures produced by summing up our employment measures (that is, $L^{\mathrm{a}}$ and $L^{\mathrm{p}}$) differ from industry-wide aggregates for a few reasons; the BLS reports employment of 254,000 and 618,000, respectively, in the final assembly and parts industries. First, our construction captures only employment associated with vehicles which are purchased or leased domestically. For this reason, final vehicles which are exported (accounting for approximately 15 percent of production according to total production as reported by International Organization of Motor Vehicle Manufacturers (2023) and exports reported by US International Trade Administration (2025)) and parts which are exported for assembly into vehicles purchased outside of the US are counted in simple aggregate employment figures but are intentionally excluded from our calculations. Within the parts industry specifically, some portion of the industry's output is used for purposes other than producing new vehicles (e.g. replacement and aftermarket parts). Additionally, our registration data cover only a subset of the overall light-duty vehicle market; while we capture the vast

industry (4-digit NAICS code 3363) accounts for over half of all intermediate goods that the automobile manufacturing industry purchases. The largest other suppliers of intermediate goods to auto manufacturing industries include retail, wholesale, engines, glassware, unspecified machinery, etc., and other industries that are likely supplying auto parts. Additionally, the Use table does not distinguish EV versus GV manufacturing, and EV manufacturing was even less common in 2017.

majority (over 80 percent) of non-commercial light-duty registrations (our panel, once annualized by multiplying by six, counts 10.56 million registrations as compared to the true value of 12.95 million), there are an additional 2.5 million fleet sales which we exclude from our analysis (National Automobile Dealers Association 2023). Finally, we suspect that the fleet vehicles may be disproportionately US-made, in which case our aggregate employment estimates would undershoot relative to industry totals.

# B Descriptive Facts Appendix

## B.1 Texas Registration Data Analysis

To examine the frequency with which EV buyers took advantage of the "loophole within a loophole" (that is, first leasing a vehicle ineligible for the purchase credit, obtaining advantageous lease terms due to the leasing loophole, then proceeding to buy out the lease), we combined two data sources with VIN-level information. The first is Cox Automotive's data on new vehicle transactions, which allows us to identify VINs associated with leases. The second is Texas registration data, which allows us to track the same VIN across time (as we can observe repeated registrations for the same VIN). This is not possible using the California data, which are provided at the VIN-prefix level. Because the California registration data lack the full VIN to identify a particular vehicle across time, they do not let us identify lease buyouts in California. On the other hand, the Cox data alone do not have information on re-registrations of vehicles or of lease buyouts directly. We also note that, while the Texas market is not as large as California's, it is the second largest in the country and makes up a substantial share of the US market.

Combining the Cox and Texas registration data, we computed the early buyout rate in Texas by filtering to vehicles which were recorded as leased in the Cox data and identifying vehicles that were re-registered to a different address within three months of their initial registration. We use the three-month window given the comments dealerships conveyed about rapidly converting a lease to a purchase for consumers who hoped to purchase a vehicle but obtain the lease subsidy. We use re-registration as a proxy for buyouts, as leased vehicles appear to be almost exclusively recorded as being initially registered to a leasing company; in our data, over 90 percent of leases are initially registered to addresses with at least 100 leased vehicles over the time period. These addresses are either out-of-state, P.O. boxes, or known addresses of leasing companies.

This analysis finds that levels of estimated early buyouts remained low among EVs and GVs both before and after the new lease credit rules went into effect. In late 2022, EVs and GVs had early buyout rates of around half a percent of leases. In early 2023 EVs had an early buyout rate of around 1.4 percent. The estimated GV early buyout rate remained unchanged.

# C Event Study Appendix

## C.1 Event Study Regression Tables

Appendix Table A2 presents the regression estimates used in the calibration of the empirical model in Section 5. Appendix Table A3 presents regression estimates documenting the lack of substantial price changes after EVs lost eligibility in August 2022 and April 2023.

## C.2 Event Studies with Purchase Prices, Lease Prices, and Registration Quantities

This appendix presents event studies of changes in purchase prices and in total registrations (including both purchases and leases) around the 30D vehicle eligibility changes.

### Purchase Prices

The fixed-weight purchase price indices in Panel (a) of Appendix Figure A14 suggest limited changes in purchase prices coincident with credit eligibility. The red and orange vertical lines indicate the date of eligibility change for the eligibility group of the corresponding color from Figure 1. For the vehicles that lost credit eligibility in August 2022, shown in red, prices change relatively little in the several months before and after those vehicles lost eligibility in mid-August 2022, although prices drop more substantially in early 2023 as part of the price cuts described in Appendix Figure A1. For the Excluded/Reduced April 2023 group in orange, prices also drop temporarily in early 2023, but the index stays in the range of $63,000 to $65,000. For all other EVs in black, which are mostly Tesla, prices drop by more than $10,000 between mid-2022 and the end of 2023. The price trends in late 2022 may reflect supply conditions, while we interpret patterns in Spring and Summer 2023 as less constrained.

The event study coefficients $\gamma_s^e$ from equation (2) for each eligibility group in Panel (b) of Appendix Figure A14 corroborate this finding, as they also suggest limited effects from 30D credit eligibility changes on purchase prices. The confidence intervals rule out price drops of more than $500 to $1,000 between September and December 2022 among the vehicles excluded from credit eligibility in August 2022. They also rule out price drops of more than $2,000 among the group of vehicles that lost or had decreased credit eligibility in April 2023. We do not consider event studies for the groups that changed eligibility in January 2023 because those coincide with the large Tesla-led price cuts, reflecting broader contemporaneous market trends. Regression estimates that restrict the coefficients to be equal across the submodels that experienced an eligibility change in August 2022 and April 2023 also suggest that prices did not substantially fall following these eligibility changes. These estimates reject the hypothesis that prices dropped by more than $625 in the three months after losing eligibility versus the three months before; see Appendix Table A3.

**Relative Lease Prices**

Panel (a) of Figure 4 presents the fixed-weight indexes of relative lease price for the two groups of EVs that changed eligibility, all other EVs, and all GVs. Relative lease price for GVs dropped moderately in both 2022 and 2023. As the market weakened, relative lease prices for EVs also decreased in the latter half of 2022. For the Excluded August 2022 group, lease prices decreased more sharply in 2023. For the Included January 2023 group, relative lease prices increased temporarily in December 2022, because Tesla cut purchase prices in that month but did not correspondingly cut lease prices until the next month. The group's relative lease prices were relatively flat in 2023 until Tesla started to offer a $7,500 lease rebate late in the year, and the blue line correspondingly drops in November and December 2023. For all other EVs, relative lease prices also decrease more than those of GVs over the course of 2023.

Panel (b) of Figure 4 presents the event study estimates. We define the "event" as the start of 45W lease credits in January 2023. The patterns match Panel (a), except that they adjust for the comparison to GVs, where relative lease prices also decreased. The left sub-panel shows that for the Excluded August 2022 group compared to GVs, relative lease prices decreased by about $5,000 compared to December 2022, and by about $7,000 relative to their level in July through November 2022. The middle sub-panel shows that with the exception of the December 2022 blip, the Included January 2023 group's relative lease prices trended slightly upward relative to GVs during 2023 until the November 2023 Tesla lease price reduction. The right sub-panel shows that for all other EVs, relative lease prices dropped by around $3,000 relative to GVs by the end of 2023.

Also notable is the heterogeneity in EV relative lease price changes across firms; see Appendix Figure A8. By July-August 2023 relative to October-December 2022, Kia, Volvo, Volkswagen, and Hyundai had dropped relative lease prices by close to $7,500; Jeep, BMW, Toyota, and Ford had dropped relative lease prices by about $1,000 to $4,000; and Tesla and GM had not reduced relative lease prices.

While the Excluded August 2022 group's relative lease price decrease in 2023 is consistent with substantial or full pass-through of the $7,500 lease credit, two other results are not. First, while Tesla eventually offered lease rebates, the Tesla and GM relative lease price decreased by much less than $7,500, and the decrease only occurred after an 11-month delay. Second, the Excluded August 2022 group's relative lease price did not increase in late 2022 after losing eligibility. An industry insider suggested a possible explanation—the IRA might have caused a change in economic incidence. Perhaps the lease credit was not passed through before the IRA, but media coverage of tax credits and the leasing loophole could have raised the salience of leasing and the tax credit, inducing firms to compete harder for leases.

**Registration Quantities**

The IRA should affect total registrations in several ways. First, changes in vehicle eligibility for Section 30D credits should shift demand among income-eligible consumers. Second, the January 2023 income eligibility restriction for Section 30D credits should reduce EV demand among con-

sumers who do not want to lease. Third, the January 2023 availability of EV lease credits under 45W should increase overall EV demand. Appendix Figure A15 presents the fixed-weight indexes and event study estimates. The figures illustrate substantial market trends that predate, and are thus likely unrelated to, changes in EV credit eligibility. The Excluded August 2022 group saw a significant registration decrease in July and August 2022 (before they lost 30D eligibility), driven by decreases for Hyundai and Kia. The Included January 2023 group saw a steady increase in 2022 and 2023 (before they regained 30D eligibility), as Tesla demand grew steadily. The Excluded/Reduced April 2023 group saw decreases in registrations in the first few months of 2023 (again, before they lost 30D eligibility).

Comparing against the final month before the eligibility change (month $-1$ on the x-axis), the figures show no statistically detectable evidence of responses to credit eligibility. For the Excluded August 2022 group, the 95 percent confidence intervals rule out registration decreases of more than about 20 percent. For the Included January 2023 group, the confidence intervals rule out registration increases of more than about 20-30 percent. For the Excluded/Reduced April 2023 group, the confidence intervals rule out registration decreases of more than about 20-30 percent in the ensuing four months. However, especially given the evidence of other market trends before the eligibility changes, we do not know what would have happened but for those eligibility changes.

**Comparison to Model**

The equilibrium model does not target the observed pass-through of credits to prices from Section 4.2, but the simulated pass-through in the model following the removal of all EV credits indicates that they are reasonably aligned. First, as a comparison to the purchase price patterns in Appendix Figure A14 Panel (b), we look at pass-through of credits to purchase prices. Among EVs that previously had credits in this simulation, the model indicates that purchase prices fall by $219 on average per $7,500 in credits lost, suggesting that the incidence of the purchase credits mostly falls on consumers, consistent with our descriptive findings and within the confidence interval of Appendix Table A3. Second, as a comparison to the lease price patterns in Figure 4 Panel (b, left), we look at pass-through of credits to lease prices (relative to purchase prices). Among EVs in the same Excluded August 2022 group, the model indicates more than complete pass-through of $8,883 to consumers, overshifting by 18.4 percent above the $7,500 amount and somewhat above the confidence interval in Appendix Table A2 column (2). The fact that our model is able to generate overshifting is a consequence of the consumer heterogeneity in our model, which leads to more flexible demand curvature (Miravete, Seim, and Thurk 2023). Of course the regressions and model could differ for many reasons, most notably supply chain conditions in the pre-IRA period.

## C.3 Doubly-Robust Event Studies

This appendix presents "doubly robust" estimates where the GV control group is reweighted to match the EV pre-IRA average price. We compute weights with entropy balancing (Hainmueller 2012). This method computes weights such that the reweighted sample matches a set of target

moments, while maintaining maximal "closeness" (in an entropy sense) to a set of researcher-defined initial weights. In our case, these initial weights are the monthly average registrations of each GV, and the targeted moment will be average pre-IRA purchase price. Since EVs have a higher average price, this will have the effect of upweighting more expensive GVs.

Because we can only compute new weights for GVs which are present in the pre-period sample, we lose 7 percent of overall registrations after switching to these weights. To ensure that the aggregate EV-GV balance is approximately the same between the primary specification and this alternative specification, the entropy-balanced weights are normalized to sum to one, then multiplied by the monthly average registrations across all GVs.

Appendix Figures A10–A12 present the reweighted event study estimates.

# D Equilibrium Model Appendix

## D.1 Quantity Demanded and Consumer Surplus

There are $M$ consumers in the market indexed by $i$. We define consumer types $h(i)$ with representative income $y_{h(i)}$, with a continuum of consumers receiving indirect utility from choice $j$ with price $p_j$ and demand subsidy, if income-eligible, $\tau_{h(i)j}$ equal to

$$U_{ih(i)j} = \xi_j - \alpha_{h(i)} \left(p_j - \tau_{h(i)j}\right) + \boldsymbol{\beta}_{h(i)}^{'} \boldsymbol{X}_j + \epsilon_{ij}$$

where $\xi_j$ is a consumption utility common to all consumers and $\epsilon_{ij}$ is an idiosyncratic preference unique to each consumer distributed type-1 extreme value (specified with mean zero and unit scale). In the nested logit, each $\epsilon_{ij}$ draw is independent across individuals but, for a given individual, is correlated across nests as in equation (4). We normalize the common part of utility of the outside option to zero, so $U_{i0} = \epsilon_{i0}$. Consumers have heterogeneous price sensitivities and preferences for specific vehicle attributes in $\boldsymbol{X}_j$ across types according to $\alpha_{h(i)}$ and $\boldsymbol{\beta}_{h(i)}^{'}$. Specifically,

$$\begin{aligned} \alpha_{h(i)} &= \exp(\alpha_0 + \alpha_y y_{h(i)}) \\ \boldsymbol{\beta}_{h(i)}^{'} \boldsymbol{X}_j &= (\beta_0 \times 1_j\{\text{Inside good}\} + \beta_{EV} \times 1_j\{\text{EV}\} + \beta_{Lease} \times 1_j\{Lease\}) \times y_{h(i)} \end{aligned}$$

Integrating over consumers of a given type $h$, the market share of choice $j$ is

$$s_j^h = \int 1\{U_{ihj} \geq \max_{k \in \mathcal{J}} U_{ihk}\} dF(\boldsymbol{\epsilon_i})$$

To express the market shares of each choice under the nested logit assumption for $F(\cdot)$, it is useful to first define the following inclusive values, which represent the expected utility of a choice within a nest conditional on selecting that nest:

$$
\begin{aligned}
I_k^h &= (1-\sigma^k)\ln\sum_{j\in\mathcal{J}_k}\exp\left(\frac{\xi_j-\alpha_h\left(p_j-\tau_{hj}\right)+\boldsymbol{\beta}_h'\boldsymbol{X}_j}{1-\sigma^k}\right)\\
I_c^h &= (1-\sigma^c)\ln\sum_{k\in\mathcal{K}_c}\exp\left(\frac{I_k^h}{1-\sigma^c}\right)\\
I_g^h &= (1-\sigma^g)\ln\sum_{c\in\mathcal{C}_g}\exp\left(\frac{I_c^h}{1-\sigma^g}\right)\\
I^h &= \ln\left(1+\sum_{g\in EV,GV}\exp I_g^h\right)
\end{aligned}
\tag{14}
$$

In the above notation, $\mathcal{J}$ is the set of (1,356) submodel options available to purchase or lease, plus the outside option; $\mathcal{J}_k$ is the partition of $\mathcal{J}$ corresponding to submodel $k$ in the set of (758) submodels $\mathcal{K}$, plus the outside option; $\mathcal{K}_c$ is the partition of $\mathcal{K}$ corresponding to class $c$ in the set of (9) classes $\mathcal{C}$, plus the outside option; and $\mathcal{C}_g$ is the partition of $\mathcal{C}$ corresponding to powertrain $g \in \{\text{EV,GV}\}$, plus the outside option. Our normalization of the mean outside option value to zero and placement of the outside option in a nest all by itself imply that in each of the first three lines, $I_{h0} = 0$. This is the source of the 1 in the final inclusive value.

Then, unconditional choice probabilities for $j$ are given by the product of conditional probabilities within a nest and the overall choice probability of powertrain, $g$, which can in turn be expressed in terms of the inclusive values above:

$$
\begin{aligned}
s_j^h &= s_{j|k(j)}^h \times s_{k(j)|c(j)}^h \times s_{c(j)|g(j)}^h \times s_{g(j)}^h \\
&= \frac{\exp\left(\frac{\xi_j-\alpha_h\left(p_j-\tau_{hj}\right)+\boldsymbol{\beta}_h'\boldsymbol{X}_j}{1-\sigma^k}\right)}{\exp\left(\frac{I_{k(j)}^h}{1-\sigma^k}\right)} \times \frac{\exp\left(\frac{I_{k(j)}^h}{1-\sigma^c}\right)}{\exp\left(\frac{I_{c(j)}^h}{1-\sigma^c}\right)} \times \frac{\exp\left(\frac{I_{c(j)}^h}{1-\sigma^g}\right)}{\exp\left(\frac{I_{g(j)}^h}{1-\sigma^g}\right)} \times \frac{\exp I_{g(j)}^h}{\exp I^h}
\end{aligned}
\tag{15}
$$

McFadden (1978) provides a full derivation given the joint CDF of $\epsilon_{ij}$. Total registrations come from aggregating these within-type market shares, given each type is a share $w_h$ of the total population of $M$ consumers

$$q_j = M\sum_h w_h s_j^h$$

The substitution between any two choices, $j$ and $r$, given by $\frac{\partial q_j}{\partial p_r}$ in equation (6), can be derived using equations (14) and (15).

In the nested logit extension to the Small and Rosen (1981) log-sum consumer surplus formula, total consumer surplus is given by

$$CS = M\sum_h w_h\frac{I^h}{\alpha_h} \tag{16}$$

### D.2 Estimation and Equilibrium

The demand side parameters are estimated using a nested fixed-point approach: the outer loop uses a gradient-based optimization over the eight structural parameters $\boldsymbol{\alpha} = \{\alpha_0, \alpha_y\}$, $\boldsymbol{\beta} = \{\beta_0, \beta_{EV}, \beta_{Lease}\}$, and $\boldsymbol{\sigma} = \left\{\sigma^k, \sigma^c, \sigma^g\right\}$ to match our eight data moments while the inner loop solves for $\boldsymbol{\xi}$ using the Berry (1994) contraction mapping to match observed market shares. The contraction mapping is adapted to the nested logit using Grigolon and Verboven (2014), who show that the contraction at each iteration $(t)$ must be dampened by $1 - \max\left\{\sigma^k, \sigma^c, \sigma^g\right\}$

$$\xi_j^{(t+1)} \leftarrow\!\!\shortmid \xi_j^{(t)} + \left(1 - \max\left\{\sigma^k, \sigma^c, \sigma^g\right\}\right)\left(\log s_j^{obs} - \log s_j^{(t)}\right) \tag{17}$$

where $s_j^{obs}$ are observed market shares in the data and $s_j^{(t)}$ are implied market shares in the model given prices and the current values of outer loop parameters and $\xi_j^{(t)}$. In the outer loop, we minimize the $L^2$-norm between our model moments, $m_n(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\sigma})$, and our estimated data moments, $\hat{m}_n$, $n = 1, ..., 8$

$$(\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\sigma}}) = \arg\min Q(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\sigma}) = \sqrt{\sum_n \left(m_n(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\sigma}) - \hat{m}_n\right)^2} \tag{18}$$

The eight moments we match are (1) a market share-weighted model-level own-price demand elasticity, (2) the change in lease shares from a simulated decrease in lease prices for the subset of vehicles excluded from credits in August 2022, (3) the share of EV owners who choose another EV as a second choice if their first choice was unavailable, (4) the share of EV owners who would choose another EV in the same vehicle segment as their first choice if their first choice was unavailable, (5) the difference in average transaction price for high- and low-income households, (6) the difference in EV choice shares for high- and low-income households, (7) the difference in lease choice shares for high- and low-income households, and (8) the ratio of inside choice shares for high- to low-income households. Since we have as many moments as parameters to estimate, we match the targeted moments exactly.

The values of each data moment and their sources are presented in Table 4. For the own-price elasticity, since our model is at the level of a submodel-by-purchase option while the data elasticity moment is aggregated to the model-level, we simulate the model-level elasticity by raising and lowering prices by 0.5 percent and taking the average percent change in registrations as a central difference approximation. We do this for every model and take a registration-weighted average. For simulating the lease share change, we raise the price of the lease options for the August 2022 excluded group by \$5,538 (Table A2)—running the event study estimate in reverse to the higher pre-IRA baseline—and calculate the new choice probabilities for each option. We compute the change in the share of registrations for each submodel that are leases and then compare the registration-weighted average change in lease shares across submodels.

To calculate the second choice moments in the model, we simulate removing each submodel $k$ from the choice set and compute the new choice probabilities for all remaining products. Define

$s_{r\backslash k}$ as the market share of submodel $r$ when $k$ is unavailable. The second choice share is defined by the ratio

$$\frac{s_{r\backslash k} - s_r}{s_k} \tag{19}$$

since any new registrations of $r$ when $k$ is no longer available must be from consumers who originally had $k$ but then had $r$ as their next best option. The NVES only surveys consumers who registered a new vehicle and we can only compute second choice shares among respondents who provided one, so in practice we actually compute $s_{r\backslash k,0}$ for $r, k = 1, ..., K$, where $\backslash k, 0$ removes $k$ and the outside option from the choice set.[41] The share of EV consumers whose second choice is also an EV is then given by

$$D_{EV\to EV} = \frac{\sum_{k\in EV}\sum_{r\in EV, r\neq k}\left(s_{r\backslash k,0} - s_{r\backslash 0}\right)}{\sum_{k\in EV} s_{k\backslash 0}} \tag{20}$$

Similarly, the own-class share among EV owners is given by a weighted average across classes, $c$, within EVs:

$$D_{EV-class\to EV-class} = \frac{\sum_{c\in EV} s_{c\backslash 0}\left(\frac{\sum_{k\in c}\sum_{r\in c, r\neq k}\left(s_{r\backslash k,0} - s_{r\backslash 0}\right)}{\sum_{k\in c} s_{k\backslash 0}}\right)}{\sum_{c\in EV} s_{c\backslash 0}} \tag{21}$$

The differences in transaction prices, lease propensity, EV propensity, and the inside choice ratio for high- versus low-income households are computed by calculating the type-specific values and aggregating over types according to their share of the total vehicle market. To compute the inside choice share in the survey data, we combine the NVES data with information about the population distribution from the IRS SOI. Using the provided survey weights, which are meant to reflect the composition of the overall new vehicle market, we compute aggregate vehicle sales among high- and low-income households in the NVES and divide that number by the count of households in the IRS SOI data. This gives us an estimate of the average number of vehicles acquired per household. The ratio of that statistic between households above and below $300,000 of reported income measures differences in inside choice propensities.

While we only target the overall difference between high- and low-income households, Figure A17 reports that we closely match overall patterns across the full income distribution.

Estimation of the supply side consists of inverting the Nash-Bertrand first-order condition in equation (8). In particular, stacking the system of $J$ equations gives the markup equation

$$\boldsymbol{\mu} = -\left(\tilde{\boldsymbol{\Omega}}_{In}\right)^{-1}\boldsymbol{q} \tag{22}$$

where $\boldsymbol{\mu}$ is a $J$-vector of markups, $\mu_j = p_j - c_j + \kappa_j$ and $\tilde{\boldsymbol{\Omega}}_{In}$ is the $J \times J$ Jacobian matrix of inside demand derivatives but with zeros in elements whenever a firm does not own products $j$

[41] The proportional substitution in the nested logit to the outside option implies $s_{k\backslash 0} = s_k/(1 - s_0)$. Over half of respondents (58.5 percent) did not consider a second choice.

and $r$. That is, if $f(j)$ returns the identity of the firm that produces $j$, then each $j, r$ element is given by $\left[\tilde{\boldsymbol{\Omega}}_{jr}\right] = \frac{\partial q_r}{\partial p_j} \times 1_{f(r)=f(j)}$ for $j, r = 1, ..., J$. Given the estimated demand parameters, we can compute $\tilde{\boldsymbol{\Omega}}_{In}$ and $\boldsymbol{q}$ to back out markups and—given prices and subsidies—marginal costs.

To compute alternative price equilibria in our counterfactuals that change the policy vectors $\boldsymbol{\tau}$ and $\boldsymbol{\kappa}$, we take our estimated marginal costs and demand parameters as given and then follow the standard approach found in Morrow and Skerlos (2011) to solve a fixed-point problem closely related to equation (22). The solutions to the two problems coincide at an equilibrium point. As discussed in Section 5, there is no guarantee that the pricing equilibrium will be unique, but we do not encounter multiple equilibria in practice. Appendix Figure A19 presents the distribution of implied marginal costs in the first panel, and the distribution of equilibrium markups under a counterfactual policy environment with no EV subsidies.

### D.3 Computation of Standard Errors and Confidence Intervals

We compute the asymptotic variance of our model parameters using the delta method. Call $\theta \in \mathbb{R}^P$ the $P$-vector of model parameters to estimate. The model-implied moments, $m(\theta; z)$, are a function of these parameters and other data inputs, $z \in \mathbb{R}^Z$. In our main specification, $P = 8$ and $Z = 1$.[42] Given estimates of data moments, $m$, to target, denote the full set of data inputs $d = (z, m)$. Our minimum-distance estimator is

$$Q(\theta) = \|g(\theta; d)\|^2, \qquad g(\theta; d) \equiv m(\theta; z) - m$$

First, we will derive the covariance of the distance metric at a known $\theta$. To make this explicit, define this function by $h(d) \equiv g(\theta; d)$. The Jacobian of $h(d)$ with respect to the data moments, $d$, is

$$J_d = \left[ \begin{array}{cc} J_z & -I_P \end{array} \right]$$

where $J_z = \partial h(d)/\partial z$ and $I_P$ is the identity matrix of dimension $P$. Let the joint covariance of the estimators for the data moments be

$$\Sigma_d = \left[ \begin{array}{cc} \Sigma_{zz} & \Sigma_{zm} \\ \Sigma_{mz} & \Sigma_{mm} \end{array} \right]$$

Then, by the delta method, the asymptotic distribution of $h(d)$ is

$$\sqrt{n}\left(g(\theta; \hat{d}) - g(\theta; d_0)\right) = \sqrt{n}\left(h(\hat{d}) - h(d_0)\right) \xrightarrow{d} \mathcal{N}(0, \underbrace{J_d \Sigma_d J_d'}_{S_g})$$

[42] In Section D.2 we omitted $z$ from our notation for brevity, but in implementation $z$ is the change in relative lease prices that we simulate to compute the semi-elasticity of lease shares among the August-excluded group of vehicles. The semi-elasticity is the model moment we target; the relative lease price change is a data input. To compute these standard errors, we take into account the covariance between these two moments in the data from the joint regression reported in Appendix Table A2.

Expanding the block product gives the covariance of the distance metric, $S_g$,

$$\begin{aligned} S_g &= \begin{bmatrix} J_z & -I_P \end{bmatrix} \begin{bmatrix} \Sigma_{zz} & \Sigma_{zm} \\ \Sigma_{mz} & \Sigma_{mm} \end{bmatrix} \begin{bmatrix} J_z' \\ -I_P \end{bmatrix} \\ &= J_z \Sigma_{zz} J_z^{'} - J_z \Sigma_{zm} - \Sigma_{mz} J_z^{'} + \Sigma_{mm} \end{aligned}$$

Second, we will propagate the uncertainty in the distance metric to uncertainty in the parameter estimates through the implicit function theorem. Define the implicit function $\theta(d)$ that solves for the parameter values that equate the model-implied moments to the provided data moments

$$g(\theta(d), d) = 0$$

In the just-identified case where $P = \dim(\theta)$ and the Jacobian of $g(\theta, d)$ with respect to $\theta$, denoted $J_\theta = \partial g(\theta, d)/\partial \theta'$, is nonsingular when evaluated at the true values of $\theta$ and $d$, we can totally differentiate the above equation to get

$$\frac{\partial g}{\partial \theta'} d\theta + \frac{\partial g}{\partial (z,m)'} d(z,m) = 0 \implies \frac{d\theta}{d(z,m)'} = -J_\theta^{-1} J_d$$

Therefore, by the delta method applied to the estimator $\theta(d)$,

$$\sqrt{n}(\hat{\theta} - \theta_0) = \sqrt{n}(\theta(\hat{d}) - \theta(d_0)) \xrightarrow{d} \mathcal{N}(0, \underbrace{J_\theta^{-1} J_d \Sigma_d J_d^{'} J_\theta^{-1'}}_{S_\theta}) \tag{23}$$

This gives us an expression for the asymptotic covariance of the parameter estimates

$$S_\theta = J_\theta^{-1} J_d \Sigma_d J_d^{'} J_\theta^{-1'} = J_\theta^{-1} S_g J_\theta^{-1'}$$

We compute empirical standard errors by numerically approximating $\hat{J}_\theta$ and $\hat{J}_z$ at the estimated parameter values along with estimating the covariance of the moments $\hat{\Sigma}_d$ in our auxiliary data. We can estimate this covariance only within, and not across, each individual block of data moments: the own-price elasticity, the lease regression estimates, and the second choice data. We therefore construct $\hat{\Sigma}_d$ in two parts.

First, within each block: (1) since Grieco, Murry, and Yurukoglu (2024) do not report uncertainty around their own-price elasticity, we assume the elasticity target is known with certainty (below, we discuss relaxing this assumption); (2) we compute the covariance between our regression estimates of the relative lease price and lease share change in the joint regression reported in Appendix Table A2; and (3) we bootstrap the covariance matrix of the six NVES second choice and income moments by resampling with replacement the original survey responses. We assume the population distribution in the IRS SOI is known with certainty. The resulting standard errors when the covariance between blocks is zero are presented in our baseline results in Table 5.

Second, for the covariance between blocks, rather than assuming zero joint covariance between

different datasets, we follow Cocci and Plagborg-Møller (2025), who derive sharp upper bounds for standard errors that are valid under any correlation structure consistent with the within-block covariances. Let $D_\theta \equiv -J_\theta^{-1} J_d$ denote the sensitivity of the parameter estimates to the data inputs. For any scalar statistic $r(\theta)$—for example, a parameter or a welfare outcome—we can define its loading on the data inputs as $a_r = D_\theta' \nabla_\theta r(\theta)$ where $\nabla_\theta r(\theta)$ is the scalar's gradient with respect to the $P$-vector of parameters (when $r(\theta)$ is one of the parameters, this is just a vector with one in position $p$ and zero elsewhere). A final application of the delta method then gives

$$\sqrt{n}\left(r(\hat{\theta}) - r(\theta_0)\right) \approx a_r' \sqrt{n}(\hat{d} - d_0)$$

to first order. We now partition $d$ and $a_r$ into three blocks corresponding to the moment blocks $b \in \{\text{elas}, \text{lease}, \text{nves}\}$, and let $\Sigma_b$ denote the covariance submatrix of $\Sigma_d$ within block $b$. Cocci and Plagborg-Møller (2025) show that when the covariances across blocks are unknown, the sharp worst-case standard error is

$$\overline{\text{se}}(r(\hat{\theta})) = \sum_b \sqrt{a_{r,b}' \Sigma_b a_{r,b}}, \tag{24}$$

which follows from the Cauchy-Schwarz inequality. When $r(\theta)$ is a welfare object, since outcomes are complex nonlinear functions of the parameters, we numerically approximate $\nabla_\theta$ via central finite differences. Since the elasticity block is a single scalar, denote $\Sigma_{\text{elas}}$ by $\sigma_{\text{elas}}^2$, where $\sigma_{\text{elas}}$ is the standard error of the elasticity moment. When $\sigma_{\text{elas}} = 0$, we have $\overline{\text{se}}(r(\hat{\theta})) = \sqrt{a_{r,\text{lease}}' \Sigma_{\text{lease}} a_{r,\text{lease}}} + \sqrt{a_{r,\text{nves}}' \Sigma_{\text{nves}} a_{r,\text{nves}}}$.

Alternatively, we assess sensitivity to treating the own-price elasticity target as not known with certainty. Grieco, Murry, and Yurukoglu (2024) do not report a standard error on their own-price elasticity, but they do report standard errors for their structural parameter estimates. In particular, they report an estimate of -3.112 with a standard error of (1.124) on their mean price sensitivity coefficient. We therefore calibrate a standard error $\sigma_{\text{elas}}$ for the elasticity block so that the worst-case $t$-statistic on our baseline price coefficient, $\alpha_0$, matches their corresponding $t$-statistic: $\frac{|\hat{\alpha}_0|}{\overline{\text{se}}(\hat{\alpha}_0; \sigma_{\text{elas}})} = \frac{|-3.112|}{1.124}$. This calibration gives $\sigma_{\text{elas}} = 1.187$. Since the elasticity block is a scalar, its contribution to the worst-case standard error of any statistic is simply a linear addition of $|a_{r,\text{elas}}|\sigma_{\text{elas}}$ in equation (24), which makes it simple to compute.

We report both sets of standard error bounds on our parameter estimates in Appendix Table A5. We use these standard errors to construct confidence intervals with at least 95 percent coverage on our counterfactual outcomes in Appendix Tables A6 and A7, reporting the more conservative $\sigma_{\text{elas}} = 1.187$ version alongside our baseline confidence intervals.[43] Our parameter estimates remain significantly different from zero under this calibration. The statistical significance of our baseline results in the counterfactual tables is mostly robust to the conservative bound, though a few worst-case confidence intervals, such as for US surplus under the domestic SCC under some

[43] The confidence intervals under $\sigma_{\text{elas}} = 0$ are very similar to the conventional confidence intervals, reflecting the fact that the standard error bounds are nearly the baseline values. Therefore, we omit reporting these intermediate confidence intervals.

counterfactuals, cross zero.

### D.4 Additional Evidence on Model Fit

This subsection discusses eight additional types of evidence on model fit. Appendix Table A10 compares moments that the model does not formally target against reference values.[44] Along many of these dimensions, and several more that we discuss below, model fit is good though not perfect. Together, we believe this evidence suggests that the model realistically captures important aspects of the setting we study.

First, Table 5 shows that our demand system is consistent with utility maximization. In estimation, we do not impose the requirement that $1 > \sigma^k \geq \sigma^c \geq \sigma^g \geq 0$, but our main estimates satisfy this condition with tight standard errors and non-overlapping confidence intervals. This does not always occur. For example, when we estimate versions of the model that move the EV/GV nests from the top level to the second level, estimation pushes $\sigma^g < 0$, which rejects that alternative nest ordering.[45] This occurs both when the top nest is vehicle class and when it is domestic/foreign brands. We take this as evidence that powertrain-related preferences are strong enough to support our nesting assumptions.

Second, Appendix Figure A17 shows that we capture heterogeneity throughout the income distribution, even though in estimation we only focus on the simple difference between households above and below the 30D income-eligibility threshold.

Third, we only target second-choice moments from the EV side of the market, but the GV second choice shares from the model and the data also align. Our estimated model implies that 95.8 percent of GV owners choose another GV as a second choice, compared to 96 percent in the NVES. Figure 6 shows the aggregated substitution matrix between our nine-by-nine different powertrain-segment combinations in the model and the data. The correlation between the two is 0.9 when weighting cells by first-choice market share. When restricting to buyers whose first and second choices are EVs, the weighted correlation is 0.995. In particular, our model-implied second choices for those with an EV as their first choice (the lower-left portion of the right panel) match closely the patterns in the NVES data: the highest density is for substitution to within-class EVs, and the off-diagonal pattern is qualitatively quite similar.

As an example of the model's predictions, we compare second choices between the model and NVES for two prominent but different EVs. For this exercise, we restrict the NVES reports to vehicles also in our model choice set. Consider the Chevrolet Bolt EV, a budget subcompact hatchback. The model predicts that the most popular second choices for the Bolt are the Tesla Model 3, the Tesla Model Y, and the Chevrolet Bolt EUV. In the NVES data, the Tesla Model 3 is also the most common second choice, with the Tesla Model Y in fifth, and the Bolt EUV in eighth. Both the model and the NVES show that sedans are a closer substitute for the Bolt than

[44] We evaluated these other moments after fully completing model development.

[45] As these estimates are inconsistent with random utility maximization, we do not report anything further on them.

SUVs are. Second, consider the Ford Mustang Mach-E, a premium crossover SUV. The leading model-implied second choices for the Mustang Mach-E are the Tesla Model Y, the Tesla Model 3, the Jeep Wrangler PHEV, and the Volkswagen ID.4. Similarly, the NVES reports the Tesla Model Y as the closest second choice, the Volkswagen ID.4 in third, and the Tesla Model 3 in fourth.[46]

Fourth, to benchmark the correlation with environmental outcomes, we calculate the average miles per gallon equivalent (MPGe) for GVs and EVs of second-choice vehicles. In the NVES, among GV buyers who consider a second choice, the second choice has an MPG of 27.6, compared to our model-estimated mean MPG of 29.6. The corresponding second choice for EV buyers in the NVES has mean MPGe of 66.1, compared to our model-estimated mean MPGe of 71.9. While these statistics suggest good fit, this moment is closely related to choice of powertrain and mean MPGe within each powertrain.

Fifth, we compare the second choices in the NVES to the model-implied second choices along US firm ownership and domestic assembly shares. In the NVES, among buyers of vehicles made by US firms, the second choice share of US firms is 62.2 percent, while the model-implied share is 40.1 percent. Domestic assembly share of second choices, conditional on the domestic assembly of the first choice, is 58 percent in the NVES and 42.4 percent in the model, though this is closely related to firm ownership. We acknowledge that the fit along these dimensions is imperfect, so Appendix Table A9 reports results replacing the vehicle class nest in the nested logit structure with a US or foreign firm nest. That demand system is able to replicate these reference values. Column 3 of Appendix Table A9 shows how this alternative changes counterfactual conclusions. The signs of the headline effects and the rankings across policy counterfactuals are stable and magnitudes are largely consistent with the baseline model. Section 6 discusses how we interpret this model uncertainty.

Sixth, our estimates also imply an aggregate new vehicle demand elasticity of -0.95 and an aggregate GV demand elasticity of -1.1. Berry, Levinsohn, and Pakes (2004), Grieco, Murry, and Yurukoglu (2024) and the empirical evidence in Allcott et al. (2026) suggest comparable values ranging from -0.9 to -1.3. The similarity is not a high bar given that we match their mean price elasticity by design. Our own-price elasticity of demand for EVs is -2.1, slightly more inelastic than the value of -2.8 that Xing, Leard, and Li (2021) estimate.

Seventh, we estimate positive marginal costs for all submodels. While our methodology allows the leased and purchased versions of a submodel to have different marginal costs, we estimate a correlation of 0.97 between the marginal costs of purchased and leased options of the same submodel.[47] We estimate that EVs have higher average marginal costs than GVs as well as greater dispersion across firms. We reiterate that our estimates reflect a combination of positive static production costs and negative dynamic benefits, and our data do not allow us to distinguish between

[46]NVES respondents rarely mention the Jeep Wrangler PHEV as a second choice for the Mach-E, though the Ford Bronco, a direct competitor to the Jeep Wrangler PHEV, is eighth as the top non-BEV vehicle.

[47]This statistic describes the 76 EV submodels that have both purchase and lease options. The 0.97 correlation is weighted by registrations; the unweighted correlation is 0.95. The mean difference in estimated marginal costs between the purchase and lease options is $857, with lease costs higher. When including GVs in these statistics, the weighted and unweighted correlations are 0.97 and 0.96.

the two.[48,49] The EV markups also disproportionately represent Tesla, which had high margins and a majority of the US EV market in this period. Consequently, to the extent that there is overlap in the distribution of costs between EVs and GVs, we cannot say whether this overlap reflects similar static marginal production costs with small returns to learning-by-doing, or higher EV production costs with large returns to learning-by-doing. In either case, given marginal costs, we estimate share-weighted mean markups of $11,200 for EVs (21 percent of price) and $8,200 for GVs (19 percent of price). This has similar magnitude to the Grieco, Murry, and Yurukoglu (2024) estimate that vehicle markups average 22 percent of price in 2018, their most recent year, though again this is unsurprising since we target their mean price elasticity. The dispersion of implied markups across submodels ranges from $5,700 to $18,800, with the largest markups for Tesla; Appendix Figure A19 shows the full distribution.

Eighth, we can benchmark our estimated percentage markups to industry reports of gross profit margins, which measure a firm's accounting profit after subtracting out the cost of goods sold from sales revenue. Model-implied markups may differ from accounting profit for many reasons, including fixed costs of investment, capital depreciation, research and development activity, shadow costs of plant capacity, dynamic benefits from learning-by-doing, and tax loss harvesting. Still, we estimate that Tesla's markup as a share of revenue is 24.9 percent, which is in the range of Tesla's reported gross margin of its automotive business in its 10-K filing for 2022–2023 (19.4 to 28.5 percent). Similarly, when ranking firms by their profit margin for EVs, the ordering from the model matches real-world reporting. At the extremes, the model indicates Tesla as the most profitable and Rivian the least profitable; in the middle, the model ranks GM's electric vehicle division as more profitable than Ford's (Boudette 2024).

### D.5 Sensitivity Analysis for MCPF, SCC, and $CO_2$ Emissions

Appendix Table A8 revisits the calculation of constrained optimal uniform subsidies under alternative assumptions. Under the global SCC, panel (e) shows a constrained optimal uniform EV subsidy of $13,478, and a mean constrained optimal differentiated subsidy of $18,685. Although Figure 7, panel (a), implies that the optimal policy applied to all vehicles and under the global SCC would be a tax, we find that the constrained optimal policy applied only to the 30D-eligible vehicles is a subsidy. This sign change between tax and subsidy occurs because the constrained optimal policy does not tax GVs, and encourages sufficient substitution from GVs to EVs that subsidizing EVs becomes optimal, highlighting the role of the substitution terms of our constrained optimal subsidy formulas. Appendix Table A8 also revisits results using an MCPF of 1.4, reflecting the deadweight loss of taxation. Here the constrained optimal uniform EV subsidy is $1,834 with a

[48] Our static model also omits costs of introducing a new technology that are neither fixed costs nor static marginal costs, such as variable costs of educating dealers and training service networks. If part of the measured EV markup compensates for these unmeasured costs, our estimates overstate EV producer surplus, and the producer surplus gains we attribute to the credits are an upper bound over the medium run.

[49] Manufacturers in mid-2022 faced severe supply constraints, and we therefore do not use data from this period for the model. The model seeks to describe the medium run, which is why we use a static Nash-Bertrand framework applied to data from mid-2023, by which time supply constraints had somewhat relaxed.

domestic SCC and $3,604 with a global SCC, or about a fifth and a quarter, respectively, of what Table 8 finds assuming an MCPF of 1.

We also compute the lifetime externalities from EVs using estimates of long-run marginal $CO_2$ emissions from the MidCase scenario of the National Renewable Energy Laboratory's Cambium model. We choose these estimates as an alternative to our baseline estimates since they are explicitly designed with policy evaluation and "long-term forward-looking decision-making" in mind (Gagnon et al. 2024). Additionally, they are available at a suitably disaggregated level to allow for close comparability with our baseline estimates. As with all forecasts, however, these estimates take as given a particular policy landscape; most relevant to our analysis is their assumption that the IRA credits for qualifying generation will remain in place. These long-run estimates imply carbon emissions from driving a BEV that are approximately 40 percent of our baseline values. Nevertheless, due in large part to the importance of externalities from both accidents and manufacturing, this translates to a shift of only a few hundred dollars in the constrained optimal uniform subsidy under a domestic SCC. The primary welfare gains come in the form of reduced environmental damages, as the long-run estimates imply a larger reduction in emissions from replacing a GV with an EV.

Appendix Table A11 applies the decomposition from equation (13) of the constrained optimal uniform subsidies in Table 8. It decomposes these subsidies into four components—price distortions (markups and externalities); indirect substitution; profit shifting; and tax distortion. We also show the net value of US jobs gained for the planner that places weight on that outcome.[50] We find that the profit shifting motive is large, regardless of the SCC and MCPF assumptions.[51] First consider a setting with a global SCC and MCPF of 1, which Appendix Table A8 panel (e) shows has a constrained optimal uniform subsidy of $13,478. This setting provides perhaps the largest scope for environmental benefits relative to non-environmental concerns. Appendix Table A11 shows that here the profit shifting component is $3,657, while distortions account for the remaining $9,821. In this scenario, reducing distortions (market power and externalities) has around 2.5 times the importance of profit shifting in guiding optimal policy. At the other extreme, consider a scenario with a domestic SCC and MCPF of 1.4, which in Appendix Table A8 panel (a) has a constrained optimal uniform subsidy of $1,834. This setting gives the least scope for environmental concerns and further adds in the fiscal distortion from taxation. Here Appendix Table A11 shows that the profit shifting component of constrained optimal subsidies is $3,640 while the externality and markup distortions account for $2,643. Here, reducing distortions has nearly similar importance to profit shifting. However, the tax distortion component is -$4,449, which significantly lowers the

[50] The tax distortion component exceeds zero only when MCPF>1, so this term does not appear in the main text equation (13) that assumes MCPF=1, but does appear in the appendix formulas (34) and (39) that allow MCPF>1. Likewise, the jobs value component is a subset of the price distortion and indirect substitution components, modeled as part of $\phi_j$, though we present it separately in Appendix Table A11.

[51] All three counterfactual constrained optimal subsidies decrease foreign producer surplus relative to the IRA baseline. In each counterfactual, producer surplus falls most in Japan, followed by South Korea, Germany, then Sweden. These four countries have among the highest baseline EV registrations in the US. For example, the differentiated subsidy using the global SCC decreases Japanese producer surplus by $1.3 billion annually.

optimal scale of the policy.

### D.6 Sensitivity Analysis to Demand and Supply Assumptions

Alternative model assumptions on demand and supply suggest several findings (Appendix Table A9). First, several qualitative conclusions hold across every alternative. The credits raise EV registrations, shift profits from foreign toward domestic firms, and increase domestic auto employment. Second, across these different model assumptions, the variation in results like lease and US assembly shares, $CO_2$ emissions, MVPF, and cost per additional EV has a range somewhat similar to the standard confidence intervals reported in Appendix Tables A6 and A7, though somewhat below the range of the worst-case confidence intervals. This suggests that model uncertainty and statistical uncertainty are both important. Third, eliminating income heterogeneity and replacing the vehicle class nest with a foreign/domestic nest change the main results less than the other alternatives do (columns 2 and 3). The homogeneous demand model does find that the IRA increased leasing less than the baseline model does, because this sensitivity analysis mutes the incentive for higher-income households to avoid income restrictions by leasing.

Fourth, assuming an own-price elasticity of -4 attenuates the estimated impact of the IRA on several key outcomes by about 20 percent to 30 percent relative to the baseline model. For example, the estimated increase in EV registrations is 216,000 in the baseline model but about 170,000 per year in this alternative. Similarly, the increase in auto industry employment is 11,500 in the baseline model but about 8,600 workers per year in this alternative.

Fifth, assuming non-Tesla EVs price at cost changes outcomes the most—effects of the EV credits on US producer surplus fall by half, the leasing impact grows by 6 percentage points, and the impact of the credits on EV registrations and auto assembly increases (column 6). This alternative decreases markups and increases take-up of the resulting cheaper EVs. All MVPFs are below one in this alternative because producer surplus increases less than government spending does. We regard the zero-markup case as extreme. The credits' domestic benefits operate substantially through producer surplus, so assumptions that shrink profits or raise the cost of financing subsidies compress those benefits toward zero. This echoes our broader narrative that profit shifting plays a central role in evaluation of these policies; assuming away profits decreases the resulting domestic benefits.

### D.7 Impacts of IRA on Investment and Other Outcomes

To learn about how additional EV manufacturing affects investment, we consulted the Clean Investment Monitor, a project from MIT and Rhodium that tracks investments in clean technology (Rhodium Group and MIT CEEPR 2025). From active investments there, two listings stand out as large recent EV assembly plant projects with sufficient information for our purposes. The Rivian facility in Normal, Illinois, was recently expanded at the cost of $1.5 billion; the expansion is predicted to increase production from 50,000 to 215,000 for an increase of 165,000 vehicles/year, or a cost of $9,100 per additional vehicle/year (Channick 2025; Packowitz 2025; Rivian Automotive

2025). The Hyundai Metaplant near Savannah, Georgia, consisted of $7.6 billion of investment and is projected to produce up to 500,000 vehicles/year, or a cost of $15,000 per additional vehicle/year (Hyundai Motor Group 2025). Of course, a greenfield plant may require greater investment than an expansion. Taking these two estimates as representing a potential range for investment required to produce additional EVs, the additional 250,000 US-assembled EVs we estimate are demanded in response to the IRA credits would correspond to between $2.3 and $3.8 billion of additional investment in assembly plants.

These back-of-the-envelope figures benefit from additional context. Assembly plants range in capacity. The Hyundai plant mentioned above has capacity twice what we estimate to be the effect of the IRA relative to the elimination of EV credits; the Tesla Fremont plant has similar capacity; and the expanded F-150 Lightning plant can handle production of up to 150,000 EVs per year (Ford Motor Company 2023; Tesla 2023; Hyundai Motor Group 2025). Of course, the increase in demand due to the IRA EV credits is divided across many different submodels, whereas each assembly plant produces one or a few submodels, so a single plant could not supply all the additional EVs demanded due to the IRA credits. Regardless, the total number of additional EVs demanded due to the EV credits is in the ballpark of the capacity of one or possibly two additional moderately large assembly plants, assuming no baseline surplus assembly capacity.

We can use our structural model to get a sense of how much surplus firms leave on the table by not readjusting their supply chains after being determined to be 30D ineligible. One at a time, we recompute market equilibrium under simulated changes in 30D eligibility for each of the ten EV models with the most registrations in our data but that were ineligible for the full $7,500 credit. These counterfactuals reflect a scenario where manufacturers shifted their assembly or battery and mineral sourcing to meet the IRA requirements. We assume that marginal costs are unchanged, though in reality they could be larger or smaller. Our primary purpose is to compare the effects of eligibility to the upfront investment costs calculated above.

Appendix Table A12 presents the results. For the major brands (excluding Rivian), the magnitude of the producer surplus increase is 1.1 to 5.6% of brand profits, ranging from $25 to $288 million per year. The increase is less for parent companies when accounting for the profit shifting that they internalize across their brands. As we discussed above, the Hyundai Ioniq 5—which is on the high end of producer surplus increases across models at $288 million per year and 5.6% of brand profits—did gain eligibility following the opening of its $7.6 billion Georgia plant after two years of ineligibility. While the magnitude of the producer surplus increase relative to the billion dollar upfront plant investment is small, shifting production could be dynamically profitable if the firm had confidence that the surplus increase would last for some time.

The leasing loophole plays a large role in these results. While granting eligibility to these models raises the number of purchases, one-third of this increase is driven by substitution from leases of the same model. Put differently, although these models are ineligible for 30D credits, firms nonetheless recover around one-third of these forgone profits through their leases. These results may partially explain why more investment did not occur in response to the IRA credits.

# E Optimal Policy Appendix

## E.1 Optimal Differentiated Subsidy for the Global Planner

$CS$ is consumer surplus, $PS = \sum_j q_j \mu_j$ is firm surplus, $G = \eta \sum_j q_j(\tau_j + \kappa_j)$ is government spending under an assumed MCPF, and $E = \sum_j q_j \phi_j$ is negative externalities, meaning $\phi_j > 0$ would indicate damages. Recall that the outside option $j = 0$ is competitively supplied at zero markup but has associated damages. As we discuss in the main text, since statutory incidence is independent of economic incidence, $\tau_j$ and $\kappa_j$ are perfect substitutes when there are no income-eligibility distinctions and there are infinite combinations of the two that yield the same welfare. We provide results fixing $\kappa_j = 0$ and assuming all households are income-eligible.

To derive the total surplus-maximizing subsidies for the global planner, we take the derivative of $W$ in equation (9) with respect to an arbitrary good $j = 1$, where the total derivative takes into account firms' endogenous price response:

$$
\begin{aligned}
[\tau_1]: \quad \frac{dW}{d\tau_1} &= \frac{dCS}{d\tau_1} + \frac{dPS}{d\tau_1} - \frac{dG}{d\tau_1} - \frac{dE}{d\tau_1} = 0 \qquad (25)\\
&= \underbrace{q_1 - \sum_j q_j \frac{dp_j}{d\tau_1}}_{\frac{dCS}{d\tau_1}} + \underbrace{\sum_j \frac{dq_j}{d\tau_1}\mu_j + \sum_j q_j \frac{d\mu_j}{d\tau_1}}_{\frac{dPS}{d\tau_1}} - \underbrace{\eta\left(\sum_j \frac{dq_j}{d\tau_1}\tau_j + q_1\right)}_{\frac{dG}{d\tau_1}} - \underbrace{\sum_j \frac{dq_j}{d\tau_1}\phi_j}_{\frac{dE}{d\tau_1}} = 0,
\end{aligned}
$$

where the first term of the second line makes use of an Envelope condition that $\frac{\partial CS}{\partial p_j} = -q_j$, yielding:

$$
\frac{dCS}{d\tau_1} = \sum_j (-q_j)\left(\frac{dp_j}{d\tau_1} - 1\{j = 1\}\right) = q_1 - \sum_j q_j \frac{dp_j}{d\tau_1}.
$$

Firm markups are defined by $p_j = \mu_j + c_j$, so $dp_j = d\mu_j$ given constant marginal costs. Using this and canceling terms gives

$$
\eta \sum_j \frac{dq_j}{d\tau_1}\tau_j = \sum_j \frac{dq_j}{d\tau_1}\mu_j - \sum_j \frac{dq_j}{d\tau_1}\phi_j - (\eta - 1)q_1, \qquad (26)
$$

where the last term comes from the revenue-raising cost of subsidies due to the MCPF. In particular, it represents the marginal cost of raising funds at the current subsidy level that arises from inframarginal take-up of $q_1$. Demand changes with subsidies through the net effect on price faced by consumers in equilibrium:

$$
\frac{dq_r}{d\tau_j} = \sum_k \frac{\partial q_r}{\partial p_k}\left(\frac{dp_k}{d\tau_j} - 1\{j = k\}\right).
$$

Substituting in for this expression and repeating the total derivative for all goods gives a system of $J$ equations whose solution is

$$\eta \boldsymbol{\tau}^P = \underbrace{\boldsymbol{\mu}}_{\text{markup}} - \underbrace{\tilde{\boldsymbol{\phi}}}_{\text{negative externality}} + \underbrace{(\eta - 1)\left[(\boldsymbol{I} - \boldsymbol{T})\,\boldsymbol{\Omega}\right]_{In}^{-1}\boldsymbol{q}}_{\text{tax distortion}}, \tag{27}$$

where $\boldsymbol{\Omega}$ is the $J \times (J+1)$ matrix of demand partial derivatives with representative element $[\Omega]_{jr} = \frac{\partial q_r}{\partial p_j}$ for $j = 1, ..., J$ and $r = 0, ..., J$; $\boldsymbol{T}$ is the $J \times J$ matrix of equilibrium price responses to subsidies with representative element $[T]_{jr} = \frac{dp_r}{d\tau_j}$; $I$ is the identity matrix; the subscript $In$ restricts the bracketed matrix product to the $J \times J$ inside goods before being inverted; and $\tilde{\phi}_j \equiv \phi_j - \phi_0$ is the net externality of each vehicle relative to the outside option. Note that $\frac{dq_r}{d\tau_j} = -[(\boldsymbol{I} - \boldsymbol{T})\boldsymbol{\Omega}]_{jr}$.

To derive the equilibrium price response matrix, $\boldsymbol{T}$, recall that in Appendix D we show that the solution to the Nash-Bertrand pricing game is determined by the linear system

$$\boldsymbol{q} + \tilde{\boldsymbol{\Omega}}_{In}\boldsymbol{\mu} = \boldsymbol{0} \tag{28}$$

where $\tilde{\boldsymbol{\Omega}}_{In}$ is the $J \times J$ Jacobian matrix of inside demand derivatives but with zeros in elements whenever a firm does not own products $j$ and $r$. That is, $\tilde{\boldsymbol{\Omega}}_{In} = \boldsymbol{\Omega}_{In} \odot \mathcal{H}$, where $\mathcal{H}$ is the firm product-ownership matrix ($\odot$ is the Hadamard product). Define the implicit vector function $\boldsymbol{F}(\boldsymbol{p}(\boldsymbol{\tau}), \boldsymbol{\tau}) = 0$ as the solution to the pricing game in equation (28). By the implicit function theorem, we solve for $\boldsymbol{T}$ via

$$\boldsymbol{\Omega}_{In} + \boldsymbol{H}_{\Omega}\boldsymbol{\mu} = \left(\boldsymbol{\Omega}_{In} + \boldsymbol{H}_{\Omega}\boldsymbol{\mu} + \tilde{\boldsymbol{\Omega}}_{In}\right)\boldsymbol{T}' \tag{29}$$

where $\boldsymbol{T}'$ is the transpose of the price response matrix, $\boldsymbol{H}_{\Omega}$ is the (three-dimensional $J \times J \times J$) Hessian of demand, and the elements of $\boldsymbol{H}_{\Omega}\boldsymbol{\mu}$ are $[H_{\Omega}\mu]_{jr} = \sum_{k \in \mathcal{J}_{f(j)}} \frac{\partial \tilde{\Omega}_{jk}}{\partial p_r}\mu_k = \sum_{k \in \mathcal{J}_{f(j)}} \frac{\partial^2 q_k}{\partial p_j \partial p_r}\mu_k$. Therefore, $\boldsymbol{T}'$ is given by

$$\boldsymbol{T}' = \left(\boldsymbol{\Omega}_{In} + \boldsymbol{H}_{\Omega}\boldsymbol{\mu} + \tilde{\boldsymbol{\Omega}}_{In}\right)^{-1}\left(\boldsymbol{\Omega}_{In} + \boldsymbol{H}_{\Omega}\boldsymbol{\mu}\right).$$

With $\eta = 1$, equation (27) becomes the standard optimal taxation result $\tau_j^P = \mu_j - \tilde{\phi}_j$ to correct the net distortion of each good. With $\eta \neq 1$, the planner equates the total fiscal cost of the per vehicle subsidy, $\eta\tau_j^P$, with the total distortion in the economy arising from unpriced externalities and transfers to inframarginal consumers inclusive of the equilibrium price response by firms.

### E.2 Constrained Optimal Differentiated Subsidy

This subsection provides the derivation of a constrained optimal differentiated subsidy that only applies to a subset of goods $\mathcal{S}$. "Differentiated" means that the amount of the subsidy is allowed to differ across goods in $\mathcal{S}$. We allow the social planner to put no weight on some firms' surplus, for example a US planner that prioritizes domestic firms. We also allow the social planner to value the externalities associated with vehicle sales differently, for example a $\phi_j^{US}$ that uses a lower social cost of carbon or places additional weight on job creation. This nests the global planner solution when all subsidy effects are internalized.

Define $\mathcal{J}_{For}$ as the set of choices among foreign-owned firms, each of which receives producer surplus equal to quantity sold times markup, $\pi_j = q_j \mu_j$. Their surplus does not contribute to $W^{US}$. Then, taking the derivative of $W^{US}$ with respect to an arbitrary good, $j = 1$, in $\mathcal{S}$:

$$[\tau_1]: \quad \frac{dW^{US}}{d\tau_1} = \underbrace{q_1 - \sum_j q_j \frac{dp_j}{d\tau_1}}_{\frac{dCS}{d\tau_1}} + \underbrace{\sum_j \frac{dq_j}{d\tau_1}\mu_j + \sum_j q_j \frac{d\mu_j}{d\tau_1}}_{\frac{dPS}{d\tau_1}} \tag{30}$$
$$\underbrace{- \eta \left( \sum_{j \in \mathcal{S}} \frac{dq_j}{d\tau_1}\tau_j + q_1 \right)}_{\frac{dG}{d\tau_1}} - \underbrace{\sum_j \frac{dq_j}{d\tau_1}\phi_j^{US}}_{\frac{dE}{d\tau_1}} - \underbrace{\sum_{j \in \mathcal{J}_{For}} \frac{d\pi_j}{d\tau_1}}_{\frac{dPS^{For}}{d\tau_1}} = 0,$$

where the final term on the left-hand side subtracts out foreign producer surplus from domestic welfare. Canceling terms as in Section E.1 gives

$$\eta \sum_{j \in \mathcal{S}} \frac{dq_j}{d\tau_1}\tau_j = \sum_j \frac{dq_j}{d\tau_1}\mu_j - \sum_j \frac{dq_j}{d\tau_1}\phi_j^{US} - \sum_{j \in \mathcal{J}_{For}} \frac{d\pi_j}{d\tau_1} - (\eta - 1)q_1. \tag{31}$$

Partitioning the sums between $\mathcal{S}$ and $\backslash\mathcal{S}$ then combining terms,

$$\eta \sum_{j \in \mathcal{S}} \frac{dq_j}{d\tau_1}\tau_j = \sum_{j \in \mathcal{S}} \frac{dq_j}{d\tau_1}\left(\mu_j - \phi_j^{US}\right) + \sum_{j \in \backslash\mathcal{S}} \frac{dq_j}{d\tau_1}\left(\mu_j - \phi_j^{US}\right) - \sum_{j \in \mathcal{J}_{For}} \frac{d\pi_j}{d\tau_1} - (\eta - 1)q_1. \tag{32}$$

Each choice in $\mathcal{S}$ yields a first-order condition. By expressing the sums as dot products of vectors and then stacking these $S$ equations, we get the following linear system in matrix notation where $\boldsymbol{\tau}_{\mathcal{S}}$ is an $S \times 1$ vector of optimal differentiated subsidies

$$\eta \left[(\boldsymbol{I} - \boldsymbol{T})\,\boldsymbol{\Omega}\right]_{\mathcal{S}} \boldsymbol{\tau}_{\mathcal{S}} = \left[(\boldsymbol{I} - \boldsymbol{T})\,\boldsymbol{\Omega}\right]_{\mathcal{S}} \left(\boldsymbol{\mu}_{\mathcal{S}} - \boldsymbol{\phi}_{\mathcal{S}}^{US}\right) + \left[(\boldsymbol{I} - \boldsymbol{T})\,\boldsymbol{\Omega}\right]_{\backslash\mathcal{S}} \left(\boldsymbol{\mu}_{\backslash\mathcal{S}} - \boldsymbol{\phi}_{\backslash\mathcal{S}}^{US}\right) + \boldsymbol{m}_{For} + (\eta - 1)\boldsymbol{q}_{\mathcal{S}} \tag{33}$$

Here, $\boldsymbol{\Omega}$ is the $J \times (J+1)$ matrix of demand partial derivatives with representative element $[\Omega]_{jr} = \frac{\partial q_r}{\partial p_j}$, for $j = 1, ..., J$ inside goods and $r = 0, ..., J$ total choices; $\boldsymbol{T}$ is the $J \times J$ matrix of equilibrium price responses to subsidies with representative element $[T]_{jr} = \frac{dp_r}{d\tau_j}$; and $I$ is the identity matrix. The $S \times S$ submatrix $\left[(\boldsymbol{I} - \boldsymbol{T})\,\boldsymbol{\Omega}\right]_{\mathcal{S}}$ contains all elements from the bracketed matrix product with $j \in \mathcal{S}$, $r \in \mathcal{S}$. The $S \times (J + 1 - S)$ submatrix $\left[(\boldsymbol{I} - \boldsymbol{T})\,\boldsymbol{\Omega}\right]_{\backslash\mathcal{S}}$ contains all elements with $j \in S$, $r \in \backslash\mathcal{S}$. Finally, $\boldsymbol{m}_{For}$ is the vector of producer surplus impacts on foreign firms corresponding to the final term in equation (32), which has the form $\boldsymbol{m}_{For} = \boldsymbol{T}_{For}\boldsymbol{q}_{For} - \left[(\boldsymbol{I} - \boldsymbol{T})\,\boldsymbol{\Omega}\right]_{For}\boldsymbol{\mu}_{For}$, with $For$ subscripts corresponding to submatrices and subvectors as elsewhere.

Multiplying through by the inverse of $\left[(\boldsymbol{I} - \boldsymbol{T})\,\boldsymbol{\Omega}\right]_{\mathcal{S}}$ gives the constrained optimal differentiated

subsidy

$$\boldsymbol{\tau}_{\mathcal{S}}^{SB} = \underbrace{\frac{1}{\eta}\left(\boldsymbol{\mu}_{\mathcal{S}} - \boldsymbol{\phi}_{\mathcal{S}}^{US}\right)}_{\text{price distortion}} + \underbrace{\frac{1}{\eta}[(\boldsymbol{I}-\boldsymbol{T})\,\boldsymbol{\Omega}]_{S}^{-1}[(\boldsymbol{I}-\boldsymbol{T})\,\boldsymbol{\Omega}]_{\setminus S}\left(\boldsymbol{\mu}_{\setminus\mathcal{S}} - \boldsymbol{\phi}_{\setminus\mathcal{S}}^{US}\right)}_{\text{indirect substitution}} + \underbrace{\frac{1}{\eta}\left[(\boldsymbol{I}-\boldsymbol{T})\,\boldsymbol{\Omega}\right]_{S}^{-1}\boldsymbol{m}_{For}}_{\text{profit shifting}} + \underbrace{\frac{(\eta-1)}{\eta}\left[(\boldsymbol{I}-\boldsymbol{T})\,\boldsymbol{\Omega}\right]_{S}^{-1}\boldsymbol{q}_{\mathcal{S}}}_{\text{tax distortion}}. \tag{34}$$

Intuitively, the subsidy deviates from the global planner's solution by the amount of diversion (or "leakage") from the untargeted set of choices and to foreign firms' surplus.

### E.3 Constrained Optimal Uniform Subsidy

This subsection provides the derivation of Proposition 1 in the main text, the constrained optimal uniform subsidy for a subset of goods $\mathcal{S}$. "Uniform" means that we restrict the value of the subsidy, $\tau$, to be equal for all choices in $\mathcal{S}$. We proceed in a similar fashion to Section E.2. Taking the derivative of $W^{US}$ with respect to the scalar value $\tau$, which changes the subsidy level for all choices in $\mathcal{S}$, gives:

$$[\tau]: \quad \frac{dW^{US}}{d\tau} = \underbrace{\sum_{j\in\mathcal{S}} q_j - \sum_{j} q_j \frac{dp_j}{d\tau}}_{\frac{dCS}{d\tau}} + \underbrace{\sum_{j} \frac{dq_j}{d\tau}\mu_j + \sum_{j} q_j \frac{d\mu_j}{d\tau}}_{\frac{dPS}{d\tau}} - \underbrace{\eta\left(\sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}\tau^{SB,U} + \sum_{j\in\mathcal{S}} q_j\right)}_{\frac{dG}{d\tau}} - \underbrace{\sum_{j} \frac{dq_j}{d\tau}\phi_j^{US}}_{\frac{dE}{d\tau}} - \underbrace{\sum_{j\in\mathcal{J}_{For}} \frac{d\pi_j}{d\tau}}_{\frac{dPS^{For}}{d\tau}} = 0 \tag{35}$$

Canceling terms and rearranging gives

$$\eta \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}\tau^{SB,U} = \sum_{j} \frac{dq_j}{d\tau}\mu_j - \sum_{j} \frac{dq_j}{d\tau}\phi_j^{US} - \sum_{j\in\mathcal{J}_{For}} \frac{d\pi_j}{d\tau} - (\eta-1)\sum_{j\in\mathcal{S}} q_j. \tag{36}$$

Separating sums by $\mathcal{S}$ and $\setminus\mathcal{S}$ and combining terms gives

$$\eta \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}\tau^{SB,U} = \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}\left(\mu_j - \phi_j^{US}\right) + \sum_{j\in\setminus\mathcal{S}} \frac{dq_j}{d\tau}\left(\mu_j - \phi_j^{US}\right) - \sum_{j\in\mathcal{J}_{For}} \frac{d\pi_j}{d\tau} - (\eta-1)\sum_{j\in\mathcal{S}} q_j. \tag{37}$$

Dividing through gives

$$\tau^{SB,U} = \frac{\sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}(\mu_j - \phi_j^{US})}{\eta \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}} + \frac{\sum_{j\in\backslash\mathcal{S}} \frac{dq_j}{d\tau}(\mu_j - \phi_j^{US})}{\eta \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}} - \frac{\sum_{j\in\mathcal{J}_{For}} \frac{d\pi_j}{d\tau}}{\eta \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}} - \frac{(\eta-1)\sum_{j\in\mathcal{S}} q_j}{\eta \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}} \tag{38}$$

Unit demand implies $\sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau} = -\sum_{j\in\backslash\mathcal{S}} \frac{dq_j}{d\tau}$. This simplifies to equation (13) when $\eta = 1$, but takes the more general form:

$$\tau^{SB,U} = \underbrace{\frac{1}{\eta}\left(\bar{\mu}_{\mathcal{S}} - \bar{\phi}_{\mathcal{S}}^{US}\right)}_{\text{price distortion}} - \underbrace{\frac{1}{\eta}\left(\bar{\mu}_{\backslash\mathcal{S}} - \bar{\phi}_{\backslash\mathcal{S}}^{US}\right)}_{\text{indirect substitution}} - \underbrace{\frac{\sum_{j\in\mathcal{J}_{For}} \frac{d\pi_j}{d\tau}}{\eta \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}}}_{\text{profit shifting}} - \underbrace{\frac{(\eta-1)\sum_{j\in\mathcal{S}} q_j}{\eta \sum_{j\in\mathcal{S}} \frac{dq_j}{d\tau}}}_{\text{tax distortion}} \tag{39}$$

The first two terms are demand-response weighted-averages of the unpriced externality. The third term is the marginal profit shifted to foreign firms per marginal vehicle sold evaluated at the level of subsidy. The final term is the revenue-raising cost of the marginal transfer under a given MCPF. When we compute the optimal uniform subsidy for a US planner that places positive value on job creation, $\phi_j^{jobs} < 0$ (recall $\phi_j$ are negative externalities), we break out the contribution of the jobs component of the externality, reporting a "net jobs value" term equal to $-\frac{1}{\eta}\left(\bar{\phi}_{\mathcal{S}}^{jobs} - \bar{\phi}_{\backslash\mathcal{S}}^{jobs}\right)$.

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Table A1: **Share-Weighted Average Externalities by Vehicle Powertrain (Long-Run Grid)**

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Electric Vehicles | | | |
| | All EVs | Battery electric vehicles | Plug-in hybrids | Gasoline vehicles |
| Driving $CO_2$ (global SCC) | 3,676 | 2,530 | 9,225 | 13,578 |
| Total negative externality | | | | |
| Global SCC | 11,483 | 10,286 | 17,279 | 19,890 |
| Domestic SCC | 5,599 | 5,267 | 7,208 | 6,270 |

Notes: This table presents market share-weighted average lifetime externalities across submodels within a powertrain, weighting submodels by average monthly sales in months when the submodel was available. Units are $/vehicle.

Table A2: **Estimated Lease Moments**

| Model: | (1) | (2) | (3) |
|---|---|---|---|
| Dependent variables: | Leasing share (%) | Lease price relative to purchase price ($000s) | |
| *Variables* | | | |
| Excluded Aug 2022 × July-August 2023 | 39.26 | -5.538 | -4.933 |
| | (3.972) | (0.7452) | (1.153) |
| *Fixed effects* | | | |
| Submodel | Yes | Yes | Yes |
| Year-month | Yes | Yes | Yes |
| *Fit statistics* | | | |
| $R^2$ | 0.91413 | 0.72395 | 0.74424 |
| Observations | 3,951 | 3,082 | 1,766 |

Notes: This table presents regression results for the group of EVs which were excluded from 30D eligibility in August 2022. It compares outcomes for July-August 2023 with those from late 2022; columns 1 and 2 compare against the fourth quarter of 2022, whereas column 3 compares against December 2022 only. The regressions are weighted at the submodel level according to average registrations during the months the submodel was available, and standard errors are clustered at the model level. When calculating standard errors in the model following Appendix D.3, we estimate columns 1 and 2 jointly with stacked OLS equations, giving an estimated covariance of -1.615 along with the standard errors.

Table A3: **Estimated Purchase Price Effects**

| Model: | (1) |
|---|---|
| Dependent variable: | Purchase price ($000s) |
| *Variables* | |
| Excluded Aug 2022 or Excluded/Reduced Apr 2023 × post-period | 0.3679 |
| | (0.5066) |
| *Fixed effects* | |
| Submodel | Yes |
| Year-month | Yes |
| *Fit statistics* | |
| $R^2$ | 0.99024 |
| Observations | 15,971 |

Notes: This table presents regression results for the group of EVs which were excluded from 30D eligibility in August 2022 and those either excluded from 30D or that experienced reductions in purchase credits in April 2023. The pre-period is defined as the three months prior to the eligibility change (considered to be September 2022 and May 2023, respectively); the post-period is the three months following the eligibility change, and the control group is all GVs. The regression is weighted at the submodel level according to average registrations during the months the submodel was available, and standard errors are clustered at the model level.

Table A4: Phase-In and Phase-Out for Popular EVs

| Submodel | Phase-in months | Phase-out months | Total months |
|---|---|---|---|
| VOLV XC90 PLUS (PHEV) | 7 | 0 | 24 |
| KIA SORENTO SX PRESTIGE (PHEV) | 6 | 0 | 24 |
| MITS OUTLANDER SEL (PHEV) | 5 | 0 | 18 |
| VOLK ID.4 PRO (BEV) | 5 | 0 | 23 |
| BMW IX XDRIVE50 (BEV) | 4 | 0 | 23 |
| CHEV BOLT EUV PREMIER (BEV) | 4 | 0 | 23 |
| RIVA R1S ADVENTURE (BEV) | 3 | 0 | 11 |
| JEEP WRANGLER 4XE (PHEV) | 3 | 0 | 13 |
| TOYT BZ4X XLE (BEV) | 3 | 0 | 16 |
| CHEV BOLT EUV LT (BEV) | 3 | 0 | 22 |
| CHEV BOLT EV 2LT (BEV) | 3 | 0 | 22 |
| LEXS RZ 450E (BEV) | 2 | 0 | 10 |
| CADI LYRIQ LUXURY (BEV) | 2 | 0 | 13 |
| RIVA R1T ADVENTURE (BEV) | 2 | 0 | 17 |
| HYUN TUCSON LIMITED (PHEV) | 2 | 0 | 22 |
| AUDI Q5 E PREMIUM PLUS 55 (PHEV) | 2 | 0 | 23 |
| BMW I4/I4 EDRIVE40 -/EDRIVE 40 (BEV) | 1 | 2 | 16 |
| MERZ EQS SUV 450+ (BEV) | 1 | 2 | 16 |
| RIVA R1S LAUNCH EDITION (BEV) | 1 | 2 | 17 |
| VOLV C40 RECHARGE ULTIMA (BEV) | 1 | 2 | 24 |
| HYUN IONIQ 6 LIMITED (BEV) | 1 | 0 | 10 |
| HYUN IONIQ 6 SEL (BEV) | 1 | 0 | 10 |
| BMW I4/I4 EDRIVE35 -/EDRIVE 35 (BEV) | 1 | 0 | 11 |
| NISS ARIYA ENGAGE (BEV) | 1 | 0 | 11 |
| NISS ARIYA EVOLVE+ (BEV) | 1 | 0 | 13 |
| JEEP GRAND CHEROKEE OVERLAND 4XE (PHEV) | 1 | 0 | 15 |
| BMW I4/I4 M50 -/M50 (BEV) | 1 | 0 | 16 |
| JEEP GRAND CHEROKEE SUMMIT 4XE (PHEV) | 1 | 0 | 16 |
| MERZ EQS SUV 580 4MATIC (BEV) | 1 | 0 | 16 |
| AUDI Q4 E-TRON PREMIUM PLUS (BEV) | 1 | 0 | 17 |
| MERZ EQB 300 4MATIC (BEV) | 1 | 0 | 17 |
| FORD F150 LIGHTNING PRO (BEV) | 1 | 0 | 19 |
| CHEV BOLT EV 1LT (BEV) | 1 | 0 | 21 |
| BMW 330XE - (PHEV) | 1 | 0 | 22 |
| HYUN IONIQ 5 LIMITED (BEV) | 1 | 0 | 24 |
| TESL MODEL X - (BEV) | 1 | 0 | 24 |
| RIVA R1T LAUNCH EDITION (BEV) | 0 | 11 | 19 |
| JEEP WRANGLER UNLIMI SAHARA 4XE (PHEV) | 0 | 11 | 21 |
| JEEP WRANGLER UNLIMI RUBICON 4XE (PHEV) | 0 | 9 | 19 |
| MERZ EQS SEDAN 450+ (BEV) | 0 | 5 | 17 |
| KIA NIRO S (BEV) | 0 | 3 | 9 |
| BMW 330E - (PHEV) | 0 | 2 | 17 |
| NISS LEAF S (BEV) | 0 | 2 | 24 |
| HYUN IONIQ 5 SE (BEV) | 0 | 1 | 24 |

Notes: This table reports the number of months considered phase-in and phase-out months among EV submodels which are above the median for total registrations. Submodels with no phase-in or phase-out period are omitted.

Table A5: **Parameter Estimates**

| Parameter | Description | Value | Standard error | | |
|---|---|---|---|---|---|
| | | | Baseline | Bound $\sigma_{\text{elas}} = 0$ | Bound $\sigma_{\text{elas}} = 1.187$ |
| $\alpha_0$ | Price response parameter | -0.514 | (0.019) | (0.019) | (0.186) |
| $\alpha_y$ | Price response × income interaction | -0.374 | (0.011) | (0.012) | (0.148) |
| $\beta_0$ | Inside good × income interaction | 0.299 | (0.056) | (0.058) | (0.123) |
| $\beta_{EV}$ | EV × income interaction | 0.498 | (0.015) | (0.015) | (0.030) |
| $\beta_{Lease}$ | Lease × income interaction | 0.043 | (0.003) | (0.004) | (0.009) |
| $\sigma^g$ | EV-GV nest parameter | 0.499 | (0.009) | (0.009) | (0.011) |
| $\sigma^c$ | Class nest parameter | 0.613 | (0.007) | (0.007) | (0.007) |
| $\sigma^k$ | Submodel nest parameter | 0.914 | (0.014) | (0.015) | (0.044) |

Notes: This table presents parameter estimates from the demand estimation in Section 5. Baseline standard errors are calculated using the delta method assuming no covariance across our three blocks of data moments. Relative to Table 5, we add columns reporting sharp worst-case bounds on standard errors following Cocci and Plagborg-Møller (2025), which are valid under any correlation structure across our moment blocks that is consistent with the within-block covariances we can estimate; see Appendix D.3 for details. The first column treats the own-price elasticity target of Grieco, Murry, and Yurukoglu (2024) as known with certainty ($\sigma_{\text{elas}} = 0$). The second column instead calibrates $\sigma_{\text{elas}} = 1.187$, the value at which the worst-case $t$-statistic on our price coefficient $\alpha_0$ matches the $t$-statistic on the mean price sensitivity reported by Grieco, Murry, and Yurukoglu (2024).

Table A6: **Effects of Counterfactual Policies on Market Outcomes**

| | | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|---|
| | | | Repeal IRA | | Modify Trade Restr. | | |
| | | IRA (baseline) | Eliminate EV Credits | Return to Pre-IRA 30D with phaseout | IRA, add trade restrictions on leases | IRA, remove trade restrictions | IRA, remove income threshold |
| | **Panel (a): Market Aggregates under Counterfactual Scenario** | | | | | | |
| 1. | Vehicle registrations (000s/year) | 10,556 | 10,458<br>[10,453, 10,463]<br>[10,418, 10,498] | 10,529<br>[10,525, 10,532]<br>[10,519, 10,539] | 10,539<br>[10,537, 10,541]<br>[10,530, 10,548] | 10,609<br>[10,605, 10,613]<br>[10,584, 10,634] | 10,557<br>[10,557, 10,557]<br>[10,556, 10,558] |
| 2. | US firms (000s/year) | 3,952 | 3,748<br>[3,743, 3,754]<br>[3,683, 3,814] | 3,669<br>[3,662, 3,675]<br>[3,572, 3,766] | 3,961<br>[3,959, 3,964]<br>[3,952, 3,971] | 3,893<br>[3,890, 3,897]<br>[3,868, 3,919] | 3,961<br>[3,960, 3,962]<br>[3,948, 3,974] |
| 3. | Foreign firms (000s/year) | 6,604 | 6,710<br>[6,703, 6,716]<br>[6,677, 6,742] | 6,860<br>[6,850, 6,869]<br>[6,765, 6,955] | 6,578<br>[6,573, 6,582]<br>[6,560, 6,595] | 6,716<br>[6,709, 6,723]<br>[6,667, 6,765] | 6,596<br>[6,595, 6,597]<br>[6,584, 6,608] |
| 4. | EV registrations (000s/year) | 1,184 | 968<br>[958, 977]<br>[875, 1,060] | 1,136<br>[1,127, 1,145]<br>[1,121, 1,151] | 1,142<br>[1,136, 1,149]<br>[1,118, 1,167] | 1,304<br>[1,296, 1,312]<br>[1,249, 1,359] | 1,193<br>[1,192, 1,193]<br>[1,183, 1,202] |
| 5. | US firms (000s/year) | 835 | 590<br>[584, 595]<br>[508, 671] | 542<br>[537, 548]<br>[447, 638] | 836<br>[835, 837]<br>[832, 840] | 797<br>[795, 800]<br>[781, 814] | 847<br>[846, 848]<br>[831, 864] |
| 6. | Foreign firms (000s/year) | 349 | 378<br>[369, 387]<br>[361, 395] | 594<br>[581, 607]<br>[494, 694] | 307<br>[300, 314]<br>[280, 334] | 507<br>[496, 517]<br>[437, 576] | 346<br>[345, 346]<br>[338, 353] |
| 7. | EV market share (%) | 11.2 | 9.3<br>[9.2, 9.3]<br>[8.4, 10.1] | 10.8<br>[10.7, 10.9]<br>[10.7, 10.9] | 10.8<br>[10.8, 10.9]<br>[10.6, 11.1] | 12.3<br>[12.2, 12.4]<br>[11.8, 12.8] | 11.3<br>[11.3, 11.3]<br>[11.2, 11.4] |
| 8. | Lease share, within EVs (%) | 29.3 | 9.6<br>[7.0, 12.2]<br>[6.3, 12.9] | 10.2<br>[7.4, 13.0]<br>[7.2, 13.2] | 12.4<br>[10.6, 14.2]<br>[9.5, 15.2] | 13.8<br>[12.0, 15.7]<br>[10.1, 17.6] | 27.5<br>[27.1, 28.0]<br>[24.9, 30.1] |
| 9. | US assembly share, within EVs (%) | 70.5 | 60.3<br>[59.6, 60.9]<br>[56.5, 64.0] | 50.0<br>[49.3, 50.8]<br>[41.9, 58.1] | 73.4<br>[72.9, 73.9]<br>[71.6, 75.2] | 60.8<br>[60.2, 61.3]<br>[56.8, 64.7] | 71.0<br>[70.9, 71.0]<br>[70.2, 71.7] |
| 10. | Cost per additional EV ($000s/EV) | - | 32.1<br>[30.5, 33.7]<br>[18.5, 45.7] | 49.2<br>[40.2, 58.3]<br>[21.7, 76.8] | 38.4<br>[32.4, 44.4]<br>[15.2, 61.7] | 13.8<br>[13.1, 14.5]<br>[10.9, 16.8] | 59.9<br>[54.3, 65.6]<br>[-7.30, 127] |
| | **Panel (b): Employment and Battery Production Effects Relative to IRA Baseline (000s/year)** | | | | | | |
| 11. | ΔUS auto manufacturing jobs | - | -11.5<br>[-11.9, -11.1]<br>[-17.2, -5.8] | -14.4<br>[-14.9, -13.9]<br>[-22.2, -6.7] | 0.7<br>[0.5, 0.9]<br>[-0.1, 1.5] | -4.3<br>[-4.6, -4.0]<br>[-6.6, -1.9] | 0.4<br>[0.4, 0.4]<br>[-0.2, 1.0] |
| 12. | ΔUS auto parts jobs | - | -7.3<br>[-7.5, -7.0]<br>[-10.9, -3.7] | -9.2<br>[-9.6, -8.9]<br>[-14.2, -4.2] | 0.5<br>[0.4, 0.6]<br>[0.0, 1.0] | -2.8<br>[-3.0, -2.6]<br>[-4.3, -1.3] | 0.3<br>[0.2, 0.3]<br>[-0.1, 0.7] |
| 13. | ΔUS auto assembly jobs | - | -4.2<br>[-4.3, -4.1]<br>[-6.3, -2.1] | -5.2<br>[-5.3, -5.0]<br>[-7.9, -2.4] | 0.2<br>[0.2, 0.3]<br>[0.0, 0.5] | -1.5<br>[-1.6, -1.4]<br>[-2.3, -0.7] | 0.1<br>[0.1, 0.1]<br>[-0.1, 0.3] |

*Continued on next page*

| *Table A6 (continued)* | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | | Repeal IRA | | Modify Trade Restr. | | |
| | IRA (baseline) | Eliminate EV Credits | Return to Pre-IRA 30D with phaseout | IRA, add trade restrictions on leases | IRA, remove trade restrictions | IRA, remove income threshold |
| 14. ΔUS-assembled EV batteries | - | -241<br>[-247, -235]<br>[-320, -162] | -284<br>[-289, -279]<br>[-383, -185] | 13<br>[10.7, 14.6]<br>[3.92, 21.4] | -45<br>[-48.3, -41.9]<br>[-66.1, -24.2] | 10<br>[9.40, 11.6]<br>[-4.56, 25.5] |
| 15. Cost per additional US job ($000s/job) | - | 604<br>[587, 621]<br>[318, 890] | 165<br>[158, 172]<br>[32.5, 297] | -2,141<br>[-2,726, -1,556]<br>[-4,493, 211] | -388<br>[-410, -366]<br>[-514, -262] | 1,314<br>[1,133, 1,496]<br>[-879, 3,508] |
| **Panel (c): Allocation of EV Credits** | | | | | | |
| 16. Share of EVs with credits (%) | 80.0 | 0.00<br>[0.00, 0.00]<br>[0.00, 0.00] | 66.3<br>[65.7, 67.0]<br>[57.9, 74.8] | 66.7<br>[66.3, 67.1]<br>[64.2, 69.2] | 87.9<br>[87.4, 88.5]<br>[85.6, 90.2] | 86.5<br>[86.5, 86.6]<br>[85.9, 87.1] |
| 17. Among HHs >$300k (%) | 44.2 | 0.00<br>[0.00, 0.00]<br>[0.00, 0.00] | 64.7<br>[64.3, 65.2]<br>[56.7, 72.7] | 22.4<br>[19.9, 24.9]<br>[19.4, 25.4] | 40.4<br>[39.1, 41.8]<br>[33.6, 47.3] | 77.0<br>[76.1, 78.0]<br>[74.3, 79.8] |

Notes: This table reproduces Table 6 and adds two sets of 95% confidence intervals below each estimate in brackets. The first interval reports conventional 95% confidence intervals calculated using the delta method corresponding to our baseline standard errors. The second interval reports confidence intervals with at least 95% coverage corresponding to the sharp worst-case standard errors reported in the final column of Appendix Table A5. Following Cocci and Plagborg-Møller (2025), these confidence intervals are valid under any correlation structure across our three blocks of data moments that is consistent with the within-block covariances we can estimate; see Appendix D.3. For these reported bounds, we calibrate the standard error of the price elasticity moment to be $\sigma_{\text{elas}} = 1.187$, the value at which the worst-case $t$-statistic on our price coefficient $\alpha_0$ matches the $t$-statistic on the mean price sensitivity reported by Grieco, Murry, and Yurukoglu (2024), since they do not report a standard error on the elasticity. We omit reporting the bounds assuming $\sigma_{\text{elas}} = 0$ since these are very similar to the baseline intervals.

Table A7: **Effects of Counterfactual Policies on Welfare Outcomes**

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | | Repeal IRA | | Modify Trade Restr. | | |
| | IRA (baseline) | Eliminate EV Credits | Return to Pre-IRA 30D with phaseout | IRA, add trade restrictions on leases | IRA, remove trade restrictions | IRA, remove income threshold |
| **Panel (a): Surplus Effects Relative to IRA Baseline ($billion/year)** | | | | | | |
| 1. ΔUS consumer surplus | - | -4.86<br>[-5.05, -4.68]<br>[-5.48, -4.25] | -0.63<br>[-0.87, -0.39]<br>[-1.17, -0.10] | -1.16<br>[-1.33, -0.98]<br>[-1.52, -0.79] | 2.61<br>[2.44, 2.77]<br>[2.21, 3.01] | 0.60<br>[0.57, 0.64]<br>[0.52, 0.68] |
| 2. ΔGlobal producer surplus | - | -1.66<br>[-1.70, -1.62]<br>[-1.81, -1.51] | -1.60<br>[-1.64, -1.55]<br>[-2.12, -1.07] | 0.19<br>[0.16, 0.22]<br>[0.07, 0.31] | -0.29<br>[-0.33, -0.25]<br>[-0.54, -0.04] | 0.03<br>[0.02, 0.04]<br>[-0.05, 0.11] |
| 3. ΔUS producer surplus | - | -2.62<br>[-2.70, -2.54]<br>[-2.99, -2.26] | -3.97<br>[-4.06, -3.88]<br>[-4.58, -3.36] | 0.41<br>[0.34, 0.48]<br>[0.30, 0.53] | -1.23<br>[-1.30, -1.16]<br>[-1.49, -0.98] | 0.18<br>[0.16, 0.20]<br>[0.03, 0.33] |
| 4. ΔForeign producer surplus | - | 0.96<br>[0.90, 1.03]<br>[0.71, 1.21] | 2.38<br>[2.29, 2.46]<br>[2.26, 2.49] | -0.22<br>[-0.26, -0.18]<br>[-0.32, -0.12] | 0.94<br>[0.88, 1.00]<br>[0.83, 1.06] | -0.15<br>[-0.16, -0.14]<br>[-0.22, -0.08] |
| 5. Δ$CO_2$ emissions (million tons/year) | - | 4.88<br>[4.69, 5.06]<br>[2.96, 6.80] | 2.61<br>[2.46, 2.76]<br>[1.82, 3.40] | 0.37<br>[0.28, 0.46]<br>[0.10, 0.64] | -2.05<br>[-2.18, -1.93]<br>[-2.90, -1.21] | -0.28<br>[-0.31, -0.25]<br>[-0.64, 0.08] |
| 6. ΔGlobal negative externalities (global SCC) | - | 1.99<br>[1.91, 2.07]<br>[1.21, 2.77] | 1.03<br>[0.96, 1.11]<br>[0.70, 1.36] | 0.18<br>[0.13, 0.22]<br>[0.05, 0.31] | -0.86<br>[-0.92, -0.80]<br>[-1.20, -0.51] | -0.10<br>[-0.11, -0.09]<br>[-0.24, 0.03] |
| 7. ΔUS negative externalities (domestic SCC) | - | 0.95<br>[0.91, 0.99]<br>[0.58, 1.32] | 0.48<br>[0.44, 0.52]<br>[0.31, 0.64] | 0.10<br>[0.07, 0.12]<br>[0.02, 0.17] | -0.42<br>[-0.45, -0.39]<br>[-0.58, -0.25] | -0.04<br>[-0.05, -0.04]<br>[-0.10, 0.01] |
| 8. ΔForeign negative externalities (foreign SCC) | - | 1.04<br>[1.00, 1.08]<br>[0.63, 1.45] | 0.56<br>[0.52, 0.59]<br>[0.39, 0.73] | 0.08<br>[0.06, 0.10]<br>[0.02, 0.14] | -0.44<br>[-0.47, -0.41]<br>[-0.62, -0.26] | -0.06<br>[-0.07, -0.05]<br>[-0.14, 0.02] |
| 9. ΔUS subsidy spending | - | -6.94<br>[-7.00, -6.88]<br>[-7.10, -6.78] | -2.37<br>[-2.41, -2.34]<br>[-3.02, -1.73] | -1.60<br>[-1.61, -1.58]<br>[-1.64, -1.55] | 1.66<br>[1.61, 1.71]<br>[1.24, 2.08] | 0.51<br>[0.45, 0.58]<br>[0.42, 0.61] |
| 10. ΔGas tax revenue | - | 0.76<br>[0.73, 0.79]<br>[0.45, 1.07] | 0.27<br>[0.24, 0.30]<br>[0.19, 0.36] | 0.11<br>[0.09, 0.13]<br>[0.04, 0.18] | -0.30<br>[-0.33, -0.28]<br>[-0.44, -0.17] | -0.03<br>[-0.04, -0.03]<br>[-0.08, 0.01] |
| 11. ΔGlobal surplus | - | -0.82<br>[-1.07, -0.57]<br>[-1.30, -0.34] | -0.61<br>[-0.86, -0.36]<br>[-0.92, -0.30] | 0.57<br>[0.38, 0.76]<br>[0.34, 0.80] | 1.21<br>[1.05, 1.38]<br>[0.81, 1.61] | 0.19<br>[0.15, 0.22]<br>[0.03, 0.34] |
| 12. ΔUS total surplus (global SCC) | - | -1.78<br>[-1.97, -1.59]<br>[-2.37, -1.19] | -2.99<br>[-3.15, -2.82]<br>[-3.20, -2.77] | 0.79<br>[0.64, 0.94]<br>[0.55, 1.02] | 0.27<br>[0.16, 0.38]<br>[-0.11, 0.65] | 0.33<br>[0.30, 0.37]<br>[0.13, 0.54] |
| 13. ΔUS total surplus (domestic SCC) | - | -0.74<br>[-0.90, -0.57]<br>[-1.64, 0.16] | -2.43<br>[-2.57, -2.29]<br>[-2.68, -2.18] | 0.87<br>[0.74, 1.00]<br>[0.62, 1.11] | -0.17<br>[-0.25, -0.08]<br>[-0.66, 0.32] | 0.28<br>[0.24, 0.31]<br>[0.14, 0.41] |
| 14. ΔForeign total surplus (foreign SCC) | - | -0.08<br>[-0.16, 0.01]<br>[-0.72, 0.56] | 1.82<br>[1.70, 1.94]<br>[1.55, 2.09] | -0.30<br>[-0.36, -0.24]<br>[-0.45, -0.15] | 1.38<br>[1.30, 1.46]<br>[1.09, 1.67] | -0.09<br>[-0.10, -0.08]<br>[-0.11, -0.07] |

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| *Table A7 (continued)* | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | | Repeal IRA | | Modify Trade Restr. | | |
| | IRA (baseline) | Eliminate EV Credits | Return to Pre-IRA 30D with phaseout | IRA, add trade restrictions on leases | IRA, remove trade restrictions | IRA, remove income threshold |
| **Panel (b): MVPF of Higher-Spending versus Lower-Spending Scenario** | | | | | | |
| 15. Global MVPF | - | 1.11<br>[1.07, 1.14]<br>[1.04, 1.17] | 1.23<br>[1.14, 1.32]<br>[1.07, 1.39] | 0.67<br>[0.55, 0.78]<br>[0.54, 0.80] | 1.62<br>[1.55, 1.69]<br>[1.28, 1.95] | 1.34<br>[1.25, 1.43]<br>[1.06, 1.62] |
| 16. US MVPF (global SCC) | - | 1.23<br>[1.21, 1.26]<br>[1.14, 1.32] | 2.13<br>[2.09, 2.17]<br>[1.83, 2.42] | 0.54<br>[0.45, 0.63]<br>[0.40, 0.67] | 1.14<br>[1.09, 1.19]<br>[0.92, 1.36] | 1.61<br>[1.50, 1.72]<br>[1.23, 1.99] |
| 17. US MVPF (domestic SCC) | - | 1.10<br>[1.07, 1.12]<br>[0.97, 1.22] | 1.92<br>[1.88, 1.95]<br>[1.72, 2.11] | 0.49<br>[0.42, 0.57]<br>[0.35, 0.63] | 0.91<br>[0.87, 0.96]<br>[0.68, 1.15] | 1.50<br>[1.39, 1.61]<br>[1.25, 1.75] |
| **Panel (c): Welfare Cost per Ton of $CO_2$ Abated** | | | | | | |
| 18. US cost/ton $CO_2$ abated ($/ton) | - | -124<br>[-154, -93.1]<br>[-356, 109] | -904<br>[-929, -879]<br>[-1,178, -630] | 2,361<br>[1,442, 3,279]<br>[602, 4,119] | 109<br>[63.5, 155]<br>[-108, 327] | -965<br>[-1,127, -804]<br>[-1,973, 42.1] |
| 19. Global cost/ton $CO_2$ abated ($/ton) | - | 73.3<br>[26.5, 120]<br>[-75.2, 222] | 6.60<br>[-77.7, 90.9]<br>[-119, 133] | 1,769<br>[880, 2,659]<br>[195, 3,343] | -349<br>[-401, -297]<br>[-707, 9.07] | -427<br>[-570, -283]<br>[-965, 112] |

Notes: This table reproduces Table 7 and adds two sets of 95% confidence intervals below each estimate in brackets. The first interval reports conventional 95% confidence intervals calculated using the delta method corresponding to our baseline standard errors. The second interval reports confidence intervals with at least 95% coverage corresponding to the sharp worst-case standard errors reported in the final column of Appendix Table A5. Following Cocci and Plagborg-Møller (2025), these confidence intervals are valid under any correlation structure across our three blocks of data moments that is consistent with the within-block covariances we can estimate; see Appendix D.3. For these reported bounds, we calibrate the standard error of the price elasticity moment to be $\sigma_{\text{elas}} = 1.187$, the value at which the worst-case $t$-statistic on our price coefficient $\alpha_0$ matches the $t$-statistic on the mean price sensitivity reported by Grieco, Murry, and Yurukoglu (2024), since they do not report a standard error on the elasticity. We omit reporting the bounds assuming $\sigma_{\text{elas}} = 0$ since these are very similar to the baseline intervals.

Table A8: **Counterfactual Simulation Results: Sensitivity Analysis**

| | | (1) IRA | (2) US-optimal uniform EV subsidy with 30D restrictions | (3) US-optimal differentiated EV subsidy with 30D restrictions |
|---|---|---|---|---|
| | **Panel (a): Marginal Cost of Public Funds = 1.4, Social Cost of Carbon = $28** | | | |
| 1. | Mean EV subsidy | $7,328 | $1,834 | $2,332 |
| 2. | ΔGlobal negative externalities | -0.95 | -0.16 | -0.42 |
| 3. | ΔGlobal surplus | -3.30 | -0.28 | -0.24 |
| 4. | ΔUS total surplus | -2.34 | 0.10 | 0.27 |
| 5. | Cost per additional EV ($000s/EV) | 32.1 | 27.2 | 31.1 |
| 6. | US MVPF (global SCC) | 1.10 | 1.48 | 1.56 |
| | **Panel (b): Marginal Cost of Public Funds = 1.4, Social Cost of Carbon = $241** | | | |
| 1. | Mean EV subsidy | $7,328 | $3,604 | $4,789 |
| 2. | ΔGlobal negative externalities | -1.99 | -0.74 | -1.66 |
| 3. | ΔGlobal surplus | -2.26 | -0.36 | -0.13 |
| 4. | ΔUS total surplus | -1.30 | 0.39 | 0.80 |
| 5. | Cost per additional EV ($000s/EV) | 32.1 | 28.6 | 33.7 |
| 6. | US MVPF (global SCC) | 1.23 | 1.54 | 1.60 |
| | **Panel (c): Social Cost of Carbon = $100** | | | |
| 1. | Mean EV subsidy | $7,328 | $10,997 | $14,576 |
| 2. | ΔGlobal negative externalities | -1.30 | -1.89 | -3.46 |
| 3. | ΔGlobal surplus | 0.13 | 0.13 | 0.59 |
| 4. | ΔUS total surplus | 1.09 | 2.54 | 3.56 |
| 5. | Cost per additional EV ($000s/EV) | 32.1 | 33.4 | 40.0 |
| 6. | US MVPF (global SCC) | 1.14 | 1.19 | 1.19 |
| | **Panel (d): Social Cost of Carbon = $200** | | | |
| 1. | Mean EV subsidy | $7,328 | $12,731 | $17,460 |
| 2. | ΔGlobal negative externalities | -1.79 | -3.23 | -6.06 |
| 3. | ΔGlobal surplus | 0.62 | 0.53 | 1.43 |
| 4. | ΔUS total surplus | 1.58 | 3.35 | 4.94 |
| 5. | Cost per additional EV ($000s/EV) | 32.1 | 34.4 | 42.1 |
| 6. | US MVPF (global SCC) | 1.21 | 1.20 | 1.19 |
| | **Panel (e): Social Cost of Carbon = $241** | | | |
| 1. | Mean EV subsidy | $7,328 | $13,478 | $18,685 |
| 2. | ΔGlobal negative externalities | -1.99 | -3.92 | -7.41 |
| 3. | ΔGlobal surplus | 0.82 | 0.73 | 1.87 |
| 4. | ΔUS total surplus | 1.78 | 3.74 | 5.62 |
| 5. | Cost per additional EV ($000s/EV) | 32.1 | 34.8 | 42.9 |
| 6. | US MVPF (global SCC) | 1.23 | 1.20 | 1.20 |

Notes: This table presents sensitivity analysis for counterfactual simulation results presented in Tables 8 and 9. All results are relative to the scenario with no EV credits. For each scenario, in column 2 we simulate the uniform EV subsidy subject to Section 30D trade restrictions that maximizes US total surplus and, in column 3, the choice-specific differentiated EV subsidy that maximizes US total surplus. The defaults are an MCPF of 1 and an SCC of $241. The mean of subsidies is computed only on EVs which are eligible to receive subsidies among income-eligible households as implied by the model. The EVs eligible to receive subsidies in all scenarios are those vehicles already eligible under 30D in the IRA. Uniform subsidy scenarios do not impose any household income restrictions whereas the IRA does. "US total surplus" equals "[global] total surplus" minus foreign automakers' producer surplus.

Table A9: **Counterfactual Simulation Results: Model Robustness**

| | | (1) Main specification | (2) Homogeneous demand | (3) Foreign/ domestic nest | (4) Elasticity of -4 | (5) Nocke-Schutz restriction | (6) Non-Tesla EVs priced at cost |
|---|---|---|---|---|---|---|---|
| | **Panel (a): Market Aggregates under Counterfactual Scenario** | | | | | | |
| 1. | ΔVehicle registrations (000s/year) | 97.9 | 132 | 100.0 | 76.3 | 135 | 119 |
| 2. | ΔEV registrations (000s/year) | 216 | 240 | 221 | 170 | 246 | 257 |
| 3. | ΔEV market share (pp) | 2.0 | 2.16 | 2.00 | 1.54 | 2.22 | 2.26 |
| 4. | ΔLease share, within EVs (pp) | 19.7 | 17.4 | 19.6 | 20.2 | 17.3 | 26.2 |
| 5. | ΔUS assembly share, within EVs (pp) | 10.3 | 10.1 | 10.0 | 9.29 | 10.9 | 11.9 |
| | **Panel (b): Surplus Effects Relative to No EV Subsidy Baseline ($billion/year)** | | | | | | |
| 6. | ΔUS consumer surplus | 4.86 | 5.18 | 4.89 | 5.52 | 5.30 | 5.84 |
| 7. | ΔGlobal producer surplus | 1.66 | 2.00 | 1.65 | 1.49 | 1.65 | 0.58 |
| 8. | ΔUS producer surplus | 2.62 | 2.85 | 2.59 | 2.75 | 2.51 | 1.38 |
| 9. | ΔForeign producer surplus | -0.96 | -0.85 | -0.94 | -1.26 | -0.86 | -0.80 |
| 10. | ΔGlobal neg. externalities (global SCC) | -1.99 | -2.02 | -2.04 | -1.61 | -2.10 | -2.00 |
| 11. | ΔUS neg. externalities (dom. SCC) | -0.95 | -0.96 | -0.98 | -0.77 | -1.00 | -0.96 |
| 12. | ΔForeign neg. externalities (for. SCC) | -1.04 | -1.06 | -1.07 | -0.84 | -1.10 | -1.03 |
| 13. | ΔUS subsidy spending | 6.94 | 7.38 | 6.93 | 6.88 | 7.38 | 8.56 |
| 14. | ΔGas tax revenue | -0.76 | -0.82 | -0.77 | -0.60 | -0.84 | -0.86 |
| 15. | ΔGlobal surplus | 0.82 | 1.01 | 0.88 | 1.13 | 0.83 | -1.00 |
| 16. | ΔUS total surplus (global SCC) | 1.78 | 1.85 | 1.82 | 2.39 | 1.69 | -0.21 |
| 17. | ΔUS total surplus (dom. SCC) | 0.74 | 0.80 | 0.76 | 1.55 | 0.59 | -1.24 |
| 18. | ΔForeign total surplus (for. SCC) | 0.08 | 0.21 | 0.13 | -0.42 | 0.24 | 0.24 |
| | **Panel (c): Impacts on $CO_2$ Relative to No EV Subsidy Baseline** | | | | | | |
| 19. | Δ$CO_2$ emissions (million tons/year) | -4.88 | -4.96 | -5.00 | -3.94 | -5.15 | -4.85 |
| | **Panel (d): Employment and Battery Production Effects Relative to No EV Subsidy Baseline (000s/year)** | | | | | | |
| 20. | ΔUS auto manufacturing jobs | 11.5 | 12.7 | 11.3 | 8.60 | 13.8 | 14.5 |
| 21. | ΔUS auto parts jobs | 7.3 | 8.08 | 7.18 | 5.44 | 8.83 | 9.14 |
| 22. | ΔUS auto assembly jobs | 4.2 | 4.57 | 4.17 | 3.15 | 5.00 | 5.34 |
| 23. | ΔUS-assembled EV batteries | 241 | 249 | 242 | 206 | 260 | 280 |
| | **Panel (e): MVPF versus No EV Subsidy Baseline** | | | | | | |
| 24. | Global MVPF | 1.11 | 1.12 | 1.11 | 1.15 | 1.10 | 0.89 |
| 25. | US MVPF (global SCC) | 1.23 | 1.23 | 1.24 | 1.32 | 1.21 | 0.98 |
| 26. | US MVPF (domestic SCC) | 1.10 | 1.10 | 1.10 | 1.21 | 1.07 | 0.87 |
| 27. | Cost per additional EV ($000s/EV) | 32.1 | 30.7 | 31.4 | 40.4 | 30.0 | 33.3 |
| 28. | Cost per additional US job ($000s/job) | 604 | 583 | 611 | 801 | 533 | 591 |
| 29. | US-optimal uniform EV subsidy with 30D restrictions | $9,825 | $8,712 | $9,855 | $12,209 | $7,855 | $3,267 |

Notes: This table presents simulation results comparing outcomes under the IRA against a counterfactual with no EV subsidies. Column 1 is the main specification also reported in Tables 6 and 7. Column 2 reports a specification that restricts all coefficients interacted with income to be zero and does not account for income-eligibility. Column 3 modifies the nested logit assumption by replacing the vehicle class nest in the second level of the tree in Appendix Figure A16 with the US- or foreign-firm classification. Column 4 targets an alternative, more inelastic own-price model-level elasticity. Column 5 modifies the supply side assumptions under the same homogeneous demand model in column 2 as discussed in Section 5. Column 6 restricts all firms except Tesla to pricing their EVs at marginal cost.

Table A10: **Moments Not Formally Targeted: Measures of Model Fit**

| Statistic | Model Value | Reference Value |
|---|---|---|
| **Panel (a): Second Choice Measures** | | |
| Share of GV buyers with GV second choice | 95.8% | 96% (NVES) |
| Average MPG(e) of second choice, EV buyers | 71.9 | 66.1 (NVES) |
| Average MPG(e) of second choice, GV buyers | 29.6 | 27.6 (NVES) |
| US firm share of second choices, US firm first choice | 40.1% | 62.2% (NVES) |
| Domestically assembled share of second choices, domestically assembled first choice | 42.4% | 58% (NVES) |
| Correlation (share-weighted), class-powertrain aggregate second choices | 0.9 (Figure 6) | |
| Correlation (share-weighted), class-powertrain aggregate second choices, EV first and second choice | 0.995 (Figure 6) | |
| **Panel (b): Price Measures** | | |
| Aggregate new vehicle market demand elasticity<br>Aggregate GV demand elasticity | -0.95<br>-1.1 | -1 (Berry, Levinsohn, and Pakes 2004)<br>-1.29 (Grieco, Murry, and Yurukoglu 2024)<br>-0.92 (Allcott et al. 2026) |
| Aggregate EV demand elasticity | -2.1 | -2.76 (Xing, Leard, and Li 2021) |
| Pass-through of lease credits to lease prices relative to buy prices | \$8,883 | \$4,077 to \$6,999<br>(Appendix Table A2) |
| Pass-through of IRA credits to purchase prices | \$7,281 | \$6,875 to \$8,861<br>(Appendix Table A3) |
| Share-weighted average markup | 19% (GV)<br>21% (EV) | 22%<br>(Grieco, Murry, and Yurukoglu 2024) |

Notes: This table presents statistics on data moments that we do not formally target, as discussed near the end of Section 5.2. All second choice statistics are weighted by the market share (or survey share) of the first choice unless otherwise specified. The ranges of reference values for pass-through are the 95 percent confidence intervals of our regression estimates. The pass-through of IRA credits to purchase prices is measured per \$7,500 credit: we take \$7,500 minus the average change in equilibrium purchase prices for affected vehicles, scaling (dividing) each submodel's price change by 0.5 for a partial credit and 1 for a full credit.

Table A11: **Decomposition of Constrained Optimal Uniform Subsidies**

| | MCPF = 1.0, SCC = \$28, Job = \$0 | MCPF = 1.4, SCC = \$28, Job = \$0 | MCPF = 1.0, SCC = \$241, Job = \$0 | MCPF = 1.4, SCC = \$241, Job = \$0 | MCPF = 1.0, SCC = \$28, Job = \$10.7k | MCPF = 1.0, SCC = \$241, Job = \$10.7k |
|---|---|---|---|---|---|---|
| Price distortion | \$5,687 | \$3,945 | -\$2,957 | -\$2,317 | \$5,705 | -\$2,936 |
| Indirect substitution | \$86 | -\$1,302 | \$12,778 | \$7,083 | \$149 | \$12,871 |
| Net jobs value | – | – | – | – | \$479 | \$476 |
| *Subtotal* | *\$5,774* | *\$2,643* | *\$9,821* | *\$4,766* | *\$6,333* | *\$10,411* |
| Profit shifting | \$4,052 | \$3,640 | \$3,657 | \$3,461 | \$3,994 | \$3,603 |
| Tax distortion | – | -\$4,449 | – | -\$4,623 | – | – |
| Uniform subsidy | \$9,825 | \$1,834 | \$13,478 | \$3,604 | \$10,327 | \$14,014 |

Notes: This table presents the decomposition of constrained optimal uniform subsidies for a US social planner under given values of the marginal cost of public funds and social cost of carbon. Each component corresponds to the respective term in Proposition 1, using equation (39) in the appendix for the more general case of an MCPF greater than one. The value of jobs is treated as an externality in the formula but reported separately as the value placed on net additional US manufacturing jobs. The combined externality and markup components, together with the net jobs value where applicable, sum to the subtotal displayed in row 4. Columns 1 and 5 correspond to the uniform subsidies calculated in Table 8. Columns 2-4 correspond to uniform subsidies calculated in Table A8 under, respectively, panels (a), (e), and (b). The decomposition was calculated by numerically approximating the derivatives in equation (39) at the subsidy value. Note how the price distortion terms (row 1) relate to the distribution of price distortions in Figure 7, panels (a) and (b): under a global social cost of carbon, the price distortion alone implies a tax, whereas under a domestic social cost of carbon, the price distortion alone implies a subsidy.

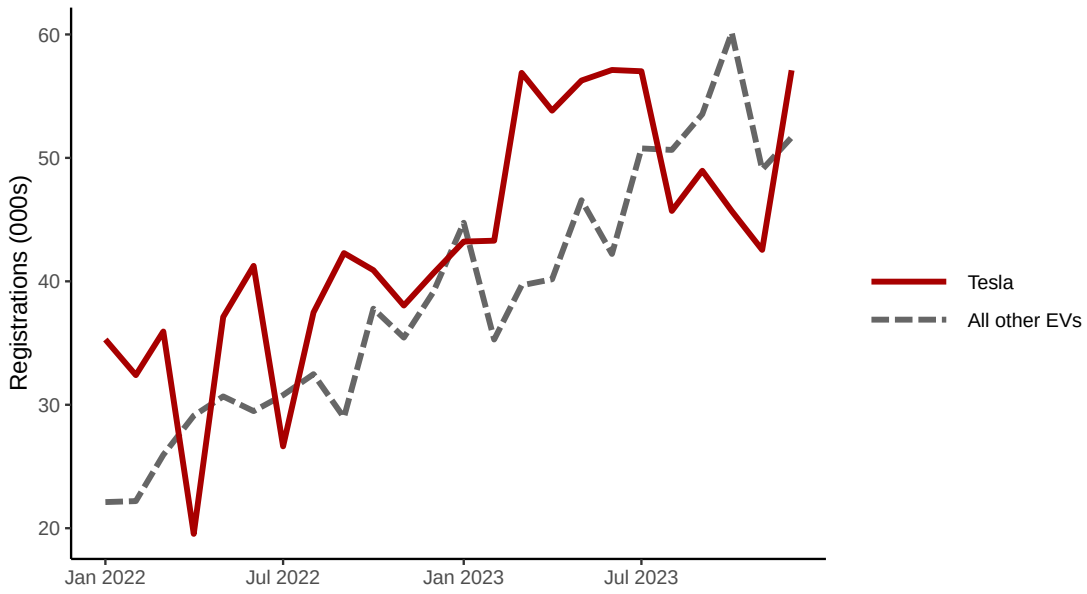
Table A12: Forgone Producer Surplus From Counterfactual 30D Eligibility

| | | | | Brand | | Parent | | |
|---|---|---|---|---|---|---|---|---|
| Model | Brand | Parent | Baseline 30D eligibility | ΔPS | %ΔPS | ΔPS | %ΔPS | ΔG |
| Wrangler PHEV | Jeep | Stellantis | Partial | \$87 | 2.7 | \$85 | 1.3 | \$73 |
| Mustang Mach-E | Ford | Ford | Partial | \$111 | 1.6 | \$111 | 1.6 | \$115 |
| Ioniq 5 | Hyundai | Hyundai | None | \$288 | 5.6 | \$266 | 2.3 | \$227 |
| RAV4 PHEV | Toyota | Toyota | None | \$253 | 2.3 | \$250 | 1.9 | \$234 |
| R1S | Rivian | Rivian | Partial | \$75 | 18.9 | \$75 | 18.9 | \$77 |
| Ioniq 6 | Hyundai | Hyundai | None | \$140 | 2.7 | \$132 | 1.2 | \$113 |
| EV6 | Kia | Hyundai | None | \$156 | 2.7 | \$144 | 1.3 | \$124 |
| ARIYA | Nissan | Nissan | None | \$87 | 2.7 | \$87 | 2.4 | \$70 |
| Sorento PHEV | Kia | Hyundai | None | \$136 | 2.4 | \$126 | 1.1 | \$130 |
| i4 | BMW | BMW | None | \$25 | 1.1 | \$25 | 1.1 | \$16 |

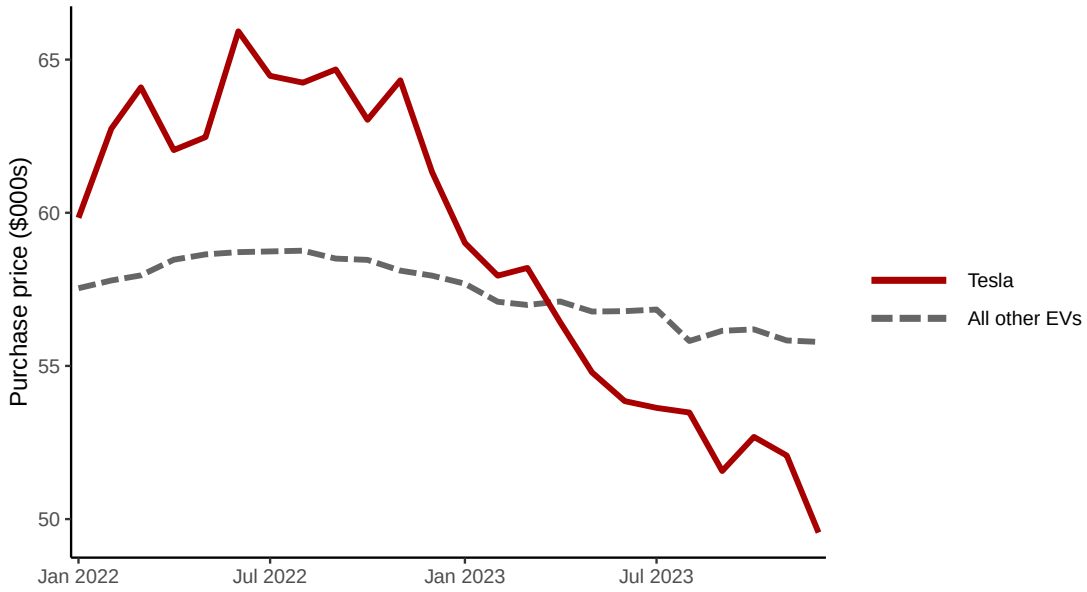
Notes: All dollar amounts in annual millions. This table reports the change in producer surplus and subsidy spending under a counterfactual equilibrium where the designated model in each row unilaterally receives full 30D eligibility "as if" it fulfilled the assembly, battery, and mineral requirements. The models shown are the ten EV models with the most registrations, in order, without full 30D eligibility, excluding constituent submodels subject to MSRP caps. Submodels over the MSRP cap remain excluded under each counterfactual. The producer surplus change is reported for the brand and for the parent firm, which internalizes substitution among parent brands. Percent change is relative to producer surplus under the IRA.

Figure A1: **Monthly Registrations and Prices of Teslas and Other EVs**

(a) **Registrations**

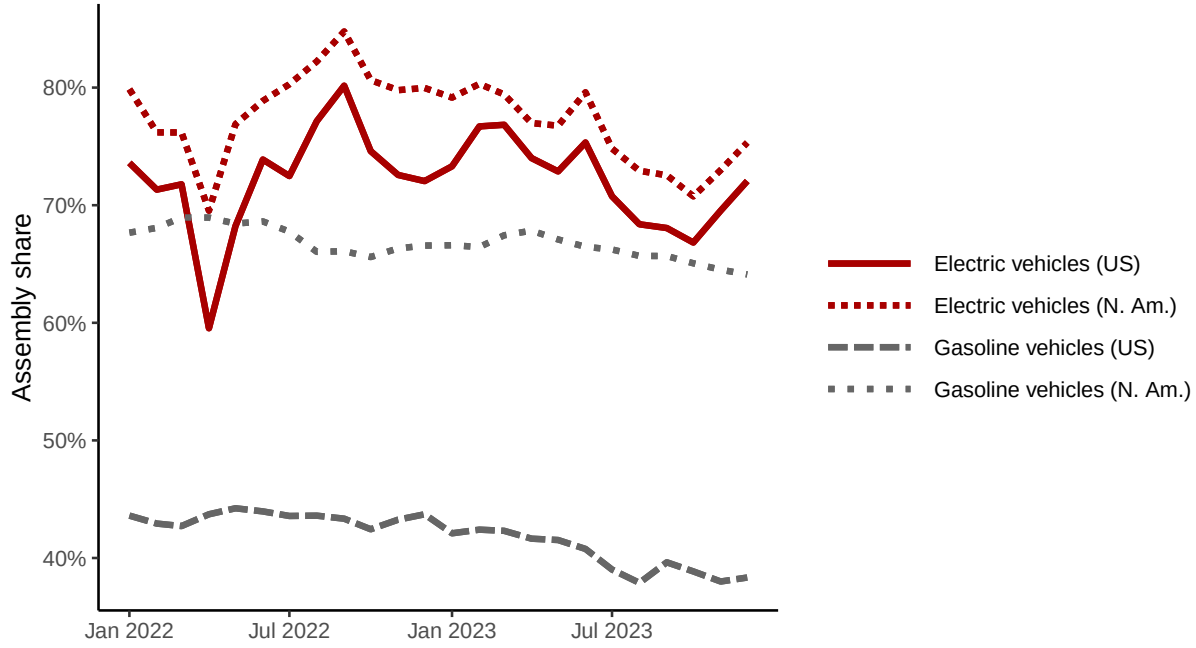



(b) **Average Purchase Prices**



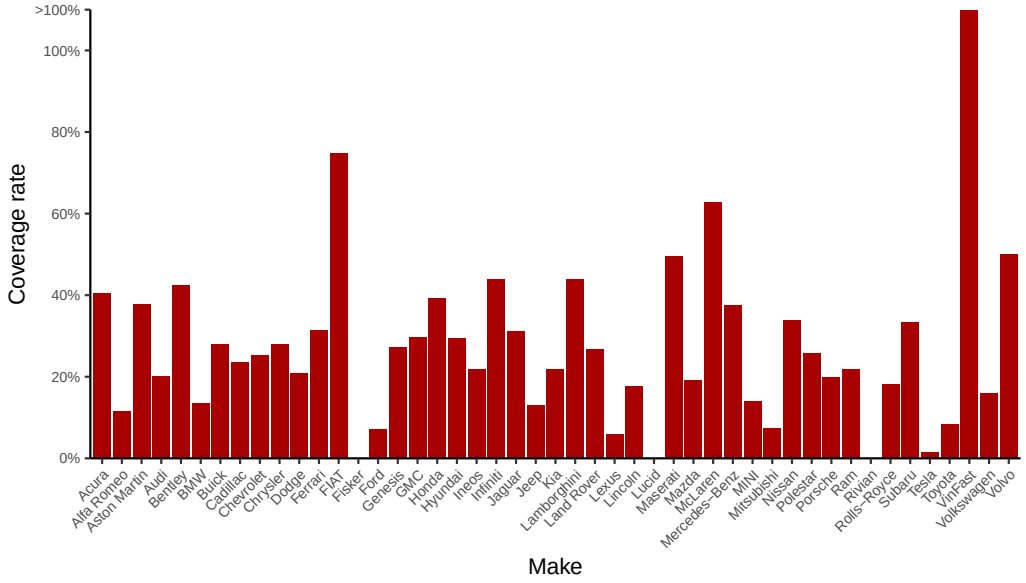

Notes: Panel (a) presents monthly registrations of Teslas and other EVs. Panel (b) presents price indexes constructed by computing the January 2023 weighted averages (weighting models by average monthly sales in months when the model was available) for Tesla and non-Tesla EVs, and then recursively adding the sales-weighted average changes for all models available in each previous or subsequent month.

Figure A2: **Shares of New Vehicle Registrations Assembled in the US or N.A.**




Notes: This figure presents the share of new vehicles registered that are assembled in the United States or in North America broadly.

Figure A3: **Cox Data Coverage Rate**



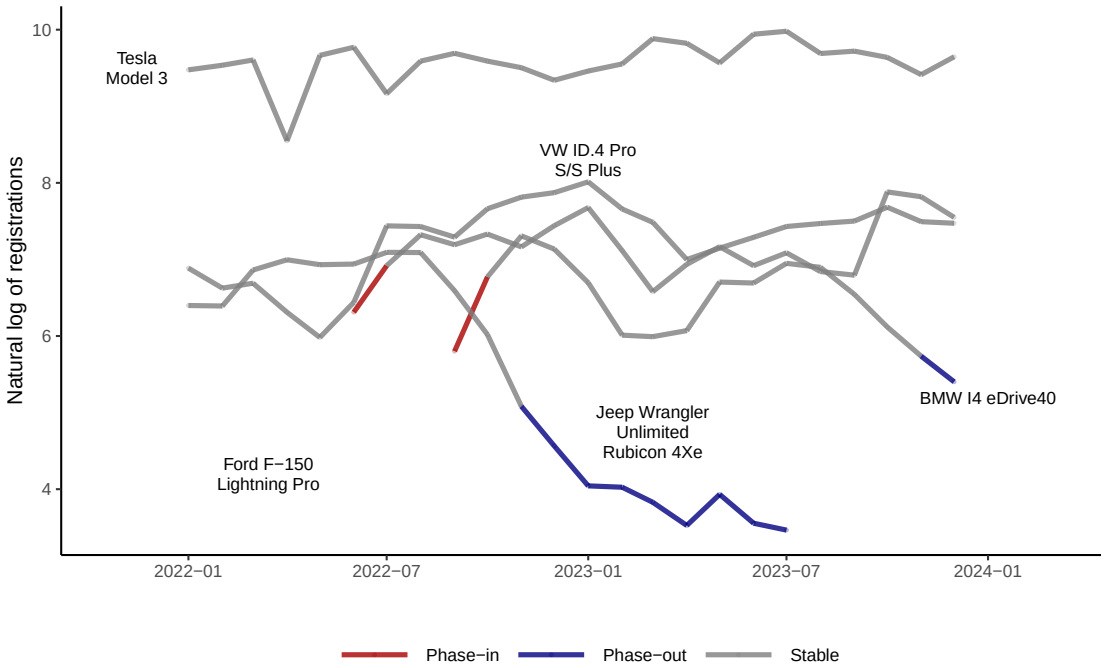

Notes: This figure shows the ratio of new vehicle transactions in the Cox dealership data to nationwide new vehicle registrations in the Experian data, for each make, aggregated over 2022 and 2023.

Figure A4: **Availability by Month for Each Electric Vehicle Submodel**



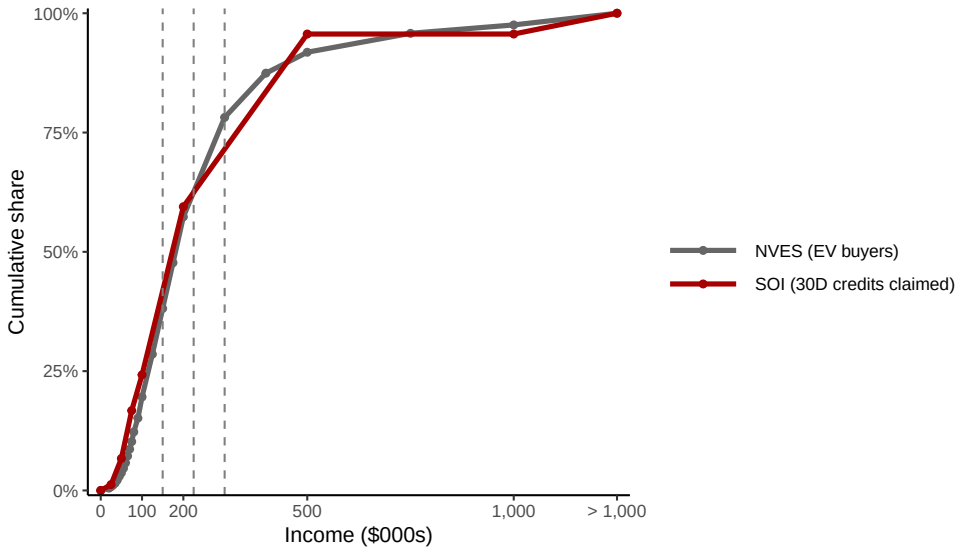

Notes: This figure has one row for each EV submodel in the data, and one column for each month of the sample. A cell is shaded if the submodel has more than 25 new registrations in that month.

Figure A5: **Phase-In and Phase-Out Periods by Submodel**



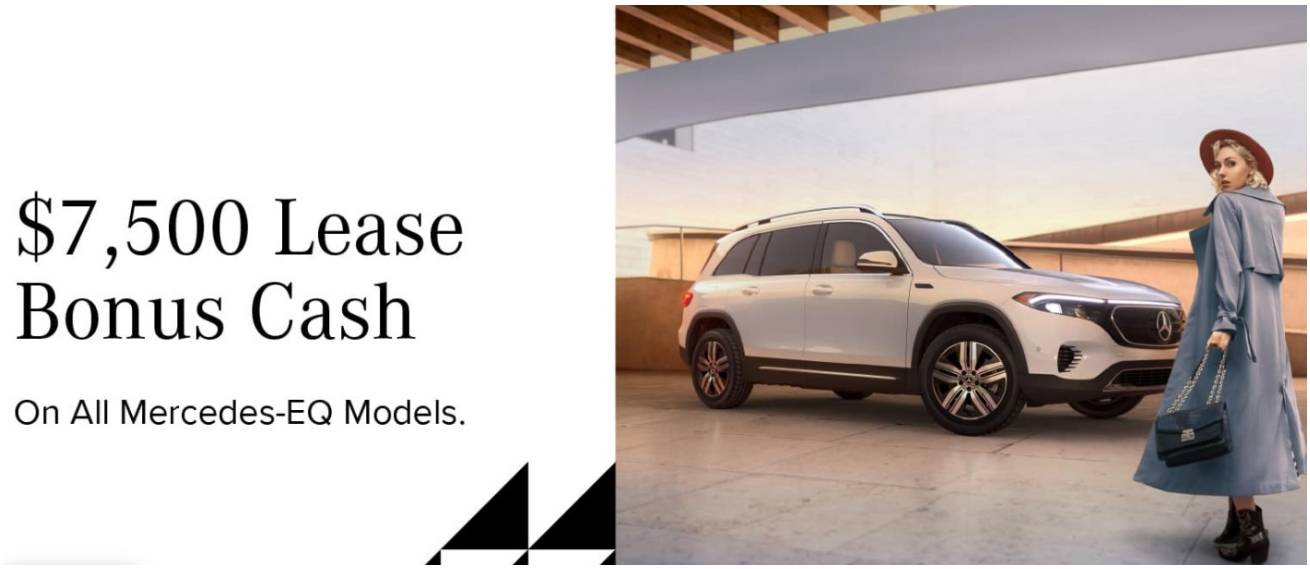

Notes: This figure presents registrations by submodel for a select few EVs for the purpose of demonstrating our classification approach. Red and blue lines represent phase-in and phase-out periods at the beginning and end of a submodel's life. To define these periods, we construct $\bar{q}_j$ as submodel $j$'s sample average monthly registrations in months with non-zero registrations. Phase-in periods are consecutive months beginning with zero registrations and ending when registrations first exceed $\bar{q}_j/2$. Analogously, phase-out periods are consecutive months ending when registrations reach zero and beginning when registrations last exceed $\bar{q}_j/2$.

Figure A6: **Income Distributions for EV Buyers in 2022/2023 and 30D Tax Credit Claimants in 2020**



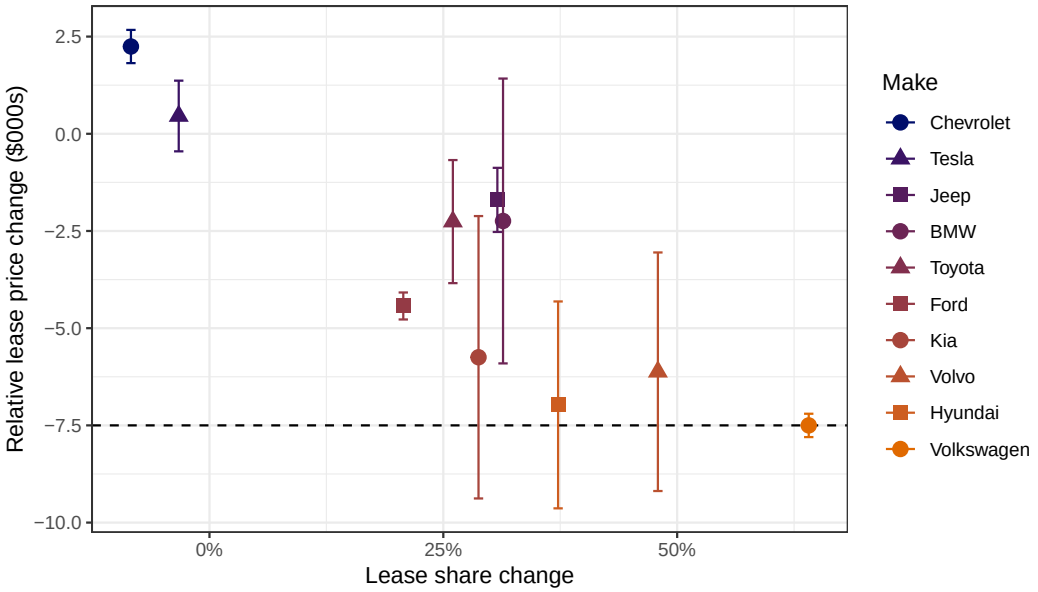

Notes: The red line shows the cumulative distribution function (CDF) of Adjusted Gross Income (AGI) for taxpayers claiming the Section 30D credit for tax year 2020, using data from the IRS Statistics of Income (2023). The grey line shows the CDF of self-reported household income for people who bought new EVs in 2022 and 2023, using data from the New Vehicle Experience Study. The IRA requires that to be eligible to claim the credit in 2023 and after, individual buyers must have AGI below $300,000 for married couples filing jointly, $225,000 for household heads, or $150,000 for all other taxpayers. The vertical lines reflect those income thresholds.

Figure A7: **Example of Promotional Material Advertising $7,500 Lease Bonus**



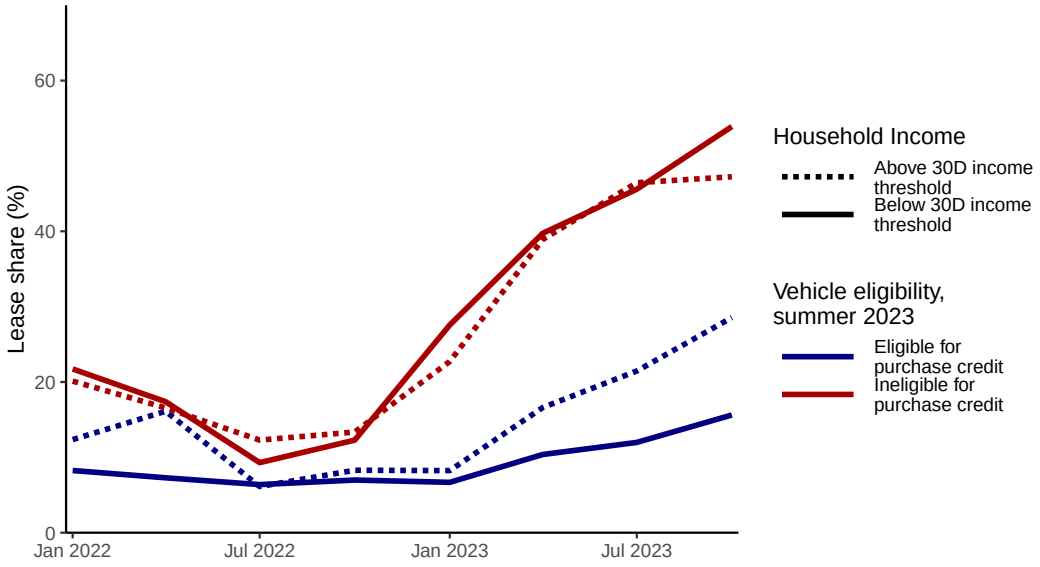

Notes: Representative screenshot showing $7,500 promotion on EV leases. Observed on the Mercedes-Benz website, August 2023.

Figure A8: **Changes in Relative Lease Prices and Lease Shares from October-December 2022 to July-August 2023 by Make**



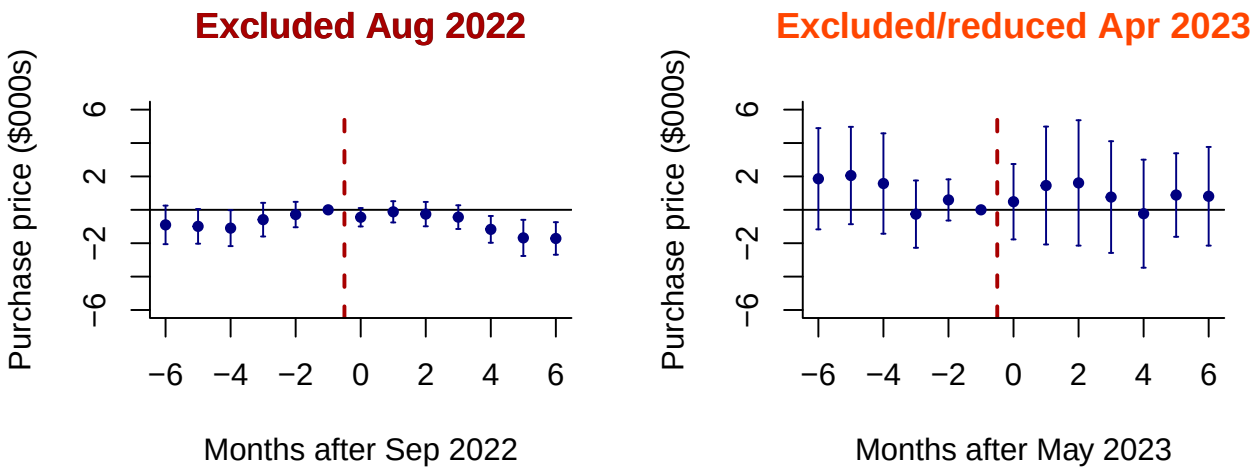

Notes: This figure presents the differences (July-August 2023 vs. October-December 2022) in relative lease prices and lease shares, for the top-10 selling EV brands.

Figure A9: **Lease Share Trends Among Income Eligibility Groups**



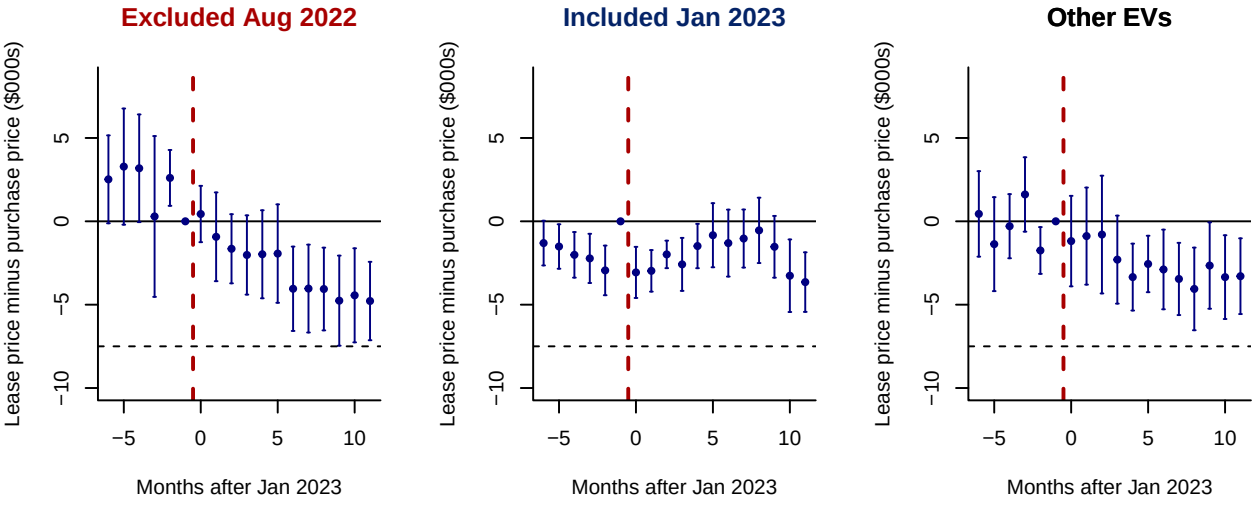

Notes: This figure presents quarterly lease shares among EVs in the NVES survey, separately shown for households above and below the income threshold for 30D credits and for vehicles which were eligible and ineligible for 30D credits in July and August 2023. This aggregation is slightly different than that shown in Figure 5, as we here group EVs only on their purchase credit eligibility as of summer 2023. Households' responses are aggregated using the survey weights and household-level eligibility is determined using self-reported household income and marital status.

Figure A10: **Purchase Price Event Study with Reweighted Controls**



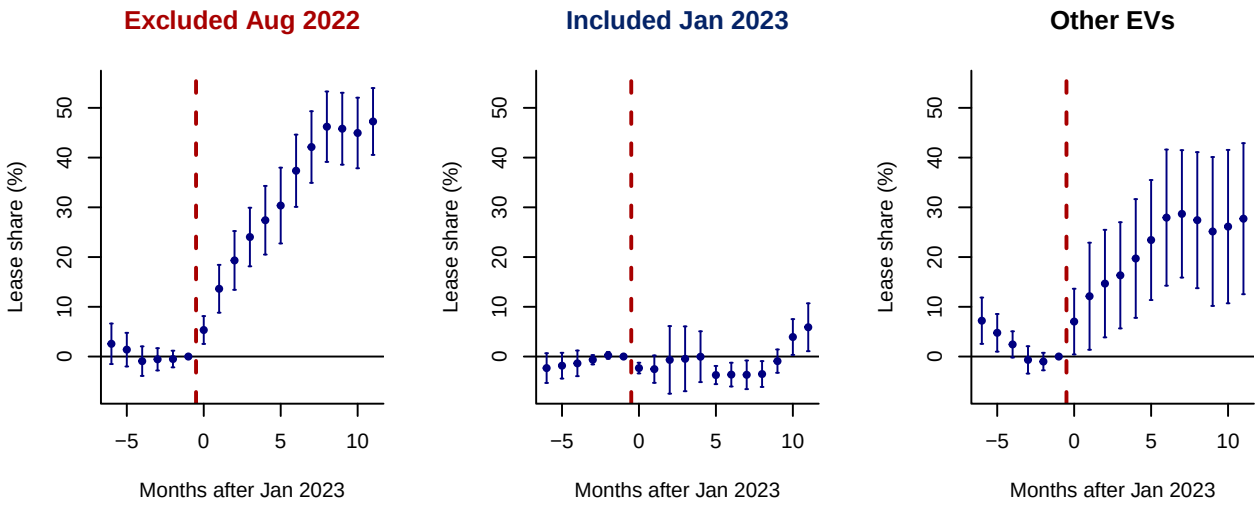

Notes: This figure presents the $\gamma_s^e$ coefficients and 95 percent confidence intervals from equation (2). Eligibility groups are described in Figure 1. This parallels Appendix Figure A14, except that we also re-weight the control observations (GVs) to match the average pre-IRA EV price.

Figure A11: **Lease Price Event Study with Reweighted Controls**




Notes: This figure presents the $\gamma_s^e$ coefficients and 95 percent confidence intervals from equation (2). Eligibility groups are described in Figure 1. This parallels Figure 4, except that we also re-weight the control observations (GVs) to match the average pre-IRA EV price.

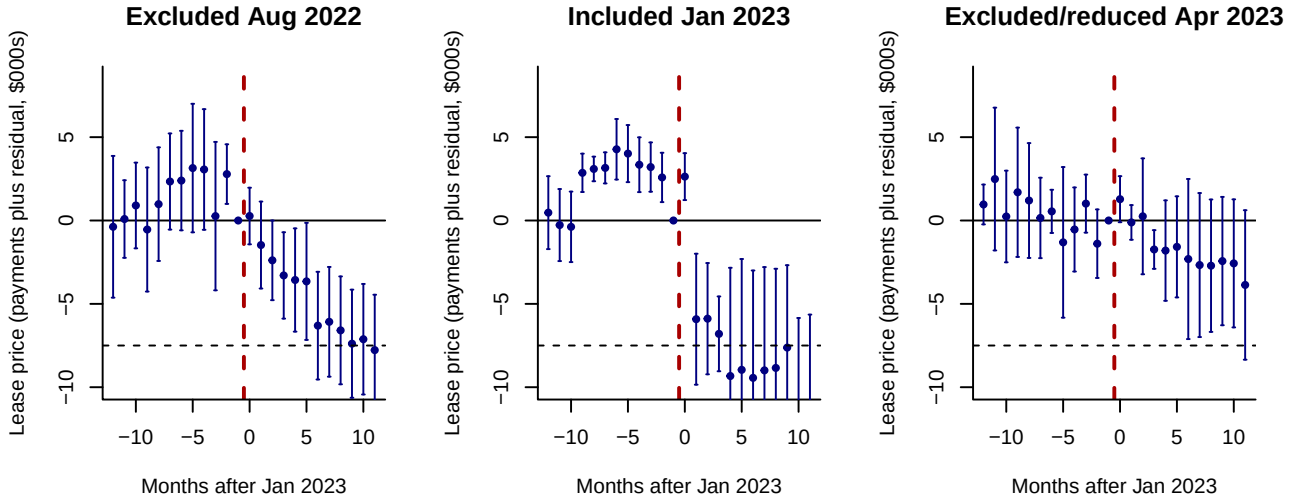
Figure A12: **Lease Share Event Study with Reweighted Controls**



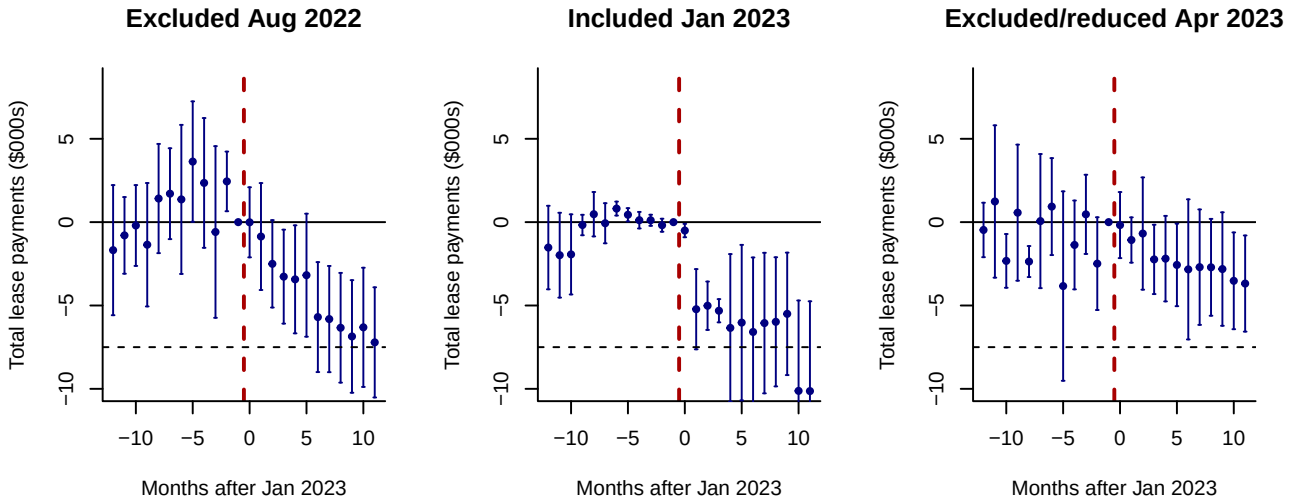

Notes: This figure presents the $\gamma_s^e$ coefficients and 95 percent confidence intervals from equation (2). Eligibility groups are described in Figure 1. This parallels Figure 5, except that we also re-weight the control observations (GVs) to match the average pre-IRA EV price.

Figure A13: **Lease Price Event Study Using Alternative Measures of Lease Prices**

(a) **Lease Price Event Study, Not Deducting Purchase Price**



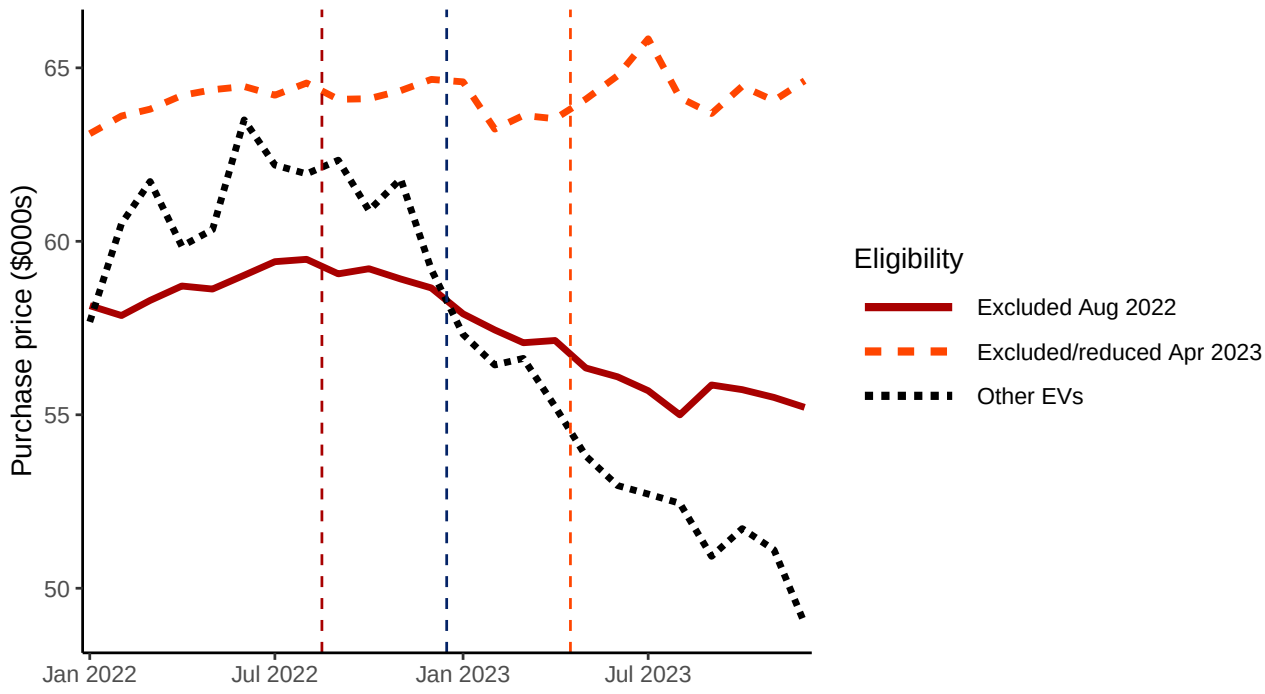

(b) **Lease Price Event Study Using Only Lease Payments**



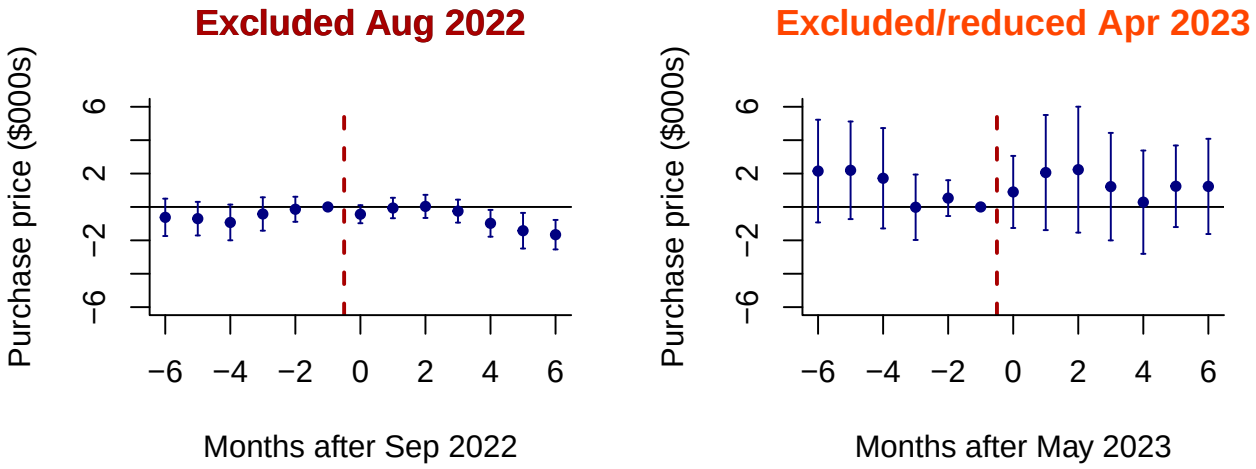

Notes: This figure presents the $\gamma_r^e$ coefficients and 95 percent confidence intervals from equation (2). Eligibility groups are described in Figure 1. This parallels Figure 4 but uses as the outcome variable the computed lease price (discounted payments plus residual value) in Panel (a) and lease payments (down payment and monthly payments) from 36-month leases in Panel (b). 36-month leases account for 65% of all leases in our Cox data. Because these measures do not difference out each vehicle's purchase price series, relative declines in lease prices seen here might reflect both the pass-through of 45W tax credits and the fall in the vehicle prices for that product group.

Figure A14: **Purchase Price Trends Associated with Eligibility Changes**

(a) **Trends by Eligibility Group**



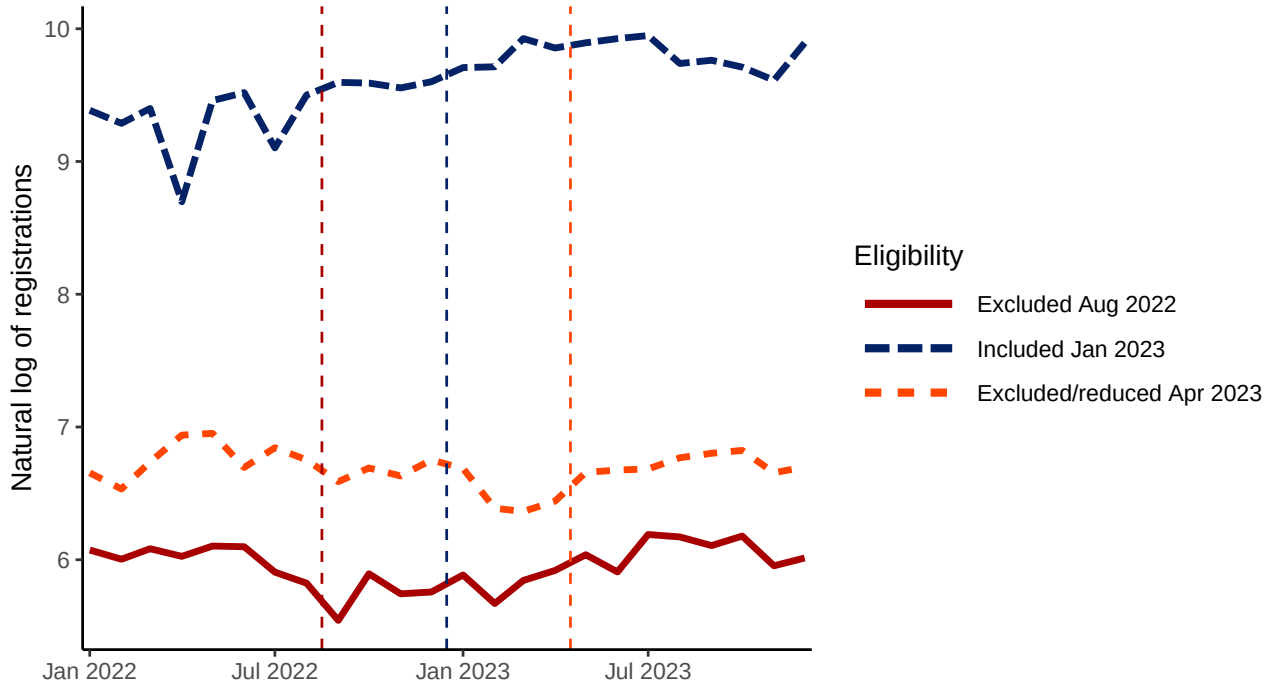

(b) **Event Study Estimates**



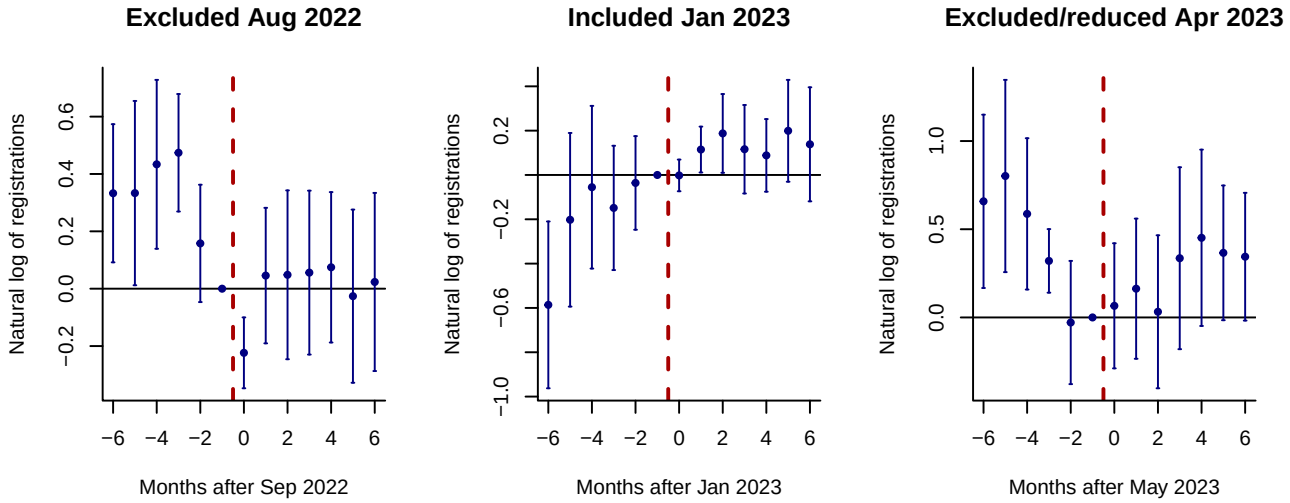

Notes: Panel (a) presents purchase price indexes constructed by computing the January 2023 weighted averages for each eligibility group and then recursively adding the sales-weighted average changes for all submodels available in each previous or subsequent month. Panel (b) presents the $\gamma_s^e$ coefficients and 95 percent confidence intervals from equation (2). Eligibility groups are described in Figure 1; the Other EVs group includes all EVs not part of either the Excluded August 2022 or the Excluded/Reduced April 2023 groups. In both panels, we weight submodels by average monthly sales in months when the submodel was available.

Figure A15: **Registration Trends Associated with Eligibility Changes**

(a) **Trends by Eligibility Group**

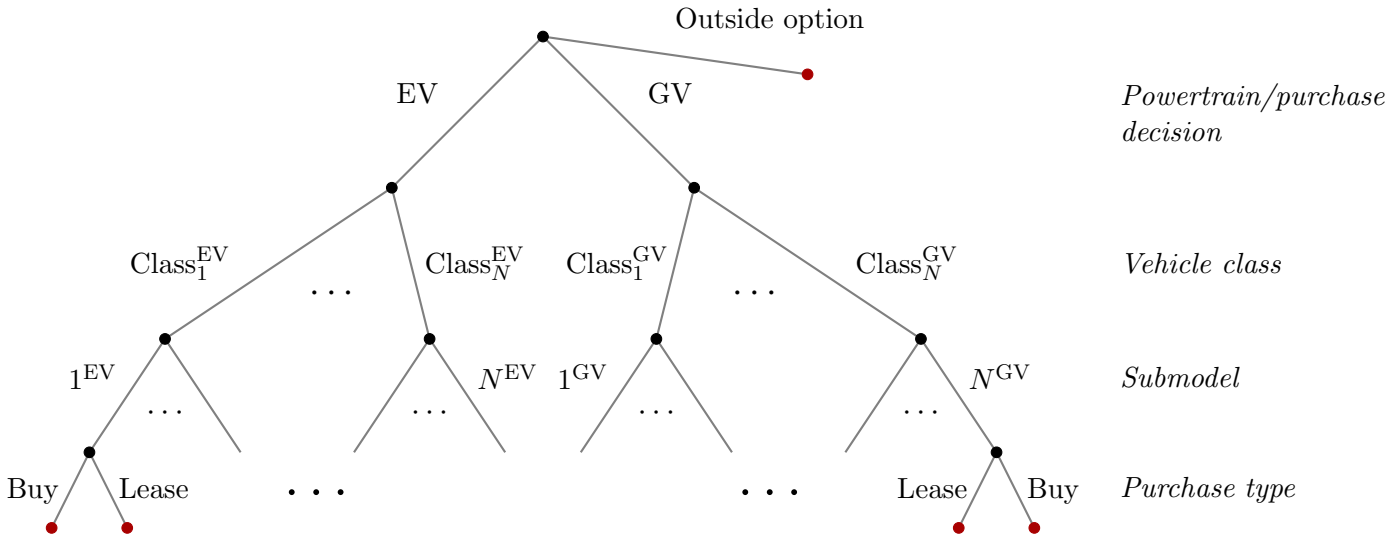



(b) **Event Study Estimates**



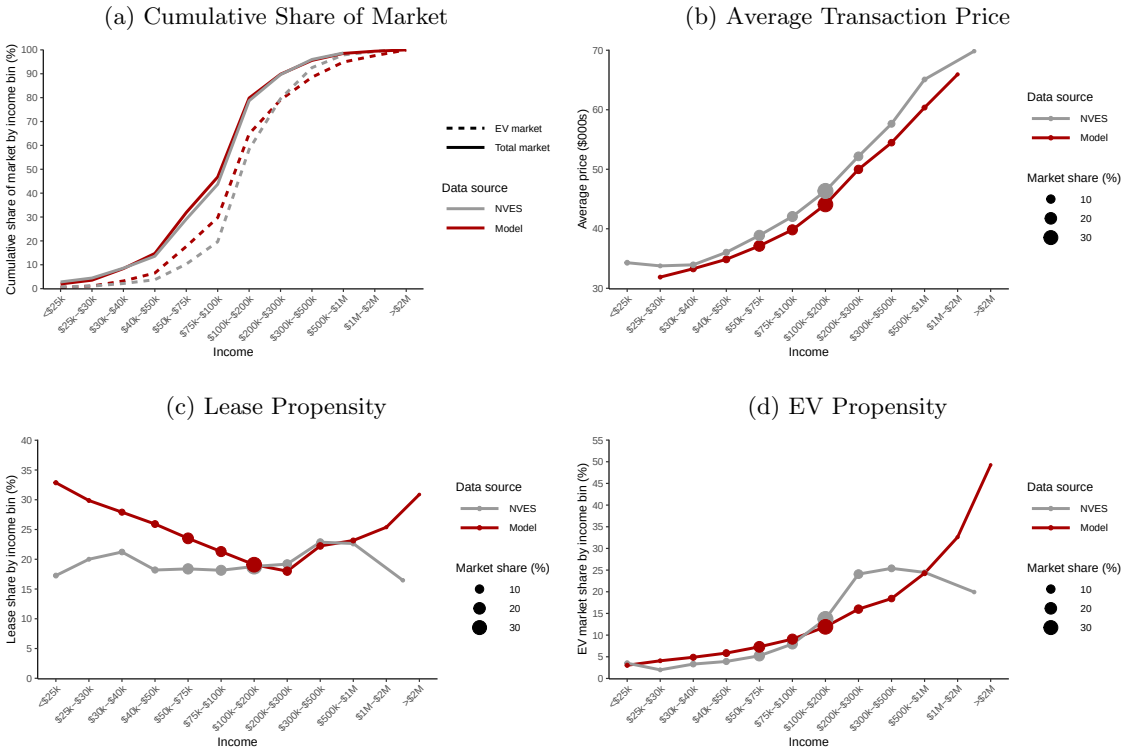

Notes: Panel (a) presents ln(registrations) indexes constructed by computing the January 2023 weighted averages for each eligibility group and then recursively adding the sales-weighted average changes for all submodels available in each previous or subsequent month. Panel (b) presents the $\gamma_s^e$ coefficients and 95 percent confidence intervals from equation (2). Eligibility groups are described in Figure 1. In both panels, we weight submodels by average monthly sales in months when the submodel was available.

Figure A16: Nested Logit Structure



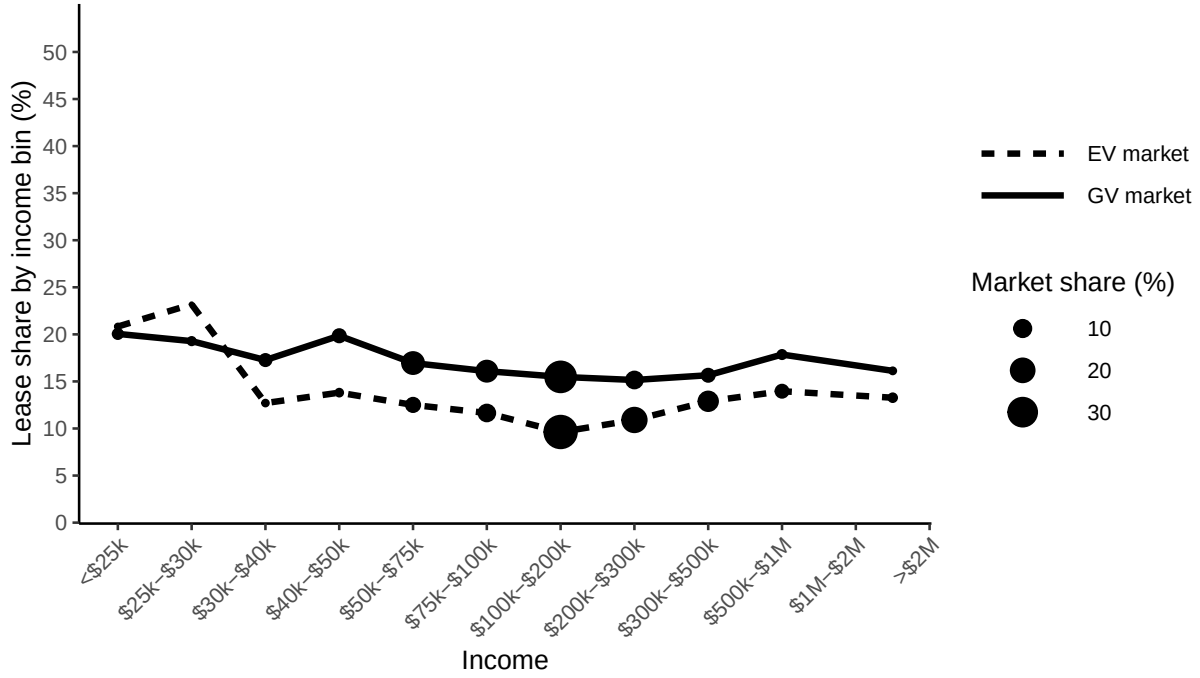

Notes: This figure presents the nested logit tree representation of the primary specification of the structural model. For inside goods, four levels of nesting correspond to powertrain, class, submodel, and purchase mode. Choices, including the outside option, are the red nodes at the extreme end of any tree segment.

Figure A17: **Model Fit along Income Heterogeneity**




Notes: This figure presents a comparison across the full type distribution between model-implied outcomes and the NVES survey data. These four outcomes correspond to the four moments used in demand estimation. We do not target the full distribution, instead collapsing the data into a comparison between households above and below $300,000 in reported income as discussed in the text. The dot sizes reflect each bin's estimated share of the vehicle market from the model.

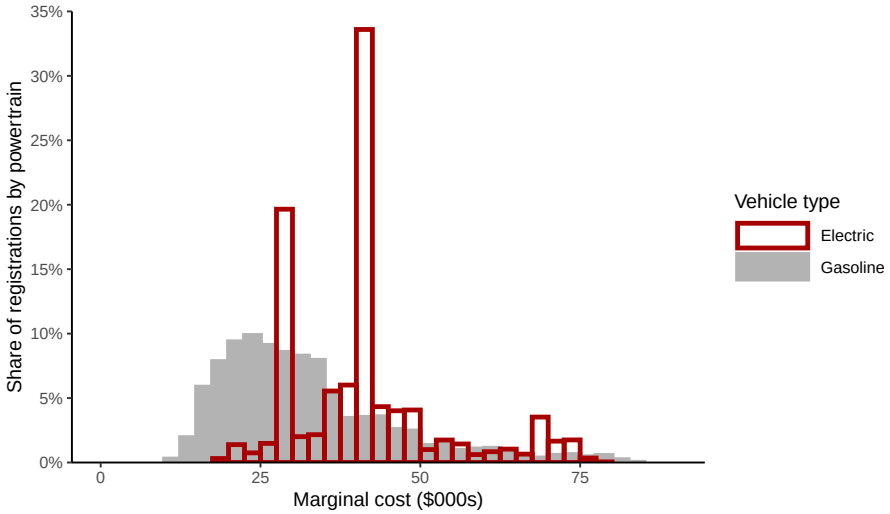
Figure A18: **Lease Propensity Heterogeneity in Income by Powertrain**



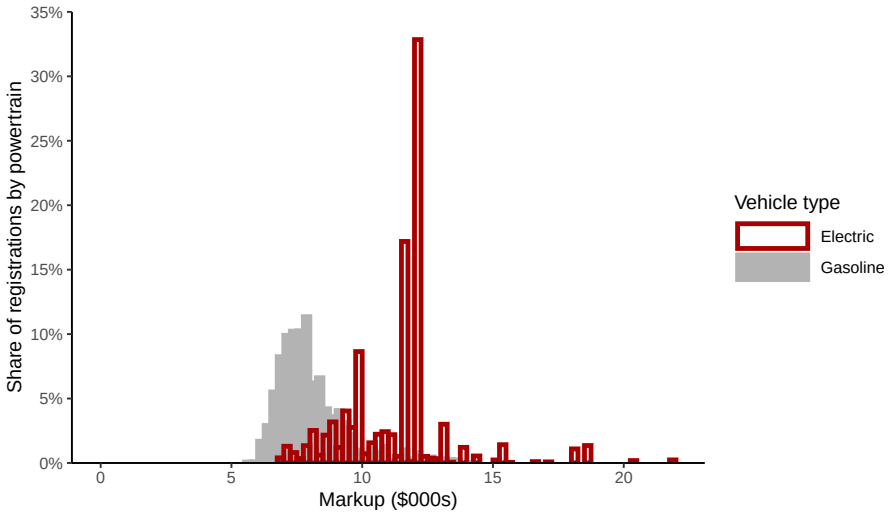

Notes: This figure presents a comparison of lease propensity across income bins in the NVES survey data for 2022, split by the powertrain of the new vehicle. The dot sizes reflect each bin's estimated share of the vehicle market from the survey weights.

Figure A19: **Distribution of Implied Marginal Cost and Markup for EVs and GVs**

(a) **Marginal Costs**




(b) **Markups**




Notes: This figure presents the distribution across submodels of purchase- and lease-specific marginal costs and markups implied by firms' first-order condition in equation (8), weighting each choice by its registrations in July and August 2023. Markups are computed in the equilibrium where all EV subsidies are set to zero.